

MATH40002 - Analysis I

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Introduction

In the introductory module (MATH40001), we constructed the number systems $\mathbb{N}$, $\mathbb{Z}$, $\mathbb{Q}$ and $\mathbb{R}$. Real analysis focuses on the real numbers, $\mathbb{R}$, formalising many properties about functions that we know from calculus. For example, an intuitive notion of a continuous function is that you can draw the function on a piece of paper without taking your pen off the page. We make concepts like these precise in this module.

Chapter 1

Countability

1.1 Countably infinite sets

Definition 1.1. A set A is countably infinite if there exists a bijection $f : \mathbb{N} \rightarrow A$.

Intuitively, this says that you can list the elements of A as $a_1, a_2, \dots$ using only the natural numbers. We say that a set is *countable* if it is finite or countably infinite. Note we will take $\mathbb{N} = \{1, 2, \dots\}$, unless stated otherwise.

Note that if there is a bijection between a given pair of sets A and B , then A is countable if and only if B is countable. This extends the natural idea of finite sets having the same number of elements if and only if there is a bijection between the sets. Formally, we define two sets having the same *cardinality* as follows:

Definition 1.2. Let A, B be sets. We say that A and B have the same cardinality if there exists a bijection $f : A \rightarrow B$.

The idea is that even in infinite settings, if we know a property about the 'size' of one set A , and there exists a bijection with another set B , then B shares this notion of 'size'. For example, as we mentioned, if such a bijection exists between two sets A, B , then A is countable if and only if B is countable. This gives rise to an equivalence relation, which we define as cardinality.

Exercise. Show that the relation $\sim$ on $\mathcal{P}(\mathcal{U})$ (where $\mathcal{U}$ is the universal set of interest) defined by $A \sim B \iff \exists f : A \rightarrow B$ such that f is a bijection, is an equivalence relation.

Theorem 1.3. If $A \subseteq \mathbb{N}$ is an infinite set, then A is countably infinite.

Proof. We construct a function $f : \mathbb{N} \rightarrow A$ inductively: since A is nonempty, and $A \subseteq \mathbb{N}$, by well-ordering of the natural numbers we can pick $a_1 = \min A$, and let $f(1) = a_1$. Assume that, for $k \geq 1$, $a_1, \dots, a_k$ have been chosen and $f(i) = a_i$ for $1 \leq i \leq k$. Since A is infinite, $A \setminus \{a_1, \dots, a_k\}$ is nonempty. Pick $a_{k+1} = \min(A \setminus \{a_1, \dots, a_k\})$, and let $f(k+1) = a_{k+1}$. Thus by construction f is strictly increasing. We claim that f is a bijection.

Injective: this holds because by construction f is strictly increasing. We saw in the introductory module that strictly increasing functions are injective.

Surjective: suppose for contradiction that f is not surjective. Then there exists $a \in A$ such that for all $n \in \mathbb{N}$, $f(n) \neq a$. By the well-ordering of the natural numbers, we can choose a to be the *smallest* element in A that is not in the image of f .

Consider the set of elements in A strictly less than a , which is $A \cap \{1, 2, \dots, a-1\}$. Since this set is bounded by a , it must be finite. Let k be the number of elements in this set.

Because a is the minimal element not in the image of f , all k elements in A that are smaller than a must be in the image of f . Furthermore, because f is strictly increasing, these k elements must be exactly $f(1), f(2), \dots, f(k)$, which correspond to our chosen $a_1, a_2, \dots, a_k$.

Following our inductive construction, the next element we pick is $a_{k+1} = \min(A \setminus \{a_1, \dots, a_k\})$. Since $\{a_1, \dots, a_k\}$ contains all elements of A strictly less than a , the smallest remaining element in the set is precisely a . Therefore, $f(k+1) = a$, which contradicts our assumption that $f(n) \neq a$ for all $n \in \mathbb{N}$. Thus, f is surjective. $\square$

Lemma 1.4. *Let S be an infinite set. There exists an injection $f : \mathbb{N} \rightarrow S$.*

Proof. We construct a function $f : \mathbb{N} \rightarrow S$ inductively: since S is infinite, it is nonempty, so we can pick $s_1 \in S$ and define $f(1) := s_1$. Suppose for $k \in \mathbb{N}$, $s_1, \dots, s_k$ have been chosen and we have defined $f(i) = s_i$ for $1 \leq i \leq k$. Since S is infinite, $S \setminus \{s_1, \dots, s_k\}$ is nonempty, so there is $s_{k+1} \in S \setminus \{s_1, \dots, s_k\}$. Define $f(k+1) := s_{k+1}$. By construction, if $i < j$ are natural numbers, then $s_j \in S \setminus \{s_1, \dots, s_{j-1}\}$ so $s_j \neq s_i$. So f is an injection. $\square$

As a result, we can always *embed* the natural numbers into any infinite set via an injection $f : \mathbb{N} \rightarrow S$ (where S is any infinite set). In this way, the natural numbers can be thought of as 'the smallest type of infinity'.

A not so surprising fact which we can prove immediately is that the set of integers, $\mathbb{Z}$, is countably infinite:

Theorem 1.5. $\mathbb{Z}$ is countably infinite.

Proof. Define $f : \mathbb{N} \rightarrow \mathbb{Z}$ by

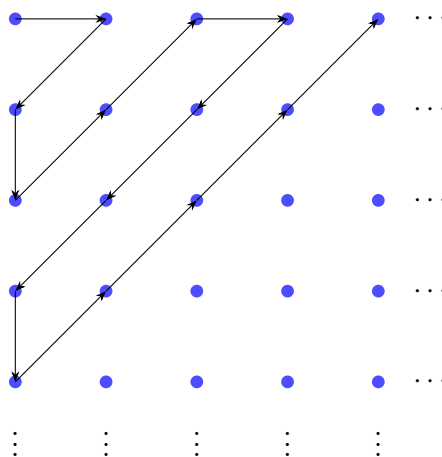
$$f(n) = \begin{cases} n/2, & \text{if } n \text{ even} \\ -\frac{(n+1)}{2}, & \text{if } n \text{ odd} \end{cases}$$

Then f is injective: suppose $f(a) = f(b)$. If a and b are of different parity then $f(a)$ and $f(b)$ are of different sign, so cannot be equal. So this case cannot happen. If a and b are even, then $a/2 = b/2$ so $a = b$. If a and b are odd, then $-(a+1)/2 = -(b+1)/2$ so $a = b$. So f is injective. Also, f is surjective: let $m \in \mathbb{Z}$ be arbitrary. If $m < 0$ then $f(-2m-1) = m$, and if $m \geq 0$ then $f(2m) = m$. Thus f is a bijection and $\mathbb{Z}$ is countable. $\square$

Remark. This proof used a useful trick called interleaving, where we listed the integers as $0, -1, 1, -2, 2, \dots$, interleaving the negative and positive integers to create the desired bijection between $\mathbb{N}$ and $\mathbb{Z}$.

A less trivial fact is that the rational numbers, $\mathbb{Q}$, is countable. This may be hard to see until observing the proof of the following fact:

Theorem 1.6. $\mathbb{N}^2$ is countably infinite.



Proof. The idea is illustrated in the above diagram. Note that the listing goes

$$(0, 0), (0, 1), (1, 0), (2, 0), (1, 1), (0, 2), \dots$$

Notice that along each diagonal, $a + b$ is constant for all pairs (a, b) in that diagonal. This, along with the triangular number formula, provides the motivation for how to construct a bijection between $\mathbb{N}^2$ and $\mathbb{N}$.

Explicitly, define $f : \mathbb{N}^2 \rightarrow \mathbb{N}$ by $f(a, b) = \frac{(a+b)(a+b+1)}{2} + b$. This is a bijection (exercise) so $\mathbb{N}^2$ is countable. Note that here we took $\mathbb{N} = \{0, 1, 2, \dots\}$. $\square$

As a consequence, we get:

Theorem 1.7. *$\mathbb{Q}$ is countably infinite.*

Proof. First we use Theorem 1.6 to show that $\mathbb{Q}_{\geq 0} := \{x \in \mathbb{Q} : x \geq 0\}$ is countably infinite. Let

$$S = \{(a, b) \in \mathbb{N}^2 : b \geq 1, \gcd(a, b) = 1\}.$$

Then $f : S \rightarrow \mathbb{Q}_{\geq 0}$, defined by $f(a, b) = a/b$ is a bijection because we know that all rational numbers in $\mathbb{Q}_{\geq 0}$ take the unique form a/b where $b \geq 1$ and $\gcd(a, b) = 1$. Since $\mathbb{N}^2$ is countably infinite by Theorem 1.6, $S \subseteq \mathbb{N}^2$ is also countably infinite, so $\mathbb{Q}_{\geq 0}$ is countably infinite.

It follows (exercise) that $\mathbb{Q}$ is countably infinite, using the same idea (interleaving) as the proof that $\mathbb{Z}$ is countably infinite. $\square$

Exercise. Finish the proof of Theorem 1.7. (*Hint: use the fact we proved above that $\mathbb{Q}_{\geq 0}$ is countably infinite, so there exists a bijection $h : \mathbb{N} \rightarrow \mathbb{Q}_{\geq 0}$.*)

The following lemma is a useful way of proving that a given set S is countably infinite without explicitly constructing a bijection $f : \mathbb{N} \rightarrow S$.

Lemma 1.8 (A criterion for countability). *Let S be an infinite set.*

1. *S is countably infinite if and only if there exists an injection $\iota : S \rightarrow \mathbb{N}$.*
2. *S is countably infinite if and only if there exists a surjection $g : \mathbb{N} \rightarrow S$.*

Proof. 1. ($\Rightarrow$) Assume S is countably infinite. By definition, there exists a bijection $f : S \rightarrow \mathbb{N}$. Since every bijection is, by definition, also an injection, we can simply let $\iota = f$. This proves the forward direction.

($\Leftarrow$) Assume there exists an injection $\iota : S \rightarrow \mathbb{N}$. The function ι restricts to a bijection between S and its image, $\iota(S) \subseteq \mathbb{N}$. Because S is an infinite set and ι is injective, $\iota(S)$ must be an infinite subset of $\mathbb{N}$. By Theorem 1.3, any infinite subset of the natural numbers is countably infinite. Therefore, there exists a bijection $h : \iota(S) \rightarrow \mathbb{N}$. Because the composition of bijections is a bijection, $h \circ \iota : S \rightarrow \mathbb{N}$ is a bijection. Thus, S is countably infinite.

2. ($\Rightarrow$) Assume S is countably infinite. By definition, there exists a bijection $f : \mathbb{N} \rightarrow S$. Since every bijection is also a surjection, we can simply let $g = f$.

($\Leftarrow$) Assume there exists a surjection $g : \mathbb{N} \rightarrow S$. We will construct an injection from S to $\mathbb{N}$ and invoke (1). Because g is surjective, for every $s \in S$, the preimage $g^{-1}(\{s\}) = \{n \in \mathbb{N} \mid g(n) = s\}$ is a non-empty subset of $\mathbb{N}$. By the well-ordering of the natural numbers, every non-empty subset of $\mathbb{N}$ has a least element. Therefore, we can define a function $\iota : S \rightarrow \mathbb{N}$ by choosing the smallest index that maps to s :

$$\iota(s) = \min(g^{-1}(\{s\}))$$

We claim ι is injective. Suppose $\iota(s_1) = \iota(s_2) = n$ for some $s_1, s_2 \in S$. This means n is the minimum element of both $g^{-1}(\{s_1\})$ and $g^{-1}(\{s_2\})$. This implies that $g(n) = s_1$ and $g(n) = s_2$. Therefore, $s_1 = s_2$, showing that ι is injective. Since S is infinite and we have found an injection $\iota : S \rightarrow \mathbb{N}$, it follows directly from (1) that S is countably infinite. $\square$

The idea for item (1) above is that if you can *embed* your set into the natural numbers via an injection $\iota : S \rightarrow \mathbb{N}$, then it must be countably infinite, as $\iota(S) \subseteq \mathbb{N}$, and we have shown infinite subsets of $\mathbb{N}$ are countable, so since ι is an injection, S has the same cardinality as $\iota(S)$, (i.e. $S \sim \iota(S)$), so S is countably infinite. The idea for item (2) is similar: if you can use the natural numbers to list all the elements of your infinite set S , then even though this surjective map $g : \mathbb{N} \rightarrow S$ may not be injective, we have nonetheless listed all the elements of S using only the natural numbers, so S must be countably infinite also.

A simple example using Lemma 1.8 is that finite products of countable sets are countable (note that we already have the tools to prove this without Lemma 1.8 by using Theorem 1.6, which is a more elementary way of proving this fact via an explicit bijection).

Before looking at the example, try and think of properties of the integers that we can use to construct an injection $\iota : A \times B \rightarrow \mathbb{N}$.

Example 1.9. If A, B are countably infinite sets, then $A \times B$ is countably infinite.

Proof. Recall from the introductory module that we define

$$A \times B := \{(a, b) : a \in A, b \in B\}.$$

Since A and B are countably infinite, let $f : \mathbb{N} \rightarrow A$, $g : \mathbb{N} \rightarrow B$ be bijections. Therefore, we can write $A = \{a_1, a_2, \dots\}$ and $B = \{b_1, b_2, \dots\}$. Define $\iota : A \times B \rightarrow \mathbb{N}$ by $\iota(a_i, b_j) = 2^i 3^j$. By the Fundamental Theorem of Arithmetic from the introductory module, ι is an injection because prime factorisations are unique, so by Lemma 1.8, $A \times B$ is countably infinite. $\square$

This trick of mapping each element of the set to a prime factorisation is useful to keep in mind, as it is a very neat way of proving countability for sets.

1.2 Uncountable sets

Not all sets are countably infinite:

Definition 1.10. An infinite set is called *uncountable* if it is not countably infinite.

For example, the set of real numbers $\mathbb{R}$ is uncountable. To show this, we first prove:

Lemma 1.11. *The set $(0, 1) := \{x \in \mathbb{R} : 0 < x < 1\}$ is uncountable.*

Proof. The main technique that we use in arguing why a given set is uncountable is *Cantor's diagonal argument*, which goes like this. Suppose, for contradiction, that $(0, 1)$ is countably infinite. Then there exists a bijection $f : \mathbb{N} \rightarrow (0, 1)$ (note here we will let $\mathbb{N} = \{1, 2, \dots\}$ for aesthetics; the argument does not change). For each $n \in \mathbb{N}$, we write $f(n) = x_n = 0.a_{1n}a_{2n}\dots$, writing each x_n in decimal form, excluding tails of the digit 9 (for example, $0.8999\dots = 0.9$). We now get a listing of the numbers in $(0, 1)$ which looks like:

$$\begin{aligned} f(1) &= 0.a_{11}a_{12}a_{13}a_{14}\dots \\ f(2) &= 0.a_{21}a_{22}a_{23}a_{24}\dots \\ f(3) &= 0.a_{31}a_{32}a_{33}a_{34}\dots \\ f(4) &= 0.a_{41}a_{42}a_{43}a_{44}\dots \\ &\vdots \end{aligned}$$

The idea is that we look at the *diagonal* elements, a_{ii} for $i \geq 1$, and construct a number $x \in (0, 1)$ such that for all $i \geq 1$, the i^{th} decimal place of x differs from the i^{th} decimal place of $f(i)$, thereby constructing a number $x \in (0, 1)$ that is not in the enumeration f of the interval $(0, 1)$, which is a contradiction to f being surjective. We construct $x \in (0, 1)$ as follows: for $j \geq 1$, let b_j be the j^{th} decimal place of x , then we define

$$b_j = \begin{cases} 2 & \text{if } a_{jj} = 1 \\ 1 & \text{if } a_{jj} \neq 1 \end{cases}$$

Then by construction, $b_i \neq a_{ii}$ for each $i \geq 1$, so in particular (by uniqueness of decimal representations under our restriction to decimal expansions with no digit 9 tails), we have $x \neq f(i)$ for all $i \geq 1$, so $x \in (0, 1)$ is not in the image of f , a contradiction to f being surjective onto $(0, 1)$. Therefore, $(0, 1)$ is uncountable. $\square$

Theorem 1.12. $\mathbb{R}$ is uncountable.

Proof. Follows from Lemma 1.11 (exercise). $\square$

Exercise. Find a bijection $f : (0, 1) \rightarrow \mathbb{R}$, and deduce that $\mathbb{R}$ is an uncountable set.

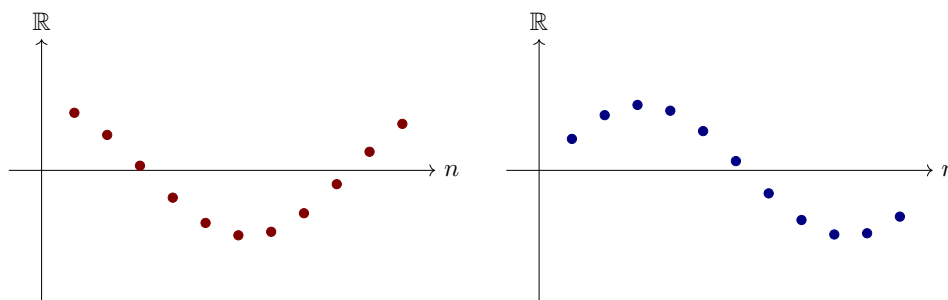
Chapter 2

Sequences and Series

2.1 Sequences

Definition 2.1. A real sequence is a function $f : \mathbb{N} \rightarrow \mathbb{R}$.

In general, $\mathbb{R}$ can be replaced by a metric space X , for example the complex numbers $\mathbb{C}$ which we will see later. It is therefore important to distinguish *real* sequences. For notation, we will write $(a_n)_{n=1}^{\infty}$ to denote a sequence $a_1, a_2, \dots$, and sometimes this will be shortened to (a_n) , or just a_n . Visually, sequences can be depicted in the following way:



2.1.1 Convergence of sequences

Intuitively, a real sequence converges to a limit $L \in \mathbb{R}$ if we can make the terms of our sequence, a_n , arbitrarily close to L , whenever we go far enough out into the sequence. Clearly, this does not suffice as a mathematical definition, so instead we define convergence of a real sequence (a_n) as follows.

Definition 2.2 (Convergence of a sequence in $\mathbb{R}$). A real sequence (a_n) converges to $L \in \mathbb{R}$ if

$$\forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall n \geq N, |a_n - L| < \varepsilon.$$

For notation, if Definition 2.2 holds for a sequence (a_n) , we will write $\lim_{n \rightarrow \infty} a_n = L$, which we will sometimes shorten to $a_n \rightarrow L$ (as $n \rightarrow \infty$) or $\lim a_n = L$, if from the context it is clear we are talking about a sequence, and not, say, a limit of a function $f(x)$ (which, as we will see, does not have to be the limit as $x \rightarrow \infty$).

Example 2.3. Use Definition 2.2 to show that the sequence (a_n) defined by $a_n = 1/n$ for each $n \in \mathbb{N}$ converges to 0.

Proof. Let $\varepsilon > 0$ be arbitrary. Choose $N \in \mathbb{N}$ such that $N > 1/\varepsilon$ (such an N exists by the Archimedean property of the real numbers from the introductory module). Then, given $n \geq N$, we have

$$|a_n - 0| = 1/n \leq 1/N < \varepsilon,$$

so $a_n \rightarrow 0$. □

Definition 2.4. A sequence *diverges* if it does not converge to any $L \in \mathbb{R}$.

Definition 2.5. A real sequence (a_n) is bounded if there exists $M > 0$ such that for all $n \in \mathbb{N}$, $|a_n| \leq M$.

Remark. Note that a real sequence can diverge but still be bounded (see below).

Example 2.6. Use Definition 2.2 to show that the sequence $a_n := (-1)^n$ diverges.

Proof. Suppose for contradiction that $a_n \rightarrow L \in \mathbb{R}$. Let $N \in \mathbb{N}$ be such that for all $n \geq N$,

$$|(-1)^n - L| < 1/2.$$

But also, by the triangle inequality,

$$2 = |(-1)^{2N} - (-1)^{2N+1}| \leq |(-1)^{2N} - L| + |L - (-1)^{2N+1}| < 1/2 + 1/2 = 1,$$

which is a contradiction. Therefore, (a_n) diverges. □

2.1.2 Basic results on convergent sequences

Unsurprisingly, if a sequence converges to a limit, then it must be unique:

Theorem 2.7. *Limits of convergent sequences are unique.*

Proof. Suppose $a_n \rightarrow L$ and $a_n \rightarrow M$, and let $\varepsilon > 0$ be arbitrary. We wish to conclude that $L = M$. By convergence of $a_n \rightarrow L$, pick $N_1 \in \mathbb{N}$ such that for all $n \geq N_1$, $|a_n - L| < \varepsilon/2$. By convergence of $a_n \rightarrow M$ let $N_2 \in \mathbb{N}$ be such that for all $n \geq N_2$, $|a_n - M| < \varepsilon/2$. Now, let $n := \max\{N_1, N_2\}$. By the triangle inequality, we have

$$|L - M| = |L - a_n + a_n - M| \leq |a_n - L| + |a_n - M| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Therefore, since ε was arbitrary, $L = M$ (to see this, if $L \neq M$ then one can pick $\varepsilon = |L - M|/2$ and get a contradiction). $\square$

Theorem 2.8 (Order Limit Theorem). *Suppose (a_n) and (b_n) are two real sequences such that $a_n \geq b_n$ for all $n \in \mathbb{N}$. Let $a_n \rightarrow a$ and $b_n \rightarrow b$. Then $a \geq b$.*

Proof. Suppose for contradiction that $a < b$. Pick $\varepsilon = \frac{b-a}{2}$. Since $a_n \rightarrow a$, there exists $N_1 \in \mathbb{N}$ such that for all $n \geq N_1$, $|a_n - a| < \varepsilon$, so in particular for $n \geq N_1$,

$$a_n < a + \varepsilon = a + \frac{b-a}{2} = \frac{a+b}{2},$$

and since $b_n \rightarrow b$ there exists $N_2 \in \mathbb{N}$ such that for all $n \geq N_2$, $|b_n - b| < \varepsilon$, so in particular for $n \geq N_2$,

$$b_n > b - \varepsilon = b - \frac{b-a}{2} = \frac{a+b}{2}.$$

Therefore, choosing $n := \max\{N_1, N_2\}$ it follows that $b_n > a_n$, which is a contradiction. $\square$

Corollary 2.9. *Suppose (a_n) is a real sequence with $a_n \rightarrow a$ such that $B \leq a_n \leq C$ for all $n \in \mathbb{N}$. Then $B \leq a \leq C$.*

Proof. Apply Theorem 2.8 using constant sequences $b_n = B$ and $c_n = C$ for all $n \in \mathbb{N}$. $\square$

Remark. If $B < a_n < C$ for all $n \in \mathbb{N}$ in the above corollary, it is not necessarily true that $B < a < C$, but we do still get the non-strict inequalities $B \leq a \leq C$. Limits only preserve non-strict inequalities.

Theorem 2.10 (Squeeze Theorem). *Let (a_n) , (b_n) and (c_n) be real sequences such that for all $n \in \mathbb{N}$, $a_n \leq b_n \leq c_n$. Suppose that $a_n \rightarrow L$ and $c_n \rightarrow L$. Then $b_n \rightarrow L$.*

Proof. Let $\varepsilon > 0$ be arbitrary. Since $a_n \rightarrow L$, we can pick $N_1 \in \mathbb{N}$ such that for all $n \geq N_1$, $|a_n - L| < \varepsilon$. Since $c_n \rightarrow L$, we can also pick $N_2 \in \mathbb{N}$ such that for all $n \geq N_2$, $|c_n - L| < \varepsilon$. Let $N := \max\{N_1, N_2\}$. Then given $n \geq N$, we have $a_n, c_n \in (L - \varepsilon, L + \varepsilon)$, i.e.

$$L - \varepsilon < a_n \leq b_n \leq c_n < L + \varepsilon,$$

so $|b_n - L| < \varepsilon$, so $b_n \rightarrow L$. $\square$

Example 2.11. Let $a_n = \frac{\sin n}{n}$. Then for all $n \in \mathbb{N}$ we have

$$-\frac{1}{n} \leq \frac{\sin n}{n} \leq \frac{1}{n},$$

since $|\sin n| \leq 1$ for all $n \in \mathbb{N}$. Therefore, since $1/n \rightarrow 0$, $a_n \rightarrow 0$ by the Squeeze Theorem $a_n \rightarrow 0$.

Theorem 2.12. *Convergent sequences are bounded.*

Proof. Let (a_n) be a convergent sequence, and let $L \in \mathbb{R}$ be such that $\lim a_n = L$. Let $N \in \mathbb{N}$ be such that for all $n \geq N$, $|a_n - L| < 1$. In other words, for all $n \geq N$ we have

$$a_n \in (L - 1, L + 1).$$

Now, pick $M := \max\{|a_1|, |a_2|, \dots, |a_{N-1}|, |L| + 1\}$. Then by construction, $|a_n| \leq M$ for all $n \in \mathbb{N}$, so (a_n) is bounded. $\square$

Remark. The idea for the above proof is that we can control (a_n) to be close to L (say, within a distance of 1 from L), past some point $N \in \mathbb{N}$, and before that, there are only finitely many terms in the sequence, so we can just take the maximum of a finite number of terms $|a_1|, \dots, |a_{N-1}|$, and then include $|L| + 1$ to complete the bound.

Similarly to the above observation, changing a finite number of terms in a convergent sequence does not affect its convergence. When dealing with convergence of sequences, we only care about the behaviour of the *tail* of the sequence. This motivates the following definition:

Definition 2.13. A sequence (a_n) *eventually* has a property P if there exists $N \in \mathbb{N}$ such that for all $n \geq N$, a_n has property P .

Remark. Modifying the hypothesis of a theorem which requires a property of a sequence to hold for all $n \in \mathbb{N}$ with *eventually* does not weaken the theorem; they are equivalent.

Example 2.14. Let the property P be 'constant'. The sequence $a_1 = 1, a_2 = 2$ and $a_n = 0$ for all $n \geq 3$ is *eventually* constant.

Exercise. Prove that if an integer sequence (a_n) (that is, $a_n \in \mathbb{Z}$ for all $n \in \mathbb{N}$) converges, then it is eventually constant.

Definition 2.15. Let (a_n) be a real sequence.

- $a_n \rightarrow \infty$ (as $n \rightarrow \infty$) if for every $R > 0$, a_n is eventually greater than R .
- $a_n \rightarrow -\infty$ (as $n \rightarrow \infty$) if for every $R < 0$, a_n is eventually less than R .

Example 2.16.

1. $a_n = n^2 \rightarrow \infty$. Indeed, let $R > 0$, and take $n > \sqrt{R}$. Then

$$a_n = n^2 > (\sqrt{R})^2 = R.$$

2. Define

$$a_n = \begin{cases} 0 & \text{if } n \text{ odd} \\ n & \text{if } n \text{ even} \end{cases}$$

This is an example of a sequence which does not diverge to ∞ , but is unbounded (as an exercise, use Definition 2.15 to prove this directly).

Theorem 2.17. Let (a_n) and (b_n) be real sequences.

1. If $a_n \rightarrow \infty$ and b_n is bounded below, then $a_n + b_n \rightarrow \infty$.
2. If $a_n \rightarrow \infty$ and $\lambda \in \mathbb{R}$, then

$$\lambda a_n \rightarrow \begin{cases} \infty, & \lambda > 0, \\ 0, & \lambda = 0, \\ -\infty, & \lambda < 0. \end{cases}$$

3. If (a_n) is bounded and $b_n \rightarrow \infty$ with $b_n \neq 0$ for all $n \in \mathbb{N}$ then $\frac{a_n}{b_n} \rightarrow 0$.
4. If $a_n \rightarrow \infty$ and there exists $B > 0$ such that eventually $b_n > B$, then $a_n b_n \rightarrow \infty$.

Proof. 1. Suppose $a_n \rightarrow \infty$ and let $M > 0$ be such that $b_n \geq M$ for all $n \in \mathbb{N}$. Let $R > 0$ be arbitrary. Since $a_n \rightarrow \infty$, let $N \in \mathbb{N}$ be such that for all $n \geq N$, $a_n > R - M$. Then given $n \geq N$, we have

$$a_n + b_n \geq a_n + M > R - M + M = R,$$

so $a_n + b_n \rightarrow \infty$.

2. Suppose $a_n \rightarrow \infty$, let $\lambda \in \mathbb{R}$. If $\lambda > 0$, let $R > 0$ be arbitrary. Then since $a_n \rightarrow \infty$ let $N \in \mathbb{N}$ be such that for all $n \geq N$, $a_n > R/\lambda$. Then for $n \geq N$, we have $\lambda a_n > \lambda(R/\lambda) = R$, so $\lambda a_n \rightarrow \infty$. If $\lambda = 0$, then λa_n is the constant zero sequence, so converges to 0. If $\lambda < 0$, let $R < 0$ be arbitrary. Since $a_n \rightarrow \infty$ let $N \in \mathbb{N}$ be such that for all $n \geq N$, $a_n > R/\lambda$. Then for $n \geq N$ we have $\lambda a_n < \lambda(R/\lambda) = R$, so $a_n \rightarrow -\infty$.
3. Let $M > 0$ be such that $|a_n| \leq M$ for all $n \in \mathbb{N}$ and suppose $b_n \rightarrow \infty$ with $b_n \neq 0$ for all $n \in \mathbb{N}$. Let $\varepsilon > 0$ be arbitrary; since $b_n \rightarrow \infty$, we can pick $N \in \mathbb{N}$ such that for all $n \geq N$, $b_n > M/\varepsilon$. Therefore, for $n \geq N$, we have

$$\left| \frac{a_n}{b_n} \right| \leq \frac{M}{b_n} < \frac{M}{M/\varepsilon} = \varepsilon,$$

so $\frac{a_n}{b_n} \rightarrow 0$.

4. Suppose $a_n \rightarrow \infty$ and let $B > 0$; suppose that $N_1 \in \mathbb{N}$ is such that for all $n \geq N_1$, $b_n > B$. Now, let $R > 0$ be arbitrary. Since $a_n \rightarrow \infty$, let $N_2 \in \mathbb{N}$ be such that for all $n \geq N_2$, $a_n > R/B$. Let $N := \max\{N_1, N_2\}$. Then given $n \geq N$ we have

$$a_n b_n > a_n B > \left(\frac{R}{B} \right) B = R,$$

so $a_n b_n \rightarrow \infty$. □

Theorem 2.18 (Growth comparisons).

1. Let $m \in \mathbb{R}$ with $m \neq 0$. Then

$$\lim_{n \rightarrow \infty} n^m = \begin{cases} \infty, & m > 0, \\ 0, & m < 0. \end{cases}$$

2. Let $\lambda \in \mathbb{R}$ with $\lambda \neq 1$. Then

$$\lim_{n \rightarrow \infty} \lambda^n = \begin{cases} \infty, & \lambda > 1, \\ 0, & |\lambda| < 1. \end{cases}$$

If $\lambda \leq -1$, then λ^n diverges.

3. Let $m > 0$ and $\lambda > 1$. Then $\lim_{n \rightarrow \infty} \frac{\lambda^n}{n^m} = \infty$ (“exponentials beat powers of n ”).

4. Let $m > 0$. Then $\lim_{n \rightarrow \infty} \frac{\log n}{n^m} = 0$ (“powers of n beat logs”).

Proof. 1. We will prove the case for $m > 0$; the case for $m < 0$ follows immediately by item 3 in Theorem 2.17. Let $R > 0$ be arbitrary, and pick $N \in \mathbb{N}$ such that $N > R^{1/m}$. Then for $n \geq N$ we have $n^m \geq N^m > (R^{1/m})^m = R$, since $x \mapsto x^m$ is an increasing function for $m > 0$, $x > 0$.

2. Suppose $\lambda > 1$. Then we have by Bernoulli’s inequality, for $n \in \mathbb{N}$,

$$\lambda^n = (1 + (\lambda - 1))^n \geq 1 + (\lambda - 1)n \rightarrow \infty,$$

since $\lambda - 1 > 0$.

Suppose $0 < |\lambda| < 1$. Then there exists $\mu > 1$ such that $|\lambda| = 1/\mu$. By the result for $\lambda > 1$ and item 3 in Theorem 2.17, we have that $|\lambda|^n = 1/\mu^n \rightarrow 0$, since $\mu^n \rightarrow \infty$ by the $\lambda > 1$ case we just proved. Because $|\lambda^n| \rightarrow 0$, it follows that $\lambda^n \rightarrow 0$. Suppose $\lambda \leq -1$. If $\lambda = -1$, the sequence is $(-1)^n$, which we have already shown diverges in an example. If $\lambda < -1$, then $|\lambda| > 1$, so by the first part, $|\lambda|^n \rightarrow \infty$. Since the absolute value of the sequence diverges to infinity, λ^n cannot converge to a finite limit, and therefore diverges.

3. Let $\lambda > 1$. Then we can write $\lambda = 1 + \delta$ for some real number $\delta > 0$. Let $k \in \mathbb{N}$ be an integer strictly greater than m . For any $n \geq k$, we can expand λ^n using the Binomial Theorem:

$$\lambda^n = (1 + \delta)^n = \sum_{j=0}^n \binom{n}{j} \delta^j.$$

Since $\delta > 0$, every term in this sum is positive. We can establish a lower bound by dropping all terms except the k -th term:

$$\lambda^n \geq \binom{n}{k} \delta^k = \frac{n(n-1) \cdots (n-k+1)}{k!} \delta^k.$$

For $n \geq 2k$, the first k terms in the factorial expansion $(n, n-1, \dots, n-k+1)$ are all greater than or equal to $n/2$. Thus, we can further bound the binomial coefficient:

$$\binom{n}{k} \geq \frac{(n/2)^k}{k!} = \frac{n^k}{2^k k!}.$$

Substituting this back into our ratio, for $n \geq 2k$:

$$\frac{\lambda^n}{n^m} \geq \frac{n^k}{n^m} \frac{\delta^k}{2^k k!} = n^{k-m} \left(\frac{\delta^k}{2^k k!} \right).$$

Notice that $\frac{\delta^k}{2^k k!}$ is a strictly positive constant independent of n . Because we chose $k > m$, the exponent $k - m$ is strictly positive. Therefore, n^{k-m} diverges to ∞ as $n \rightarrow \infty$ by item 1 in this theorem. Since $\frac{\lambda^n}{n^m}$ is bounded below by a sequence diverging to ∞ , we conclude that $\lim_{n \rightarrow \infty} \frac{\lambda^n}{n^m} = \infty$.

4. We will prove this when we rigorously define the exp and log functions in Chapter 6. Furthermore, it is much easier to prove this once we have built some more theory. $\square$

Changing a finite number of terms in a convergent sequence does not affect convergence:

Lemma 2.19. *Let $K \in \mathbb{N}$ be arbitrary. Then $(a_n)_{n=1}^\infty$ is a convergent sequence with limit $L \in \mathbb{R}$ if and only if $(a_{n+K})_{n=1}^\infty$ is a convergent sequence, with limit $L \in \mathbb{R}$.*

Proof. Suppose (a_n) converges to $L \in \mathbb{R}$. Pick $N \in \mathbb{N}$ such that for all $n \geq N$, $|a_n - L| < \varepsilon$. Since $K \geq 0$, given $n \geq N$, $n + K \geq n \geq N$, so $|a_{n+K} - L| < \varepsilon$.

Conversely, suppose (a_{n+K}) converges to $L \in \mathbb{R}$. Let $N \in \mathbb{N}$ be such that for all $n \geq N$, $|a_{n+K} - L| < \varepsilon$. Choose $N' := N + K$. Given $n \geq N'$, we have $n \geq N + K$, so for $n \geq N'$, $|a_n - L| < \varepsilon$. $\square$

2.1.3 Algebra of Limits

There is a useful theorem that allows us to avoid ε proofs and use convergence results that we have already proven:

Theorem 2.20 (Algebra of Limits). *Let (a_n) , and (b_n) be real sequences, and suppose $a_n \rightarrow a \in \mathbb{R}$, and $b_n \rightarrow b \in \mathbb{R}$. Then*

1. $a_n + b_n \rightarrow a + b$.
2. $a_n b_n \rightarrow ab$.
3. For all $\lambda \in \mathbb{R}$, $\lambda a_n \rightarrow \lambda a$.
4. If $b_n \neq 0$ for all $n \in \mathbb{N}$ and $b \neq 0$, then $1/b_n \rightarrow 1/b$.

Proof. (1) and (3) are good exercises to practice using Definition 2.2.

2. Case 1: without loss of generality, suppose that $a \neq 0$ and $b = 0$; let $\varepsilon > 0$ be arbitrary. By Theorem 2.12, there exists $M > 0$ such that for all $n \in \mathbb{N}$, $|a_n| \leq M$. By convergence of $b_n \rightarrow 0$, let $N \in \mathbb{N}$ be such that for all $n \geq N$, $|b_n| < \varepsilon/M$. Then for $n \geq N$, we have

$$|a_n b_n| = |a_n| |b_n| \leq M |b_n| < M(\varepsilon/M) = \varepsilon,$$

so $a_n b_n \rightarrow 0$.

Case 2: Suppose that $a \neq 0$ and $b \neq 0$; let $\varepsilon > 0$ be arbitrary. By Theorem 2.12 let $M > 0$ be such that for all $n \in \mathbb{N}$, $|a_n| \leq M$. By convergence of $a_n \rightarrow a$, let $N_1 \in \mathbb{N}$ be such that for all $n \geq N_1$, $|a_n - a| < \varepsilon/2|b|$. Also, by convergence of $b_n \rightarrow b$, let $N_2 \in \mathbb{N}$ be such that for all $n \geq N_2$, $|b_n - b| < \varepsilon/2M$. Choose $N := \max\{N_1, N_2\}$. Then for all $n \geq N$, we have

$$\begin{aligned} |a_n b_n - ab| &= |a_n b_n - a_n b + a_n b - ab| \\ &\leq |a_n b_n - a_n b| + |a_n b - ab| \\ &= |a_n| |b_n - b| + |b| |a_n - a| \\ &\leq M |b_n - b| + |b| |a_n - a| \\ &< M \left(\frac{\varepsilon}{2M} \right) + |b| \left(\frac{\varepsilon}{2|b|} \right) \\ &= \varepsilon, \end{aligned}$$

so $a_n b_n \rightarrow ab$.

4. Let $\varepsilon > 0$ be arbitrary. Let $N_1 \in \mathbb{N}$ be such that for all $n \geq N_1$, $|b_n - b| < |b|/2$, so by the triangle inequality we have, for $n \geq N_1$,

$$|b| = |b - b_n + b_n| \leq |b_n - b| + |b_n| < \frac{|b|}{2} + |b_n| \implies |b_n| > \frac{|b|}{2} > 0.$$

Now, let $N_2 \in \mathbb{N}$ be such that for all $n \geq N_2$, $|b_n - b| < \frac{\varepsilon|b|^2}{2}$. Choose $N := \max\{N_1, N_2\}$. For all $n \geq N$, we have

$$\left| \frac{1}{b_n} - \frac{1}{b} \right| = \frac{|b_n - b|}{|b||b_n|} < \frac{|b_n - b|}{|b|^2/2} < \frac{\varepsilon|b|^2/2}{|b|^2/2} = \varepsilon,$$

so $1/b_n \rightarrow 1/b$. $\square$

Remark. Using items 2 and 4 above, it follows that if $b \neq 0$ and $b_n \neq 0$ for all $n \in \mathbb{N}$, and $a_n \rightarrow a$, $b_n \rightarrow b$, then $a_n/b_n \rightarrow a/b$.

Example 2.21. 1. Let $a_n = \frac{n^2+4n+1}{2n^2+1}$. For $n \in \mathbb{N}$ we have

$$a_n = \frac{1 + 4/n + 1/n^2}{2 + 1/n^2}.$$

Now, using 2 in the Algebra of Limits (Theorem 2.20), $1/n^2 \rightarrow 0$ since we have shown $1/n \rightarrow 0$, and using 1, 3 and 4 in the Algebra of Limits we get $\lim a_n = \frac{1}{2}$.

2. Applying (2) inductively, since $1/n \rightarrow 0$ we get that for any $m \in \mathbb{N}$, $1/n^m \rightarrow 0$.
3. Let $a_n = \sqrt{n^2 + n} - n$. The standard trick is to multiply by the conjugate to convert the $\infty - \infty$ form into a situation where we can apply the Algebra of Limits; for $n \in \mathbb{N}$ we have

$$\begin{aligned} a_n &= \left(\sqrt{n^2 + n} - n \right) \frac{\sqrt{n^2 + n} + n}{\sqrt{n^2 + n} + n} = \frac{n}{\sqrt{n^2 + n} + n} \\ &= \frac{1}{\sqrt{1 + \frac{1}{n}} + 1} \rightarrow \frac{1}{2}, \end{aligned}$$

by the Algebra of Limits, since $1/n \rightarrow 0$.

2.1.4 Monotone Convergence Theorem

Definition 2.22. Let (a_n) be a real sequence.

1. We say that a_n is *increasing* if for all $n \in \mathbb{N}$, $a_{n+1} \geq a_n$. If the inequality is strict for all $n \in \mathbb{N}$, we say that a_n is *strictly increasing*.
2. We say that a_n is *decreasing* if for all $n \in \mathbb{N}$, $a_{n+1} \leq a_n$. If the inequality is strict for all $n \in \mathbb{N}$, we say that a_n is *strictly decreasing*.
3. A sequence a_n is *monotonic* if it is increasing or decreasing.

Example 2.23. The sequence $a_n = 1/n$ is strictly decreasing and converges to 0, the sequence $a_n = 1 - 1/n$ is strictly increasing and converges to 1, the sequence $a_n = n$ is strictly increasing and diverges to ∞ , and the sequence $a_n = (-1)^n n$ is neither strictly increasing nor strictly decreasing, and diverges. Constant sequences are both increasing and decreasing.

We now state and prove an important theorem about monotonic and bounded real sequences.

Theorem 2.24 (Monotone Convergence Theorem). *Let (a_n) be a bounded and monotone real sequence. Then (a_n) is convergent.*

Proof. We will prove the case where a_n is increasing. The case where a_n is decreasing is similar, and is left as an exercise.

Consider the set $S := \{a_n : n \in \mathbb{N}\}$. S is nonempty since $a_1 \in S$, and since a_n is bounded, the set S is bounded. By the completeness of $\mathbb{R}$, $s := \sup S$ exists. We claim that $a_n \rightarrow s$.

Let $\varepsilon > 0$. By definition of supremum, there exists $N \in \mathbb{N}$ such that $s - \varepsilon < a_N \leq s$. But since s is an upper bound, and a_n is increasing, we therefore have

$$s - \varepsilon < a_N \leq a_{N+1} \leq \cdots \leq s < s + \varepsilon,$$

i.e. for all $n \geq N$, we have $|a_n - s| < \varepsilon$, so $a_n \rightarrow s$. $\square$

Remark. This result is useful because it allows us to deduce convergence of a sequence without knowing what the limit is.

Example 2.25. Define a sequence (a_n) by $a_{n+1} = \sqrt{2 + a_n}$ for $n \in \mathbb{N}$ with $a_1 = \sqrt{2}$. We claim that (a_n) converges.

Proof. By induction, $0 < a_n < 2$ for all $n \in \mathbb{N}$; indeed, given $k \in \mathbb{N}$, if $a_k < 2$ then $a_{k+1} = \sqrt{2 + a_k} < \sqrt{2 + 2} = 2$, and the base case $a_1 = \sqrt{2} < 2$. So $0 < a_n < 2$ for all $n \in \mathbb{N}$. Therefore, a_n is bounded. Also, the sequence is strictly increasing: for $n \in \mathbb{N}$ we have

$$a_n = \sqrt{a_n a_n} < \sqrt{2 a_n} = \sqrt{a_n + a_n} < \sqrt{a_n + 2} = a_{n+1},$$

so by Theorem 2.24, (a_n) is convergent.

In fact, with recurrence relations, we can often find the limit as well, once we know that it converges: this is because $\lim a_{n+1} = \lim a_n = L$ for some $L \in \mathbb{R}$, which must satisfy, by the Algebra of Limits (Theorem 2.20), $L = \sqrt{2 + L}$, i.e.

$$L^2 - L - 2 = 0 \implies (L - 2)(L + 1) = 0,$$

but since $0 < a_n < 2$ for all $n \in \mathbb{N}$, $0 \leq L \leq 2$ by Theorem 2.8 (Order Limit Theorem), so $L = 2$. $\square$

2.1.5 Subsequences

Definition 2.26. Let (a_n) be a sequence, and let $f : \mathbb{N} \rightarrow \mathbb{N}$ be a strictly increasing function. A *subsequence* of (a_n) is a sequence defined by $b_n := a_{f(n)}$ for $n \in \mathbb{N}$.

Remark. In practice, if $\{n_k : k \in \mathbb{N}\}$ is any strictly increasing set of indices such that $n_1 < n_2 < n_3 < \dots$, the corresponding subsequence of $(a_n)_{n=1}^\infty$ is $(a_{n_k})_{k=1}^\infty$.

Example 2.27. Let $a_n = n$ for $n \in \mathbb{N}$. Then $a_{2n} = 2n$ is a subsequence of a_n . Let $a_n = 1/n$ for $n \in \mathbb{N}$. Then $a_p = 1/p$ for p prime is a subsequence of a_n . However, $(1/3, 1/2, 1, 1/4, 1/5, \dots)$ is not a subsequence of a_n (we changed the ordering of the terms).

Lemma 2.28. Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be a strictly increasing function. Then for all $k \in \mathbb{N}$, $f(k) \geq k$.

Proof. The proof is by induction on k . For the base case, clearly $f(1) \geq 1$ since the codomain is $\mathbb{N}$, and 1 is the smallest natural number. Given $m \in \mathbb{N}$, suppose that $f(m) \geq m$. Since f is strictly increasing, $f(m+1) > f(m)$ i.e. $f(m+1) \geq f(m) + 1$, since $f(m+1) - f(m) \geq 1$ (smallest distance between distinct integers is 1), so

$$f(m+1) \geq f(m) + 1 \geq m + 1,$$

by the induction hypothesis, so $f(k) \geq k$ for all $k \in \mathbb{N}$. $\square$

Theorem 2.29. Let (a_n) be a real sequence that converges to $L \in \mathbb{R}$. Then any subsequence of (a_n) converges to L .

Proof. Let (a_{n_k}) be an arbitrary subsequence of (a_n) , and let $\varepsilon > 0$. Since $a_n \rightarrow L$, pick $N \in \mathbb{N}$ such that for all $n \in \mathbb{N}$, $|a_n - L| < \varepsilon$. Let $k \geq N$. Then by Lemma 2.28, we have $n_k \geq k \geq N$, so $|a_{n_k} - L| < \varepsilon$, so $a_{n_k} \rightarrow L$ as $k \rightarrow \infty$. $\square$

Remark. In contrapositive form, Theorem 2.29 is very useful for showing that a sequence diverges. You only need to find two subsequences that converge to different limits, to prove that a sequence diverges.

Example 2.30. We now get a much easier proof for the divergence of $a_n = (-1)^n$. We have $a_{2n} = 1$ for all $n \in \mathbb{N}$ and $a_{2n-1} = -1$ for all $n \in \mathbb{N}$, so by Theorem 2.29 (the contrapositive), a_n diverges.

Theorem 2.31. *Any real sequence has a monotonic subsequence.*

Proof. Let (a_n) be any real sequence. We consider the set of *peak indices* in the sequence, $S := \{n \in \mathbb{N} : a_n \geq a_m \forall m \geq n\}$. If $k \in S$, we say that a_k is *peak*.

Case 1: S is finite (or empty). Let $m \in \mathbb{N}$ be such that $m > n$ for all $n \in S$ (note that this vacuously includes the case where S is empty). Since $k \notin S$ for all $k \geq m$, in particular for all $k \geq m$, a_k is not *peak*. We therefore construct an increasing subsequence of (a_n) by induction. Define $n_1 := m$. Since a_{n_1} is not *peak*, there exists $n_2 > n_1$ such that $a_{n_2} > a_{n_1}$. Given $k \in \mathbb{N}$, suppose that $a_{n_1} < a_{n_2} < \dots < a_{n_k}$ have been chosen. Since a_{n_k} is not *peak*, there exists $n_{k+1} > n_k$ such that $a_{n_{k+1}} > a_{n_k}$. By construction, this is a (strictly) increasing subsequence of (a_n) .

Case 2: S is infinite. By the well-ordering of the natural numbers, we may pick $n_1 := \min S$. Given $k \in \mathbb{N}$, suppose that $n_1 < n_2 < \dots < n_k$ have been chosen such that $n_i \in S$ for all $1 \leq i \leq k$. Since S is infinite, $S \setminus \{n_1, \dots, n_k\}$ is nonempty, so we may pick $n_{k+1} := \min(S \setminus \{n_1, \dots, n_k\})$. By definition of the set S , for all $k \in \mathbb{N}$, $a_{n_k} \geq a_{n_{k+1}}$, since by construction a_{n_k} is *peak* for all $k \in \mathbb{N}$, so (a_n) is a decreasing subsequence of (a_n) . $\square$

Theorem 2.32 (Bolzano-Weierstraß Theorem). *Any bounded real sequence has a convergent subsequence.*

Proof. Let (a_n) be a bounded sequence. By Theorem 2.31, (a_n) has a monotone subsequence (a_{n_k}) , which is also bounded since (a_n) is bounded, so by the Monotone Convergence Theorem (Theorem 2.24), (a_{n_k}) is convergent. $\square$

Example 2.33. The sequence $a_n = \sin(n)$ has a convergent subsequence since $|a_n| \leq 1$ for all $n \in \mathbb{N}$. Finding such a subsequence is harder and is beyond the scope of this course.

2.1.6 Cauchy sequences

Convergent sequences, we have seen, get closer and closer to some limit L . What can we say about sequences whose terms get arbitrary close together, past some fixed point $N \in \mathbb{N}$? First, we define precisely what this means:

Definition 2.34. A real sequence (a_n) is *Cauchy* if

$$\forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall m, n \geq N, |a_m - a_n| < \varepsilon.$$

A useful fact we can immediately prove (in a similar way to Theorem 2.12) is that Cauchy sequences are bounded.

Lemma 2.35. *Cauchy sequences are bounded.*

Proof. Exercise. □

In fact, in $\mathbb{R}$, the concepts of Cauchy sequences and convergent sequences are equivalent in the following sense:

Theorem 2.36 (Cauchy criterion for convergence in $\mathbb{R}$). *Let (a_n) be a real sequence. Then*

$$(a_n) \text{ is convergent} \iff (a_n) \text{ is Cauchy.}$$

Proof. Suppose that (a_n) is a convergent real sequence with limit $L \in \mathbb{R}$, and let $\varepsilon > 0$ be arbitrary. Let $N \in \mathbb{N}$ be such that for all $n \geq N$, $|a_n - L| < \frac{\varepsilon}{2}$. Then given $m, n \geq N$, we have by the triangle inequality

$$|a_m - a_n| = |(a_m - L) + (L - a_n)| \leq |a_m - L| + |a_n - L| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

so (a_n) is Cauchy.

Conversely, suppose that (a_n) is Cauchy. By Lemma 2.35, (a_n) is bounded. By the Bolzano-Weierstraß Theorem, there exists a convergent subsequence of (a_{n_k}) of (a_n) with limit $L \in \mathbb{R}$. We claim that $a_n \rightarrow L$. Let $\varepsilon > 0$ be arbitrary. Since $\lim_{k \rightarrow \infty} a_{n_k} = L$, we can let $N_1 \in \mathbb{N}$ be such that for all $n_j \geq N_1$, $|a_{n_j} - L| < \frac{\varepsilon}{2}$. Also, (a_n) is Cauchy, so we can let $N_2 \in \mathbb{N}$ be such that for all $m, n \geq N_2$, $|a_m - a_n| < \frac{\varepsilon}{2}$. Now, set $N := \max\{N_1, N_2\}$ and let $n_i \in \mathbb{N}$ be large enough so that $n_i \geq N$. Then given $n \geq N$ we have $|a_n - a_{n_i}| < \frac{\varepsilon}{2}$ and so by the triangle inequality, we get

$$|a_n - L| \leq |a_n - a_{n_i}| + |a_{n_i} - L| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

so $a_n \rightarrow L$ and (a_n) is convergent. □

The key idea for the proof above of Cauchy in $\mathbb{R} \implies$ convergent in $\mathbb{R}$ for a given sequence is that by the Bolzano-Weierstraß theorem, there exists a convergent subsequence of (a_n) , but at the same time, since (a_n) is Cauchy, the terms in the sequence are getting arbitrarily closer together, while the convergent subsequence terms are getting arbitrarily close to a limit $L \in \mathbb{R}$, so we can use this to force any sufficiently far out term in the sequence to be arbitrarily close to L as well.

Remark. It is important to note that the above equivalence, in particular Cauchy $\implies$ convergent does not hold for all sets (however, convergence implies Cauchy in any metric space). It relies on the fact that $\mathbb{R}$ is a *complete* set, that is, it has the least upper bound property, which means that, intuitively, all sequences in $\mathbb{R}$ that should converge (*Cauchy* sequences) actually have a limit within that space. For example, $\mathbb{Q}$ is not a complete set (indeed, $\mathbb{R}$ is the completion of $\mathbb{Q}$, as we saw when constructing $\mathbb{R}$ in the introductory module), so the fact that Cauchy sequences are convergent does not hold in $\mathbb{Q}$. For example, $(1 + 1/n)^n \rightarrow e \notin \mathbb{Q}$. We will prove the irrationality of e later, and we may define $e := \sup\{(1 + 1/n)^n : n \in \mathbb{N}\}$; then by the Monotone Convergence Theorem (Theorem 2.24), $\lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n = e$ (note that there are more practical ways of defining e and the $\exp : \mathbb{R} \rightarrow (0, \infty)$ function, as we will see in Chapter 6).

Remark. In addition to the above comment about completeness, we notice that the Bolzano-Weierstraß Theorem was required to prove that Cauchy sequences are convergent. In fact, there is an equivalence between the following concepts for complete sets (in particular, $\mathbb{R}$ in our case):

- Least upper bound property: every non-empty subset of $\mathbb{R}$ that is bounded above has a supremum (least upper bound) in $\mathbb{R}$.
- Cauchy completeness: Every Cauchy sequence in $\mathbb{R}$ converges to a limit in $\mathbb{R}$.
- Bolzano-Weierstraß Theorem: Every bounded sequence in $\mathbb{R}$ has a convergent subsequence.
- Monotone convergence Theorem: Every bounded, monotonic sequence in $\mathbb{R}$ converges.
- Nested interval property: Any sequence of nested, closed, and bounded intervals $[a_n, b_n]$ has a nonempty intersection.

Exercise. Prove that the above concepts are equivalent.

2.1.7 Complex sequences

As hinted at earlier, sequences need not be real. An interesting and rich area of mathematics is analysis of the complex numbers, $\mathbb{C}$, which we will see more

of in second year. Here, we build some theory on complex sequences (and later, complex series and complex power series). There will be no rigorous construction of $\mathbb{C}$ (however, it is not too bad now that we have constructed $\mathbb{R}$ because, in fact, we can construct $\mathbb{C}$ from $\mathbb{R}^2$ with some special operations, namely the usual complex addition and complex multiplication). We will assume all the usual properties of the complex numbers, in particular that $\mathbb{C}$ is a field with the usual complex addition and complex multiplication.

Definition 2.37. A complex sequence is a function $f : \mathbb{N} \rightarrow \mathbb{C}$.

Again, practically, we write (z_n) or just z_n to denote a complex sequence. For a complex sequence (z_n) it is natural to associate two real sequences $(x_n), (y_n)$ with (z_n) such that for each $n \in \mathbb{N}$, $z_n = x_n + iy_n$, i.e. $x_n = \Re(z_n)$ and $y_n = \Im(z_n)$, where $\Re$ and $\Im$ denote the real and imaginary parts of a complex number, respectively. It is therefore not too hard to prove facts about complex sequences using our results about real sequences.

Definition 2.38 (Convergence of a sequence in $\mathbb{C}$). A complex sequence (z_n) converges to $z \in \mathbb{C}$ if

$$\forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall n \geq N, |z_n - z| < \varepsilon.$$

Remark. Note that the modulus $|\cdot|$ is now the *complex* modulus defined by $|z| := \sqrt{x^2 + y^2}$ where $z = x + iy$.

First, we prove a convergence criterion for complex sequences, in relation to its real and imaginary parts:

Theorem 2.39 (Parts criterion for convergence in $\mathbb{C}$). *Let (z_n) be a complex sequence, and define real sequences $(x_n), (y_n)$ by $x_n := \Re(z_n)$ and $y_n := \Im(z_n)$ for all $n \in \mathbb{N}$. Let $z \in \mathbb{C}$ and write $z = x + iy$ for some $x, y \in \mathbb{R}$. Then*

$$z_n \rightarrow z \iff x_n \rightarrow x \text{ and } y_n \rightarrow y.$$

Proof. Suppose $z_n \rightarrow z = x + iy$, and let $\varepsilon > 0$ be arbitrary. Let $N \in \mathbb{N}$ be such that for all $n \geq N$, $|z_n - z| < \varepsilon$. Given $n \geq N$, we have

$$|x_n - x| = \sqrt{(x_n - x)^2} \leq \sqrt{(x_n - x)^2 + (y_n - y)^2} = |z_n - z| < \varepsilon,$$

and similarly

$$|y_n - y| \leq \sqrt{(x_n - x)^2 + (y_n - y)^2} = |z_n - z| < \varepsilon,$$

so $x_n \rightarrow x$ and $y_n \rightarrow y$.

Conversely, suppose $x_n \rightarrow x$ and $y_n \rightarrow y$. Let $N_1 \in \mathbb{N}$ be such that for all $n \geq N_1$, $|x_n - x| < \frac{\varepsilon}{\sqrt{2}}$, and let $N_2 \in \mathbb{N}$ be such that for all $n \geq N_2$, $|y_n - y| < \frac{\varepsilon}{\sqrt{2}}$. Let $N := \max\{N_1, N_2\}$. Given $n \geq N$ we have

$$|z_n - z| = \sqrt{(x_n - x)^2 + (y_n - y)^2} < \sqrt{\frac{\varepsilon^2}{2} + \frac{\varepsilon^2}{2}} = \varepsilon,$$

so $z_n \rightarrow z$. □

Example 2.40. 1. Let $z_n = \frac{1}{n} + in$. This does not converge because $\Im(z_n) = n \rightarrow \infty$. Geometrically, since $\Re(z_n) = \frac{1}{n} \rightarrow 0$, this sequence gets arbitrarily close to the imaginary axis in the complex plane, but diverges from the real axis.

2. For $k \in \mathbb{N}$, define the sequence $z_n = e^{\frac{2n\pi i}{k}}$; this does not converge (cycles around a regular k -gon centred at $(0, 0)$). Exercise: what are the possible subsequential limits of this sequence? (i.e. if (z_{n_k}) is a convergent subsequence of (z_n) , what are the possible limits?)

3. Fix $z_1 \in \mathbb{C}$ and for $n \in \mathbb{N}$ define $z_{n+1} = \frac{z_n + |z_n|}{2}$. We claim that this sequence converges. First, for $n \in \mathbb{N}$ we write $z_n = x_n + iy_n$, so we have

$$z_{n+1} = \frac{x_n + iy_n + |z_n|}{2},$$

and $|z_n|$ is real for all $n \in \mathbb{N}$, so we have $x_{n+1} = \Re(z_{n+1}) = \frac{x_n + |z_n|}{2}$ and $y_{n+1} = \Im(z_{n+1}) = \frac{y_n}{2}$. Therefore, by induction $y_{n+1} = \frac{y_1}{2^n}$ so $y_n \rightarrow 0$. Furthermore, we know that

$$|z_n| = \sqrt{x_n^2 + y_n^2} \geq \sqrt{x_n^2} = |x_n|,$$

so $x_{n+1} \geq x_n$ for all $n \in \mathbb{N}$. Also, by construction of (z_n) , $|z_n| \leq |z_1|$ for all $n \in \mathbb{N}$ so $|x_n| \leq |z_n| \leq |z_1|$ for all $n \in \mathbb{N}$. Therefore, (x_n) is an increasing sequence and is bounded above, so by the Monotone Convergence Theorem, x_n converges to some $L \in \mathbb{R}$. Therefore, by Theorem 2.39, $z_n \rightarrow L$.

Definition 2.41. A complex sequence (z_n) is bounded if there exists $M \in \mathbb{R}_{>0}$ such that $|z_n| \leq M$ for all $n \in \mathbb{N}$.

Theorem 2.42. *Let (z_n) be a complex sequence. If (z_n) converges, then (z_n) is bounded.*

Proof. Using the parts criterion for convergence in $\mathbb{C}$ (Theorem 2.39), $x_n := \Re(z_n)$ and $y_n := \Im(z_n)$ are convergent and so by Theorem 2.12 (x_n) and (y_n) are bounded. Let $M_1 \in \mathbb{R}_{>0}$ be such that $|x_n| \leq M_1$ for all $n \in \mathbb{N}$ and let $M_2 \in \mathbb{R}_{>0}$ be such that $|y_n| \leq M_2$ for all $n \in \mathbb{N}$. Choose $M := \max\{M_1, M_2\} \in \mathbb{R}_{>0}$; then $|x_n| \leq M$ and $|y_n| \leq M$ for all $n \in \mathbb{N}$, so for all $n \in \mathbb{N}$ we have

$$|z_n| = \sqrt{x_n^2 + y_n^2} \leq \sqrt{2}M,$$

so (z_n) is bounded. $\square$

Theorem 2.43 (Bolzano-Weierstraß Theorem in $\mathbb{C}$). *Any bounded complex sequence has a convergent subsequence.*

Proof. Let (z_n) be a bounded complex sequence. Put $x_n := \Re(z_n)$ and $y_n := \Im(z_n)$. Then for all $n \in \mathbb{N}$, $|x_n| = \sqrt{x_n^2} \leq \sqrt{x_n^2 + y_n^2} = |z_n|$ and $|y_n| \leq |z_n|$, so (x_n) and (y_n) are bounded. By the Bolzano-Weierstraß Theorem in $\mathbb{R}$ (Theorem 2.32), there exists a convergent subsequence (x_{n_k}) of (x_n) . Consider the subsequence (y_{n_k}) of (y_n) . By the Bolzano-Weierstraß Theorem in $\mathbb{R}$, there exists a convergent subsequence $(y_{n_{k_j}})$ of (y_{n_k}) . Furthermore, by Theorem 2.29, $(x_{n_{k_j}})$ is a convergent subsequence of (x_n) , so by the parts criterion for convergence in $\mathbb{C}$, $(z_{n_{k_j}})$ is a convergent subsequence of (z_n) . $\square$

Definition 2.44. A complex sequence is *Cauchy* if

$$\forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall m, n \geq N, |z_m - z_n| < \varepsilon.$$

Theorem 2.45 (Cauchy criterion for convergence in $\mathbb{C}$). *Let (z_n) be a complex sequence and let $x_n := \Re(z_n)$, $y_n := \Im(z_n)$ for all $n \in \mathbb{N}$. Then the following are equivalent:*

1. (z_n) is Cauchy in $\mathbb{C}$.
2. (x_n) and (y_n) are Cauchy in $\mathbb{R}$.
3. (z_n) converges in $\mathbb{C}$.

Proof. $1 \Rightarrow 2$: For $m, n \in \mathbb{N}$ we have $|x_m - x_n| = |\Re(z_m - z_n)| \leq |z_m - z_n|$, so (x_n) is Cauchy in $\mathbb{R}$, and similarly for (y_n) (same argument, just use imaginary part).

$2 \Rightarrow 3$: By the Cauchy Criterion for Convergence in $\mathbb{R}$ (Theorem 2.36), $x_n \rightarrow x$ and $y_n \rightarrow y$ for some $x, y \in \mathbb{R}$, so by Theorem 2.39, $z_n \rightarrow x + iy = z$.

$3 \Rightarrow 1$: Suppose $z_n \rightarrow z$ for some $z \in \mathbb{C}$, and let $\varepsilon > 0$ be arbitrary. Since $z_n \rightarrow z$, we can let $N \in \mathbb{N}$ be such that for all $n \geq N$, $|z_n - z| < \frac{\varepsilon}{2}$. Then given $m, n \geq N$ we have, by the triangle inequality,

$$|z_m - z_n| \leq |z_m - z| + |z_n - z| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

so (z_n) is Cauchy in $\mathbb{C}$. □

2.2 Series

2.2.1 Partial sums

Definition 2.46. A complex sequence (a_n) is called *summable* (or convergent as a series) if the sequence of partial sums $s_n := \sum_{i=1}^n a_i$ converges to some limit s as $n \rightarrow \infty$. We then write $\sum_{n=1}^{\infty} a_n = s$.

Some remarks:

- Note that complex sequences include real sequences.
- A series is fundamentally just a special type of sequence—the sequence of its partial sums. All theorems about series are ultimately just theorems about sequences applied to S_N .
- A series is fundamentally a *limit of finite sums*, rather than an “infinite sum”. It is dangerous to treat a series as an infinite form of addition because infinite series do not always obey the standard rules of finite arithmetic (such as commutativity and associativity), which we will see when we look at divergent series and conditionally convergent series. Because it is a limit, the *order* in which we take the limit matters, which explains why rearranging the terms of conditionally convergent series can alter the final result.
- Rather than saying a sequence (a_n) is summable, we may also say that $\sum_{n=1}^{\infty} a_n$ converges, or sometimes if from the context it is clear that we are talking about series, we may just write $\sum a_n$ converges.
- When talking about divergent series, we often abuse notation and write $\sum_{n=1}^{\infty} a_n$ diverges, even though writing $\sum_{n=1}^{\infty} a_n$ does not make sense in this case (it is not a real number, we just use it symbolically in this case).

Example 2.47. The geometric series $\sum_{n=1}^{\infty} (1/2)^n$ has partial sums $S_N = 1 - (1/2)^N$ (exercise: show this), which converge to 1.

Theorem 2.48 (Algebra of Limits for Series). *If $\sum a_n$ and $\sum b_n$ are converge, and $\lambda \in \mathbb{C}$, then $\sum(a_n + b_n)$ and $\sum \lambda a_n$ are also summable, with:*

$$\sum_{n=1}^{\infty} (a_n + b_n) = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n \quad \text{and} \quad \sum_{n=1}^{\infty} \lambda a_n = \lambda \sum_{n=1}^{\infty} a_n.$$

Proof. This is a simple application of the Algebra of Limits (Theorem 2.20) on the sequences $s_n := \sum_{i=1}^n a_i \rightarrow s$ and $t_n := \sum_{i=1}^n b_i$. In particular, set $u_n = \sum_{i=1}^n (a_i + b_i)$. Then we have

$$u_n = \sum_{i=1}^n a_i + \sum_{i=1}^n b_i = s_n + t_n \rightarrow s + t,$$

i.e. $\sum_{n=1}^{\infty} (a_n + b_n) = \sum_{n=1}^{\infty} a_n + \sum_{n=1}^{\infty} b_n$, as required. Furthermore, set $v_n = \sum_{i=1}^n \lambda a_i$. Then we have

$$v_n = \lambda s_n \rightarrow \lambda s,$$

by the Algebra of Limits for sequences again, i.e. $\sum_{n=1}^{\infty} \lambda a_n = \lambda \sum_{n=1}^{\infty} a_n$. $\square$

Lemma 2.49 (Shift summability lemma). *A complex sequence (a_n) is summable if and only if the shifted sequence (a_{n+K}) is summable for any fixed $K \in \mathbb{N}$.*

Proof. Define $b_n := \sum_{i=K+1}^{n+K} a_i = \sum_{i=1}^n a_{i+K}$. Then we have

$$b_n = \sum_{i=1}^{n+K} a_i - \sum_{i=1}^K a_i = \sum_{i=1}^{n+K} a_i - C,$$

where $C := \sum_{i=1}^K a_i$ is a constant. Now, suppose (a_n) is summable. Therefore, $\lim_{n \rightarrow \infty} \sum_{i=1}^{n+K} a_i$ exists and we have that $\lim_{n \rightarrow \infty} b_n$ exists with $\sum_{n=1}^{\infty} a_{n+K} = \sum_{n=1}^{\infty} a_n - C$, using the Algebra of Limits. Conversely, suppose that (a_{n+K}) is summable, i.e. $\lim_{n \rightarrow \infty} b_n$ exists. Therefore, from the above, $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} a_{n+K} + C$ using the Algebra of Limits. $\square$

Remark. Convergence is strictly a property of the “tail” of a sequence. Adding, removing, or changing the first million terms of a series changes its final sum, but it cannot change *whether* the series converges or diverges.

2.2.2 Convergence tests for series

Theorem 2.50 (Geometric series). *The series $\sum_{n=0}^{\infty} r^n$ converges if and only if $|r| < 1$. Moreover, if $|r| < 1$ then*

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}.$$

Proof. Let $r \in \mathbb{R}$ and put $s_n = \sum_{i=0}^n r^i$. By an exercise, $s_n = \frac{1-r^{n+1}}{1-r}$. By item 2 in Theorem 2.18, $\lim_{n \rightarrow \infty} r^n$ exists if and only if $|r| < 1$, so s_n converges if and only if $|r| < 1$. Moreover, if $|r| < 1$ then by item 2 in Theorem 2.18 we have $r^n \rightarrow 0$ as $n \rightarrow \infty$, therefore

$$\sum_{n=0}^{\infty} r^n = \frac{1}{1-r},$$

as desired. □

Theorem 2.51 (Necessary condition for summability). *If the complex sequence (a_n) is summable, then $\lim_{n \rightarrow \infty} a_n = 0$.*

Proof. Suppose (a_n) is a summable complex sequence. Then $s_n := \sum_{i=1}^n a_i$ converges. Therefore, (s_n) is Cauchy, so in particular given $\varepsilon > 0$ we can pick $N \in \mathbb{N}$ such that for all $m, n \geq N$, we have $|s_m - s_n| < \varepsilon$. Therefore, letting $n \geq N$ be arbitrary and picking $m = n + 1$, we have

$$|a_n| = |s_{n+1} - s_n| < \varepsilon,$$

so $a_n \rightarrow 0$. □

Example 2.52. 1. Let $a_n = \left(1 + \frac{1}{n}\right)^n$. Then $a_n \rightarrow e \geq 1 > 0$, so $\sum_{n=1}^{\infty} a_n$ diverges by Theorem 2.51.

2. Sometimes, $a_n \rightarrow 0$ but the series still diverges: let $s_n = \sum_{i=1}^n \log\left(\frac{i+1}{i}\right)$. Then

$$s_n = \sum_{i=1}^n (\log(i+1) - \log(i)) = \log(n+1) \rightarrow \infty,$$

so $\sum_{n=1}^{\infty} \log\left(\frac{n+1}{n}\right)$ diverges, even though $\log\left(1 + \frac{1}{i}\right) \rightarrow 0$ as $i \rightarrow \infty$.

Remark. Note that this is a necessary, but *not sufficient* condition for series convergence. The classic counterexample is the harmonic series $\sum \frac{1}{n}$. Even though $\frac{1}{n} \rightarrow 0$, the series diverges to infinity because the terms do not shrink *fast enough*. We will prove this now.

Theorem 2.53 (Divergence of the Harmonic Series). *The series $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.*

Proof. Let $s_n = \sum_{i=1}^n \frac{1}{i}$ denote the partial sums of $\frac{1}{n}$. We consider s_{2^n} :

$$\begin{aligned} s_{2^n} &= 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) + \cdots + \underbrace{\left(\frac{1}{2^{n-1}+1} + \cdots + \frac{1}{2^n}\right)}_{2^{n-1} \text{ terms}} \\ &\geq 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right) + \cdots + \left(\frac{1}{2^n} + \cdots + \frac{1}{2^n}\right) \\ &= 1 + \frac{1}{2} + 2\left(\frac{1}{4}\right) + 4\left(\frac{1}{2^3}\right) + \cdots + 2^{n-1}\left(\frac{1}{2^n}\right) \\ &= 1 + \frac{n}{2} \rightarrow \infty, \end{aligned}$$

so $s_{2^n} \rightarrow \infty$ as $n \rightarrow \infty$. Therefore, given $R > 0$, let $n \in \mathbb{N}$ be such that $s_{2^n} > R$. Since (s_n) is an increasing sequence, for $m \geq 2^n$ we have $s_m \geq s_{2^n} > R$, so $s_n \rightarrow \infty$ as $n \rightarrow \infty$. $\square$

Therefore, despite the incredibly slow pace ($\mathcal{O}(\log n)$), this sequence of partial sums eventually surpasses every positive real number.

In fact, the argument used in the previous result can be generalised:

Theorem 2.54 (Cauchy Condensation test). *Let (a_n) be a decreasing sequence of non-negative real numbers. Then*

$$\sum_{n=1}^{\infty} a_n \text{ converges} \iff \sum_{n=0}^{\infty} 2^n a_{2^n} \text{ converges}.$$

Proof. Let $s_n := \sum_{i=1}^n a_i$ and $t_n := \sum_{j=0}^n 2^j a_{2^j}$. We will show that for all $k \in \mathbb{N}$,

$$\frac{t_k}{2} \leq s_{2^k} \leq t_k,$$

which will give us the desired result. Indeed, given $k \in \mathbb{N}$ we have

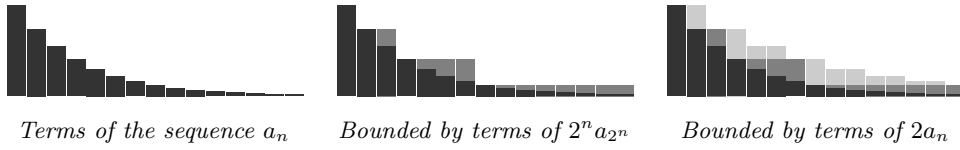
$$\begin{aligned} s_{2^k} &= a_1 + a_2 + (a_3 + a_4) + (a_5 + a_6 + a_7 + a_8) + \cdots + (a_{2^{k-1}+1} + \cdots + a_{2^k}) \\ &\geq \frac{1}{2}a_1 + a_2 + 2a_4 + 4a_8 + \cdots + 2^{k-1}a_{2^k} \\ &= \frac{1}{2} \left(a_1 + 2a_2 + 4a_4 + 8a_8 + \cdots + 2^k a_{2^k} \right) \\ &= \frac{t_k}{2}, \end{aligned}$$

and

$$\begin{aligned} s_{2^k} &= a_1 + (a_2 + a_3) + (a_4 + a_5 + a_6 + a_7) + \cdots + (a_{2^{k-1}} + \cdots + a_{2^k-1}) + a_{2^k} \\ &\leq a_1 + (a_2 + a_2) + (a_4 + a_4 + a_4 + a_4) + \cdots + (a_{2^{k-1}} + \cdots + a_{2^k-1}) + 2^k a_{2^k} \\ &= t_k, \end{aligned}$$

as desired. The result follows. $\square$

A diagram showing this inequality visually is given below. The partial sums of the series $\sum a_n$, $\sum 2^n a_{2^n}$, and $2 \sum a_n$ are represented, overlaid from left to right.



This is a powerful test and can be used to determine the convergence or divergence of a large class of infinite series. Indeed, a simple (but useful) corollary is:

Corollary 2.55 (*p*-series test). *The series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if and only if $p > 1$.*

Proof. Put $a_n = \frac{1}{n^p}$ and let $t_n = \sum_{i=0}^n 2^i a_{2^i}$. Then we have, for $n \in \mathbb{N}$,

$$t_n = \sum_{i=0}^n 2^i \frac{1}{(2^i)^p} = \sum_{i=0}^n 2^{i(1-p)}.$$

This is a geometric sum, so by Theorem 2.50, t_n converges if and only if $2^{1-p} < 1$ i.e. if and only if $p > 1$. By the Cauchy Condensation test (Theorem 2.54), it follows that $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if and only if $p > 1$. $\square$

Exercise. Determine the values of p for which the series $\sum_{n=2}^{\infty} \frac{1}{n(\log n)^p}$ converges.

Theorem 2.56 (Comparison test). *Let (a_n) and (b_n) be real sequences such that $0 \leq a_n \leq b_n$ for all n (or eventually).*

1. *If $\sum b_n$ is convergent, then $\sum a_n$ is convergent.*
2. *If $\sum a_n$ diverges, then $\sum b_n$ diverges.*

Proof. Let $s_n = \sum_{i=1}^n a_i$ and $t_n = \sum_{i=1}^n b_i$. Since $0 \leq a_i \leq b_i$ for all $i \in \mathbb{N}$, we can sum this inequality to get that $0 \leq s_n \leq t_n$ for all $n \in \mathbb{N}$. Furthermore, observe that since $a_i \geq 0$ for all $i \in \mathbb{N}$, (s_n) is an increasing sequence.

1. Suppose $\sum b_n$ is convergent. Then (t_n) is bounded, so there exists $M > 0$ such that for all $n \in \mathbb{N}$, $t_n \leq M$. Therefore, using the above observation, $s_n \leq t_n \leq M$, so (s_n) is bounded, and is increasing so by the Monotone Convergence Theorem (Theorem 2.24), (s_n) is convergent, i.e. $\sum a_n$ is convergent.
2. Suppose $\sum a_n$ diverges, i.e. (s_n) diverges. Then since $t_n \geq s_n \rightarrow \infty$, t_n diverges i.e. $\sum b_n$ diverges. $\square$

Remark. If item 1 in the above theorem holds, we say that “ $\sum a_n$ converges by comparison with $\sum b_n$ ” (and similarly for item 2).

Example 2.57. 1. For $n \in \mathbb{N}$ we have $0 \leq \frac{1}{n^2+1} \leq \frac{1}{n^2}$ and by the p -series test (Corollary 2.55), $\frac{1}{n^2}$ is summable, so $\frac{1}{n^2+1}$ is summable.

2. For $n \in \mathbb{N}$ we have $0 \leq \frac{2+\cos n}{n^3} \leq \frac{3}{n^3}$ since $|\cos n| \leq 1$ so since $\frac{1}{n^3}$ is summable by the p -series test, $\sum_{n=1}^{\infty} \frac{2+\cos n}{n^3}$ is convergent.

Definition 2.58 (Absolute and conditional convergence). A complex sequence (a_n) is *absolutely summable* if the sequence of absolute values $(|a_n|)$ is summable. If (a_n) is summable but not absolutely summable, it is called *conditionally summable*.

Remark. Absolute convergence means the series converges purely based on the rapidly decreasing magnitude of its terms, even in the worst-case scenario where every term is positive. Conditional convergence means the series only converges because of cancellation between positive and negative terms. If you take the negative terms in your sequence, take the absolute value and construct a sequence of partial sums, and similarly construct a sequence of partial sums for the positive terms, absolute convergence means that both partial sums converge. It is therefore not surprising that rearranging terms arbitrarily in your sequence does not change the value of an absolutely convergent series (we will prove this later). However, rearranging terms in a conditionally convergent series.

Example 2.59. $\sum \frac{(-1)^n}{n^2}$ is absolutely convergent. $\sum \frac{(-1)^n}{n}$ is conditionally convergent (it converges, but its absolute value is the harmonic series which we saw diverges).

Theorem 2.60. *If a real sequence (a_n) is absolutely summable, then it is summable.*

Proof. Suppose $(|a_n|)$ is summable. For $n \in \mathbb{N}$ we have $0 \leq a_n + |a_n| \leq 2|a_n|$, and $(|a_n|)$ is summable, so $(2|a_n|)$ is summable, so by the Comparison test (Theorem 2.56), $(a_n + |a_n|)$ is summable. Therefore, $a_n = (a_n + |a_n|) - |a_n|$ is summable. $\square$

Theorem 2.61 (Alternating Series test). *Let (a_n) be a sequence of non-negative real numbers that is decreasing and $a_n \rightarrow 0$ as $n \rightarrow \infty$. Then the alternating series $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$ is convergent.*

Proof. Let (a_n) be a sequence such that $a_n \geq 0$, $a_{n+1} \leq a_n$ for all $n \in \mathbb{N}$, and $\lim_{n \rightarrow \infty} a_n = 0$.

Let s_n denote the n -th partial sum, $s_n = \sum_{k=1}^n (-1)^{k+1} a_k$. We will first examine the even partial sums, s_{2n} . We can group the terms of s_{2n} as follows:

$$s_{2n} = (a_1 - a_2) + (a_3 - a_4) + \cdots + (a_{2n-1} - a_{2n})$$

Because the sequence (a_n) is monotonically decreasing, we know that $a_{2k-1} - a_{2k} \geq 0$ for all k . Therefore, s_{2n} is a sum of non-negative terms, which implies that the sequence of even partial sums (s_{2n}) is monotonically increasing: $s_{2n} \leq s_{2(n+1)}$.

Alternatively, we can rewrite the grouping of s_{2n} by isolating the first term:

$$s_{2n} = a_1 - (a_2 - a_3) - (a_4 - a_5) - \cdots - (a_{2n-2} - a_{2n-1}) - a_{2n}$$

Again, because $a_k \geq a_{k+1}$, every term in parentheses is non-negative, and $a_{2n} > 0$. Thus, we are subtracting purely non-negative quantities from a_1 , which guarantees that $s_{2n} \leq a_1$ for all n .

Since (s_{2n}) is monotonically increasing and bounded above by a_1 , it must converge to some real number s by the Monotone Convergence Theorem. That is,

$$\lim_{n \rightarrow \infty} s_{2n} = s$$

Next, we examine the odd partial sums, s_{2n+1} . We can express the $(2n+1)$ -th partial sum purely in terms of the $2n$ -th partial sum and the next sequence term:

$$s_{2n+1} = s_{2n} + a_{2n+1}$$

Taking the limit of both sides as $n \rightarrow \infty$, and using the initial hypothesis that $\lim_{n \rightarrow \infty} a_n = 0$, we find, by the Algebra of Limits:

$$\lim_{n \rightarrow \infty} s_{2n+1} = \lim_{n \rightarrow \infty} (s_{2n} + a_{2n+1}) = \lim_{n \rightarrow \infty} s_{2n} + \lim_{n \rightarrow \infty} a_{2n+1} = s + 0 = s$$

Because the sequence of even partial sums (s_{2n}) and the sequence of odd partial sums (s_{2n+1}) both converge to the exact same limit s , the sequence of all partial sums (s_n) must also converge to s . Therefore, the alternating series converges. $\square$

Theorem 2.62 (Ratio test). *Let (a_n) be a real sequence of non-zero terms. Suppose that $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = L$.*

1. *If $L < 1$, the series $\sum a_n$ is absolutely summable.*
2. *If $L > 1$, the series diverges.*
3. *If $L = 1$, the test is inconclusive.*

Proof. Let (a_n) be a real sequence with $a_n > 0$ for all $n \in \mathbb{N}$, and suppose $\lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = L$.

Case 1: $L < 1$ (convergence). Choose $r \in \mathbb{R}$ such that $L < r < 1$. Because the limit of the ratio is $L < 1$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, we have:

$$\frac{a_{n+1}}{a_n} < r$$

This implies that $a_{n+1} < ra_n$ for all $n \geq N$. By induction, for any $k \geq 1$ we get:

$$a_{N+k} < ra_{N+k-1} < r^2 a_{N+k-2} < \cdots < r^k a_N$$

We are interested in the tail of the original series, $\sum_{k=1}^{\infty} a_{N+k}$. From the inequality above, we have:

$$\sum_{k=1}^{\infty} a_{N+k} < \sum_{k=1}^{\infty} r^k a_N = a_N \sum_{k=1}^{\infty} r^k$$

Since $0 < r < 1$, the geometric series $\sum r^k$ converges. By the Direct Comparison Test, the tail $\sum a_{N+k}$ converges, and therefore the entire series $\sum a_n$ converges (by the shift summability lemma).

Case 2: $L > 1$ (divergence). Choose a real number r such that $1 < r < L$. By the definition of the limit, there exists an integer $N \in \mathbb{N}$ such that for all $n \geq N$:

$$\frac{a_{n+1}}{a_n} > r > 1$$

This means $a_{n+1} > a_n$ for all $n \geq N$. Since the sequence of terms (a_n) is strictly increasing and strictly positive for $n \geq N$, it is impossible for $\lim_{n \rightarrow \infty} a_n = 0$. By the n -th term divergence test (Theorem 2.51), the series $\sum a_n$ diverges. $\square$

Theorem 2.63 (Root test). *Let (a_n) be a real sequence. Suppose that*

$$\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = L.$$

Then

1. *If $L < 1$, the series $\sum a_n$ is absolutely summable.*
2. *If $L > 1$, the series diverges.*
3. *If $L = 1$, the test is inconclusive.*

Proof. Let (a_n) be a real sequence with $a_n \geq 0$ for all $n \in \mathbb{N}$, and suppose $\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = L$.

Case 1: $L < 1$ (convergence). Choose $r \in \mathbb{R}$ such that $L < r < 1$. By the definition of the limit, there exists an integer $N \in \mathbb{N}$ such that for all $n \geq N$:

$$\sqrt[n]{a_n} < r$$

Raising both sides to the n -th power, since both sides are non-negative, yields:

$$a_n < r^n \quad \text{for all } n \geq N$$

Because $0 \leq r < 1$, the geometric series $\sum_{n=N}^{\infty} r^n$ converges. By the Direct Comparison Test, the tail of our series $\sum_{n=N}^{\infty} a_n$ converges, which means the entire series $\sum a_n$ converges.

Case 2: $L > 1$ (divergence). Since $L > 1$, there exists an integer $N \in \mathbb{N}$ such that for all $n \geq N$:

$$\sqrt[n]{a_n} > 1$$

Raising both sides to the n -th power gives:

$$a_n > 1^n = 1 \quad \text{for all } n \geq N$$

Because the terms of the series are eventually bounded below by 1, we know that $\lim_{n \rightarrow \infty} a_n \neq 0$. By the n -th term divergence test, the series $\sum a_n$ diverges. $\square$

Example 2.64. 1. The series $\sum_{n=1}^{\infty} \frac{2^n}{n!}$ converges. Set $a_n = \frac{2^n}{n!}$ for $n \in \mathbb{N}$. Then we have

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{2^{n+1}/(n+1)!}{2^n/n!} = \frac{2}{n+1} \rightarrow 0 < 1,$$

so $\sum_{n=1}^{\infty} \frac{2^n}{n!}$ converges by the Ratio test (Theorem 2.62).

2. The series $\sum_{n=1}^{\infty} \frac{n}{2^n}$ converges. Set $a_n = \frac{n}{2^n}$ for $n \in \mathbb{N}$. Then we have

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)/2^{n+1}}{n/2^n} = \frac{1}{2} \frac{n+1}{n} \rightarrow \frac{1}{2} < 1,$$

so $\sum_{n=1}^{\infty} \frac{n}{2^n}$ converges by the Ratio test.

3. The series $\sum_{n=1}^{\infty} \left(\frac{4n^2-3n}{5n^2+1} \right)^n$ converges. Set $a_n = \left(\frac{4n^2-3n}{5n^2+1} \right)^n$ for $n \in \mathbb{N}$. Then we have

$$\sqrt[n]{a_n} = \frac{4n^2-3n}{5n^2+1} = \frac{4 - \frac{3}{n}}{5 + \frac{1}{n^2}} \rightarrow \frac{4}{5} < 1,$$

by the Algebra of Limits. Therefore, by the Root test (Theorem 2.63), $\sum_{n=1}^{\infty} \left(\frac{4n^2-3n}{5n^2+1} \right)^n$ converges.

Remark. Both the root and ratio tests examine whether the sequence in question can be bounded above ($L < 1$) or below ($L > 1$) by a convergent or divergent geometric series. If $L < 1$, it decays exponentially fast (like $(1/2)^n$), so it converges. If $L > 1$, it grows at least as fast as a divergent geometric series (intuitively), so it diverges.

Example 2.65. For $a_n = \frac{1}{n!}$, the ratio is $\left| \frac{1/(n+1)!}{1/n!} \right| = \frac{1}{n+1} \rightarrow 0 < 1$. Thus, factorials grow so fast that their reciprocals decay much faster than any geometric series, guaranteeing absolute convergence.

The following series test is useful for series with oscillating terms.

Theorem 2.66 (Dirichlet test). *Let (a_n) and (b_n) be sequences of real numbers such that:*

1. (b_n) is decreasing and $b_n \rightarrow 0$ as $n \rightarrow \infty$.
2. The partial sums of (a_n) are bounded, i.e., $\exists M > 0$ such that $|\sum_{i=1}^n a_i| \leq M$ for all $n \in \mathbb{N}$.

Then the series $\sum_{n=1}^{\infty} a_n b_n$ converges.

Proof. Let (a_n) and (b_n) be sequences of real numbers such that:

1. The partial sums $A_n = \sum_{i=1}^n a_i$ are bounded. That is, there exists a real number $M > 0$ such that $|A_n| \leq M$ for all $n \in \mathbb{N}$.
2. The sequence (b_n) is decreasing ($b_{n+1} \leq b_n$ for all $n \in \mathbb{N}$).
3. $\lim_{n \rightarrow \infty} b_n = 0$.

Let $S_n = \sum_{k=1}^n a_k b_k$. First, we rewrite S_n using summation by parts (also known as Abel's lemma). Since $a_k = A_k - A_{k-1}$ (where we define $A_0 = 0$), we can substitute this into our partial sum:

$$S_n = \sum_{k=1}^n (A_k - A_{k-1}) b_k.$$

Thus

$$S_n = A_n b_n + \sum_{k=1}^{n-1} A_k (b_k - b_{k+1}).$$

Now, because $|A_n| \leq M$ for all $n \in \mathbb{N}$, we have $-M b_n \leq A_n b_n \leq M b_n$. Since $\lim_{n \rightarrow \infty} b_n = 0$, both bounds approach zero. By the Squeeze Theorem (Theorem 2.10):

$$\lim_{n \rightarrow \infty} A_n b_n = 0.$$

We must now show that the series $\sum_{k=1}^{\infty} A_k (b_k - b_{k+1})$ converges. We have

$$\sum_{k=1}^{n-1} |A_k (b_k - b_{k+1})| = \sum_{k=1}^{n-1} |A_k| |b_k - b_{k+1}|,$$

and since (b_n) is decreasing, $b_k - b_{k+1} \geq 0$, so we can drop the absolute value. Using our bound $|A_k| \leq M$, we get:

$$\sum_{k=1}^{n-1} |A_k| (b_k - b_{k+1}) \leq \sum_{k=1}^{n-1} M (b_k - b_{k+1}) = M \sum_{k=1}^{n-1} (b_k - b_{k+1}) = M(b_1 - b_n).$$

Because $b_n \geq 0$, we know $M(b_1 - b_n) \leq M b_1$. Therefore, the partial sums of the absolute series are bounded above by the constant $M b_1$. Because it is a series of non-negative terms with bounded partial sums, $\sum_{k=1}^{\infty} |A_k (b_k - b_{k+1})|$ converges. Absolute convergence implies convergence, so the series $\sum_{k=1}^{\infty} A_k (b_k - b_{k+1})$ must converge to some real number L .

Taking the limit of our summation by parts equation as $n \rightarrow \infty$, we have, by the Algebra of Limits:

$$\lim_{n \rightarrow \infty} S_n = \lim_{n \rightarrow \infty} A_n b_n + \lim_{n \rightarrow \infty} \sum_{k=1}^{n-1} A_k (b_k - b_{k+1}) = 0 + L = L$$

Therefore, (S_n) converges, i.e. the series $\sum_{n=1}^{\infty} a_n b_n$ converges. $\square$

Example 2.67. The series $\sum_{n=1}^{\infty} \frac{e^{in\theta}}{n}$ is convergent for all $\theta \not\equiv 0 \pmod{2\pi}$. We have

$$\left| \sum_{k=1}^n e^{ik\theta} \right| = \left| \frac{e^{i\theta} (1 - e^{in\theta})}{1 - e^{i\theta}} \right| \leq \frac{2}{|1 - e^{i\theta}|},$$

since $\theta \not\equiv 0 \pmod{2\pi}$, so $1 - e^{i\theta}$ is a non-zero constant. Also, $\frac{1}{n} \rightarrow 0$ and is decreasing, so by the Dirichlet test (Theorem 2.66), $\sum_{n=1}^{\infty} \frac{e^{in\theta}}{n}$ is convergent.

2.3 Power series

Definition 2.68 (Power series). A *power series* centred at $z_0 \in \mathbb{C}$ is a series of the form

$$P(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n$$

where z is a complex variable and (a_n) is a sequence of complex coefficients.

Remark. A power series is an “infinite degree polynomial”, i.e. it is the function we get when we take the limit of finite polynomials in $\mathbb{C}$, i.e. the limit of

$$\sum_{i=0}^n a_i (z - z_0)^i = a_0 + a_1 (z - z_0) + \cdots + a_n (z - z_0)^n.$$

Note that we have a *different* sequence for *each* value of $z \in \mathbb{C}$, and therefore when $P(z)$ converges, we have a different *limit* for each value of $z \in \mathbb{C}$. Power series can therefore be seen as functions $\mathbb{C} \rightarrow \mathbb{C}$. Therefore, this motivates our interest in the radius of convergence for a power series, i.e. for which values of $z \in \mathbb{C}$ does the power series converge? Since the complex plane $\mathbb{C}$ has no natural ordering (we cannot say $z_1 < z_2$), we cannot directly apply real convergence tests to $a_n (z - z_0)^n$. Instead, we apply tests to their magnitudes $|a_n (z - z_0)^n|$, so the convergence problem is in $\mathbb{R}$. We will see why this makes sense below:

Lemma 2.69. Let (a_n) be a complex sequence and let $P(z)$ be the power series associated to (a_n) , centred at $z_0 \in \mathbb{C}$. Let $w \neq z_0$ be a complex number such that $P(w)$ is convergent. Then $P(z)$ is absolutely convergent for all $z \in \mathbb{C}$ such that $|z - z_0| < |w - z_0|$.

Proof. By assumption $P(w) = \sum_{n=0}^{\infty} a_n (w - z_0)^n$ is convergent, hence $a_n (w - z_0)^n$ has to tend to 0 and $(a_n (w - z_0)^n)$ is bounded. This means that there

exists $M \in \mathbb{R}_{>0}$ such that $|a_n(w - z_0)^n| \leq M$ for all $n \in \mathbb{N}$. Let $z \in \mathbb{C}$ be such that $|z - z_0| < |w - z_0|$.

Then

$$|a_n(z - z_0)^n| = \left| a_n(w - z_0)^n \frac{(z - z_0)^n}{(w - z_0)^n} \right| = |a_n(w - z_0)^n| \left| \frac{z - z_0}{w - z_0} \right|^n \leq M \left| \frac{z - z_0}{w - z_0} \right|^n.$$

Since $|w - z_0| > |z - z_0|$, we have $\left| \frac{z - z_0}{w - z_0} \right| < 1$, so $P(z)$ is absolutely convergent by comparison with $M \left| \frac{z - z_0}{w - z_0} \right|^n$. $\square$

Definition 2.70 (Radius of convergence). The *radius of convergence*, R , of a power series $P(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n$ centred at $z_0 \in \mathbb{C}$ is defined as:

$$R := \sup \{ |z - z_0| \in \mathbb{R}_{\geq 0} : P(z) \text{ is convergent} \}.$$

Note that the supremum does not exist if the set is not bounded above. In this case we say that the power series has infinite radius of convergence, and we write $R = \infty$. Another remark is that 0 is always in the set of numbers for which the power series is convergent, hence $R \geq 0$. If $R = 0$, this means that the power series is always divergent, except for the trivial case $z = 0$.

Remark. Power series convergence regions in $\mathbb{C}$ are circular. A power series will absolutely converge for any z strictly inside the open disk defined by $|z - z_0| < R$, and it will diverge for any z strictly outside ($|z - z_0| > R$). The boundary circle $|z - z_0| = R$ is the ambiguous zone where the series might converge, diverge, or conditionally converge depending on the specific point. This is why, as mentioned earlier, why we can apply tests to the magnitudes $|a_n z^n|$, rather than having to test individual values z in the radius of convergence. However, individual values of $z \in \mathbb{C}$ such that $|z - z_0| = R$ must be tested to determine convergence or divergence on the boundary of the convergence region.

Theorem 2.71. Let $P(z) = \sum_{n=0}^{\infty} a_n(z - z_0)^n$ be a power series with radius of convergence R . Then

1. $P(z)$ is absolutely convergent if $|z - z_0| < R$.
2. $P(z)$ is divergent if $|z - z_0| > R$.

Proof. 1. Let $z \in \mathbb{C}$ be such that $|z - z_0| < R$. Therefore, by definition of R , there exists $r \in \mathbb{R}$ such that $|z - z_0| < r \leq R$ such that $P(r)$ is convergent. The result follows from Lemma 2.69.

2. Let $z \in \mathbb{C}$ be such that $|z - z_0| > R$, and suppose for contradiction $P(z)$ is convergent. Then there exists $r \in \mathbb{R}$ such that $|z - z_0| > r > R$, but then $P(r)$ is convergent by Lemma 2.69, contradicting the definition of R . $\square$

Theorem 2.72 (Power series ratio test). *Let $P(z) = \sum a_n(z - z_0)^n$ be a power series centred at $z_0 \in \mathbb{C}$. If the limit $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$ exists (or is ∞), then the radius of convergence is given by $R = \frac{1}{L}$ (with the convention that $1/0 = \infty$ and $1/\infty = 0$).*

Proof. Case 1: $L \neq 0$. We observe that for $z \in \mathbb{C}$, we have $\left| \frac{a_{n+1}(z - z_0)^{n+1}}{a_n(z - z_0)^n} \right| \rightarrow L|z - z_0|$ as $n \rightarrow \infty$. If $L|z - z_0| < 1$ then $\sum_{n=0}^{\infty} a_n z^n$ converges by the Ratio test, and if $L|z - z_0| > 1$ then $\sum_{n=0}^{\infty} a_n z^n$ diverges, so $R = \frac{1}{L}$.

Case 2: $L = 0$. Then for any $z \in \mathbb{C}$, $\left| \frac{a_{n+1}(z - z_0)^{n+1}}{a_n(z - z_0)^n} \right| \rightarrow 0$, so $R = \infty$ (the set of values for which $P(z)$ converges is unbounded).

Case 3: $L = \infty$. Then for any $z \neq 0$, $\left| \frac{a_{n+1}(z - z_0)^{n+1}}{a_n(z - z_0)^n} \right| \rightarrow \infty$, so $(a_n(z - z_0)^n)$ is not summable for any $z \neq 0$, so $R = 0$. $\square$

Remark. Intuitively, you can think of the radius of convergence R as the outcome of an asymptotic tug-of-war between two competing forces: the growth (or decay) rate of the coefficients a_n , and the geometric growth (or decay) of $|z - z_0|^n$.

Example 2.73. 1. The power series $\sum_{n=0}^{\infty} n!(z - z_0)^n$ has convergence radius $R = 0$ because

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)!}{n!} = n+1 \rightarrow \infty.$$

2. The power series $\sum_{n=0}^{\infty} \frac{n^n}{n!}(z - z_0)^n$ has convergence radius $R = \frac{1}{e}$ because

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)^{n+1}/(n+1)!}{n^n/n!} = \frac{1}{n+1} \frac{(n+1)(n+1)^n}{n^n} = \left(1 + \frac{1}{n}\right)^n \rightarrow e.$$

3. The power series $\sum_{n=0}^{\infty} \frac{1}{n!} z^n$ has convergence radius $R = \infty$ because

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{1/(n+1)!}{1/n!} = \frac{1}{n+1} \rightarrow 0.$$

By item 3 above, the following function is well-defined:

Definition 2.74 (The complex exponential function). For any $z \in \mathbb{C}$, we define the complex exponential function as the power series:

$$\exp(z) := \sum_{n=0}^{\infty} \frac{z^n}{n!}.$$

Exercise. Prove that for any $z, w \in \mathbb{C}$, $\exp(z + w) = \exp(z) \exp(w)$. (*Hint: use the Cauchy product formula for series and the Binomial Theorem.*)

Chapter 3

Limits and Continuity

In this chapter, we move on from thinking about *sequences*, which are functions $\mathbb{N} \rightarrow \mathbb{R}$, and we start to build some theory for functions $\mathbb{R} \rightarrow \mathbb{R}$, in particular what constitutes a “*nice*” function.

What motivates this chapter is that a “*nice*” function should be able to be drawn without taking our pen off the paper, because we expect nice properties of such functions; this intuition is a property called the “*Intermediate Value Property*”, and does indeed hold for *continuous* functions (but is not exclusive to continuous functions).

However, the above discussion is by no means a definition. We will define what we mean by *continuity* of a function $f : \mathbb{R} \rightarrow \mathbb{R}$ once we understand what a *limit* of a function is. In particular, since we are working with functions whose domain is $\mathbb{R}$ now, the notion of *limit* can be made meaningful without sending $x \rightarrow \infty$ (i.e. we can define the limit of a function $f(x)$, say, as $x \rightarrow a$ where $a \in \mathbb{R}$ is some real number).

3.1 Limits of Functions

We wish to adapt the definition of limit for functions $\mathbb{R} \rightarrow \mathbb{R}$. First, we make the definition for limits as $|x| \rightarrow \infty$, which are analogous to the definition of a limit for sequences (but of course, rather than only being able to send n to positive infinity like in sequences, there is another direction, the negative numbers, so we can also define limits as $x \rightarrow -\infty$ as well).

Definition 3.1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function, and let $L \in \mathbb{R}$.

1. We say that $f(x)$ tends to L as $x \rightarrow \infty$ if

$$\forall \varepsilon > 0, \exists R \in \mathbb{R}, \forall x \geq R, |f(x) - L| < \varepsilon.$$

2. We say that $f(x)$ tends to L as $x \rightarrow -\infty$ if

$$\forall \varepsilon > 0, \exists R \in \mathbb{R}, \forall x \leq -R, |f(x) - L| < \varepsilon.$$

Definition 3.2. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function.

1. We say that $f(x)$ tends to ∞ as $x \rightarrow \infty$ if

$$\forall M > 0, \exists R \in \mathbb{R}, \forall x \geq R, f(x) > M.$$

2. We say that $f(x)$ tends to ∞ as $x \rightarrow -\infty$ if

$$\forall M > 0, \exists R \in \mathbb{R}, \forall x \leq R, f(x) > M.$$

There is an obvious variation of the above definitions for the definitions of $\lim_{x \rightarrow \infty} f(x) = -\infty$ and $\lim_{x \rightarrow -\infty} f(x) = -\infty$.

To define “ $\lim_{x \rightarrow a} f(x) = L$ ”, we need to be precise about what “for x close enough to a ” actually means. Also, it is going to matter whether we approach a from the side $x > a$ or from the side $x < a$, so writing $\lim_{x \rightarrow a} f(x) = L$ may not always make sense.

Definition 3.3. Let $a \in \mathbb{R}$ and let $f : (a, \infty) \rightarrow \mathbb{R}$ be a function. We say that $f(x)$ tends to $L \in \mathbb{R}$ as $x \rightarrow a^+$ if

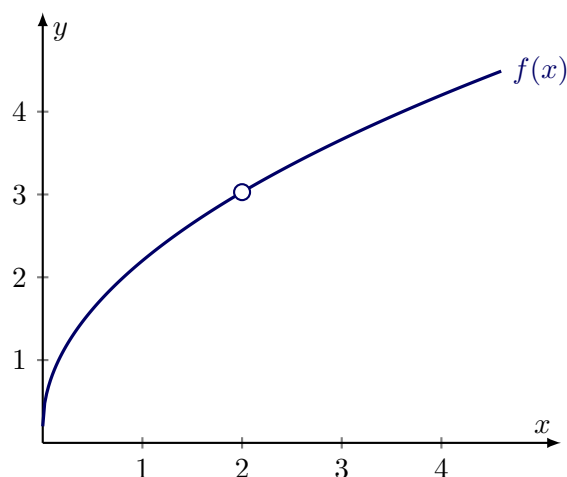
$$\forall \varepsilon > 0, \exists \delta > 0, \forall x \in (a, \infty), x \in (a, a + \delta) \implies |f(x) - L| < \varepsilon.$$

Above, we formalised “for $x > a$ sufficiently close to a ” as “ $x \in (a, a + \delta)$ for some $\delta > 0$ ”.

Definition 3.4. Let $a \in \mathbb{R}$ and let $f : (b, a) \rightarrow \mathbb{R}$ be a function for some $b < a$. We say that $f(x)$ tends to $L \in \mathbb{R}$ as $x \rightarrow a^-$ if

$$\forall \varepsilon > 0, \exists \delta > 0, \forall x \in (b, a), x \in (a - \delta, a) \implies |f(x) - L| < \varepsilon.$$

Remark. Limits are entirely about the behaviour of f near a , not at a . The value $f(a)$ does not even need to be defined. If $\lim_{x \rightarrow a} f(x) = L$ exists but $f(a)$ is undefined or doesn’t equal L , we can define a “continuous extension” $\tilde{f}$ where $\tilde{f}(x) = f(x)$ for $x \neq a$ and $\tilde{f}(a) = L$. This is a useful trick for when we want to apply some stronger theorems which only apply to, say, continuous functions.



For example, in the picture above, the function $f(x)$ has $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x) = 3$, but is not defined at $x = 2$. If we defined $f(2) = 3$, then f would be a continuous function.

Lemma 3.5. Suppose $f : (a, b) \rightarrow \mathbb{R}$ is a function, and we have

$$\lim_{x \rightarrow a^+} f(x) = L_1 \quad \text{and} \quad \lim_{x \rightarrow a^+} f(x) = L_2.$$

Then $L_1 = L_2$.

Proof. Let $\varepsilon > 0$ be arbitrary. Since $\lim_{x \rightarrow a^+} f(x) = L_1$, we can let $\delta_1 > 0$ be such that for any $x \in (a, a + \delta_1)$, $|f(x) - L_1| < \frac{\varepsilon}{2}$. Similarly, since $\lim_{x \rightarrow a^+} f(x) = L_2$, we can let $\delta_2 > 0$ be such that for any $x \in (a, a + \delta_2)$, $|f(x) - L_2| < \frac{\varepsilon}{2}$. Let $\delta = \min\{\delta_1, \delta_2\}$ and pick $x \in (a, a + \delta)$. By the triangle inequality we have

$$|L_1 - L_2| = |L_1 - f(x) + f(x) - L_2| \leq |f(x) - L_1| + |f(x) - L_2| = \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

so since $\varepsilon > 0$ was arbitrary, it follows that $L_1 = L_2$. $\square$

This lemma holds if we replace $x \rightarrow a^+$ with $x \rightarrow a^-$ and the proof is identical. Note that the above fact does not say that we must have $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x)$, even if both exist.

Now we will define a two-sided limit; note that this requires both the left and right limits to exist and $\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x)$. The function must also be defined on an interval of the form $(b, c) \setminus \{a\}$ for some $b < a < c$ (notice again that $f(a)$ need not be defined for this definition to make sense):

Definition 3.6. Let $a \in \mathbb{R}$ and let $f : (b, c) \setminus \{a\} \rightarrow \mathbb{R}$ be a function for some $b < a < c$. We say that $f(x)$ tends to $L \in \mathbb{R}$ as $x \rightarrow a$ if

$$\lim_{x \rightarrow a^-} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow a^+} f(x) = L.$$

In other words, $\lim_{x \rightarrow a} f(x) = L$ if

$$\forall \varepsilon > 0, \exists \delta > 0, \forall x \in (b, c) \setminus \{a\}, 0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon.$$

Definition 3.7. Let $I \subset \mathbb{R}$. We say that I is an *open interval* if it is a set of the form:

- an interval (a, b) for some $a < b$ in $\mathbb{R}$,
- an interval $(-\infty, b)$,
- an interval (a, ∞) ,
- $I = \mathbb{R}$.

Theorem 3.8 (Algebra of Limits). *Let $I \subset \mathbb{R}$ be an open interval, and let $a \in I$. If $\lim_{x \rightarrow a} f(x) = L$ and $\lim_{x \rightarrow a} g(x) = M$, then:*

1. $\lim_{x \rightarrow a} (f(x) + g(x)) = L + M$;
2. $\lim_{x \rightarrow a} (f(x)g(x)) = LM$;
3. $\lim_{x \rightarrow a} \frac{1}{f(x)} = \frac{1}{L}$ (provided $L \neq 0$ and $f(x) \neq 0$ near a , except possibly at a).

Proof. Exercise (compare with the proofs for the Algebra of Limits for sequences). $\square$

Remark. Let $I \subset \mathbb{R}$ be an interval, and let $a \in I$. When we say a property holds “near a ”, we mean there exists some $\delta > 0$ such that the property holds for all $x \in (a - \delta, a + \delta) \cap I$.

When we append the phrase “except possibly at a ”, we restrict x to the punctured neighborhood $0 < |x - a| < \delta$. Therefore, when talking about limits, we can concisely discuss properties of functions that only need to hold *locally*. For example, a common requirement for theorems which involve limits of functions like $\frac{1}{f(x)}$ is that $f(x) \neq 0$ near a , except possibly at a .

3.2 Continuity

Definition 3.9 (Continuity at a point). Let $I \subset \mathbb{R}$ be an open interval, and let $f : I \rightarrow \mathbb{R}$ be a function. We say that f is *continuous* at $a \in I$ if

$$\lim_{x \rightarrow a} f(x) = f(a).$$

Equivalently, f is continuous at $a \in I$ if

$$\forall \varepsilon > 0, \exists \delta > 0, \forall x \in I, |x - a| < \delta \implies |f(x) - f(a)| < \varepsilon.$$

Remark. Note that we do not need to specify that $|x - a| > 0$ since we are assuming f is defined at a now, and if $|x - a| = 0$ then $x = a$ and the inequality $|f(a) - f(a)| < \varepsilon$ is trivially satisfied for any $\varepsilon > 0$.

Exercise. Negate the above string of quantifiers to get a definition for *discontinuity* at a point $a \in I$.

Definition 3.10. Let $I \subset \mathbb{R}$ be an open interval. We say that $f : I \rightarrow \mathbb{R}$ is *continuous everywhere*, or *continuous*, if f is continuous at a for all $a \in I$.

Example 3.11. 1. Constant functions are continuous everywhere. Indeed, fix $\lambda \in \mathbb{R}$, and define $f : \mathbb{R} \rightarrow \mathbb{R}$ by $f(x) = \lambda$ for all $x \in \mathbb{R}$. Let $\varepsilon > 0$ and $a \in \mathbb{R}$ be arbitrary. Choose $\delta = 1$, and let $x \in \mathbb{R}$ satisfy $|x - a| < \delta$. Then we have

$$|f(x) - f(a)| = |\lambda - \lambda| = 0 < \varepsilon,$$

so f is continuous everywhere.

2. The function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x$ is continuous everywhere. Indeed, let $\varepsilon > 0$ and $a \in \mathbb{R}$ be arbitrary. Choose $\delta = \varepsilon$ and let $x \in \mathbb{R}$ be such that $|x - a| < \delta$. Then we have

$$|f(x) - f(a)| = |x - a| < \delta = \varepsilon,$$

so f is continuous everywhere.

3. The function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

is continuous everywhere except at $a = 0$. First we show discontinuity at $a = 0$. Pick $\varepsilon = \frac{1}{2}$, and let $\delta > 0$ be arbitrary. We know that $f(a) = 1$, so we let $x = -\frac{\delta}{2} \in (-\delta, \delta)$. Then we have

$$|f(x) - f(0)| = |0 - 1| = 1 > \frac{1}{2} = \varepsilon,$$

so f is discontinuous at $a = 0$

Now, let $\varepsilon > 0$ and $c \neq 0$ be arbitrary. Choose $\delta = \frac{|c|}{2}$, and let $x \in \mathbb{R}$ be such that $|x - c| < \delta$. Then in particular,

$$\frac{|c|}{2} < |x| < \frac{3|c|}{2},$$

so if $x < 0$, then $f(x) = f(c) = 0$, and if $x > 0$, then $f(x) = f(c) = 1$, so in either case $|f(x) - f(c)| = 0 < \varepsilon$, so f is continuous at c for every $c \neq 0$.

3.2.1 Properties of Continuous Functions

Theorem 3.12 (Algebra for Continuous Functions). *Let $I \subset \mathbb{R}$ be an open interval and let $f, g : I \rightarrow \mathbb{R}$ be continuous. Then*

1. $f + g : I \rightarrow \mathbb{R}$ is continuous;
2. $fg : I \rightarrow \mathbb{R}$ is continuous;
3. If $f(x) \neq 0$ for all $x \in I$ then the function $\frac{1}{f} : I \rightarrow \mathbb{R}$ is continuous.

Proof. All items follow from the Algebra of Limits for functions (Theorem 3.8). $\square$

Example 3.13. In Example 3.11, we showed that the identity function on $\mathbb{R}$ is continuous, and constant functions on $\mathbb{R}$ are continuous. Therefore, by induction, for every $n \in \mathbb{N}$, $x \mapsto a_n x^n$ where $a_n \in \mathbb{R}$ is a constant is continuous on $\mathbb{R}$ using item 2 in the Algebra for Continuous functions (Theorem 3.12). Using item 1 in the Algebra for Continuous Functions inductively, we deduce that *polynomials* are continuous, i.e. functions $P : \mathbb{R} \rightarrow \mathbb{R}$ of the form

$$P(x) = a_0 + a_1 x + \cdots + a_{n-1} x^{n-1} + a_n x^n,$$

where $a_0, \dots, a_n \in \mathbb{R}$ with $a_n \neq 0$ are constants, are continuous.

Furthermore, using this fact, by combining items 2 and 3 in the Algebra for Continuous functions we also deduce that functions of the form

$$R : I \rightarrow \mathbb{R}, \quad x \mapsto \frac{P(x)}{Q(x)},$$

where P, Q are polynomials and $I \subset \mathbb{R}$ is an open interval on which Q never vanishes, are continuous. Such a function is called a *rational function*.

Theorem 3.14. *Let $I \subset \mathbb{R}$ be an open interval, and let $f : I \rightarrow \mathbb{R}$ be continuous. Let $J \subset \mathbb{R}$ be an open interval such that $J \supset f(I)$, and let $g : J \rightarrow \mathbb{R}$ be continuous. Then*

$$g \circ f : I \rightarrow \mathbb{R}$$

is continuous.

Proof. Let $\varepsilon > 0$ and $a \in I$ be arbitrary. Since $g : J \rightarrow \mathbb{R}$ is continuous at $f(a) \in J$, there exists $\delta > 0$ such that for all $y \in J$, if $|y - f(a)| < \delta$ then $|g(y) - g(f(a))| < \varepsilon$. Since $f : I \rightarrow \mathbb{R}$ is continuous at $a \in I$, there exists $\delta' > 0$ such that for all $x \in I$, if $|x - a| < \delta'$ then $|f(x) - f(a)| < \delta$. Therefore, we have, for $x \in I$ with $|x - a| < \delta'$, $|f(x) - f(a)| < \delta$, so it follows that (with $y = f(x)$),

$$|g(f(x)) - g(f(a))| < \varepsilon,$$

so $g \circ f$ is continuous. $\square$

Example 3.15. Using the above Theorem 3.14 and Example 3.13 (where we deduced the continuity of polynomials and rational functions on $\mathbb{R}$), we can deduce the continuity of a large class of functions without having to use the ε - δ definition. For example, if for the moment we assume the continuity of the functions $\sin$, $\cos$ and $\exp$ (which we will show in later chapters), we can deduce that

$$x \mapsto \sin\left(\frac{x}{x^2 + 1}\right), \quad x \mapsto \sum_{k=1}^n \frac{1}{k\pi} \cos(kx), \quad x \mapsto \exp(-x^2)$$

are all continuous on $\mathbb{R}$.

3.2.2 Sequential Continuity

We can translate problems about continuity back into the familiar world of sequences with the following concept (which is a bit more general than the ε - δ definition):

Definition 3.16 (Sequential Continuity). Let $I \subset \mathbb{R}$ be an open interval. A function $f : I \rightarrow \mathbb{R}$ is *sequentially continuous* at $a \in I$ if for any sequence (x_n) contained in I with $\lim_{n \rightarrow \infty} x_n = a$, we have $\lim_{n \rightarrow \infty} f(x_n) = f(a)$.

Remark. We say that a function $f : I \rightarrow \mathbb{R}$, where $I \subset \mathbb{R}$ is an open interval, is *sequentially continuous* if it is sequentially continuous at every $a \in I$.

In $\mathbb{R}$ (and, in fact, in any metric space) the concepts of sequential continuity and ε - δ continuity are logically equivalent:

Theorem 3.17 (Sequential Characterisation of Continuity). *Let $I \subset \mathbb{R}$ be an open interval and let $f : I \rightarrow \mathbb{R}$ be a function. Then f is continuous at $a \in I$ if and only if f is sequentially continuous at $a \in I$.*

Proof. Suppose f is continuous at $a \in I$, and let $\varepsilon > 0$ be arbitrary. By definition, we can choose $\delta > 0$ such that for all $x \in I$ such that $|x - a| < \delta$, $|f(x) - f(a)| < \varepsilon$. Since $x_n \rightarrow a$ as $n \rightarrow \infty$, we can let $N \in \mathbb{N}$ be such that for all $n \geq N$, $|x_n - a| < \delta$. Therefore, $|f(x_n) - f(a)| < \varepsilon$ for $n \geq N$, so $f(x_n) \rightarrow f(a)$ as $n \rightarrow \infty$.

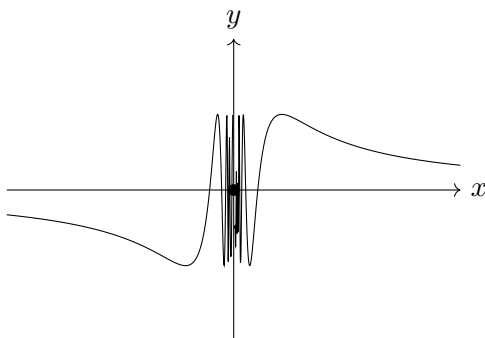
Conversely, suppose f is sequentially continuous at $a \in I$, and suppose for contradiction that f is not continuous at $a \in I$. Therefore, there exists some $\varepsilon_0 > 0$ such that for all $\delta > 0$, there exists $x \in I$ such that $|x - a| < \delta$ and $|f(x) - f(a)| \geq \varepsilon_0$. In particular, for $n \in \mathbb{N}$ we can let $\delta_n := \frac{1}{n}$, and by definition we can pick $x_n \in I$ with $|x_n - a| < \delta_n$ such that $|f(x_n) - f(a)| \geq \varepsilon_0$; by induction this defines a sequence (x_n) contained in I . However, $x_n \rightarrow a$ as $n \rightarrow \infty$, but $f(x_n) \not\rightarrow f(a)$ as $n \rightarrow \infty$, which contradicts sequential continuity of f at $a \in I$. $\square$

Remark. To prove discontinuity of $f : I \rightarrow \mathbb{R}$ at $a \in I$ where $I \subset \mathbb{R}$ is an open interval, it therefore suffices to find *one* sequence (x_n) in I such that $x_n \rightarrow a$ but $f(x_n) \not\rightarrow f(a)$.

Example 3.18. The function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} \sin\left(\frac{1}{x}\right), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

is not continuous at 0.



The idea is that for any $\delta > 0$, the function f oscillates infinitely many times on the interval $(0, \delta)$, so in particular we can take a sequence converging to zero where the function is constantly 1, which is not equal to $f(0) = 0$.

Let $x_n = \frac{1}{2\pi n + \frac{\pi}{2}}$. Then $x_n \rightarrow 0$, but $f(x_n) = \sin(2\pi n + \frac{\pi}{2}) = 1$ for all $n \in \mathbb{N}$, so $f(x_n) \rightarrow 1 \neq 0 = f(0)$, so f is discontinuous at 0.

Exercise. Show that the function $\chi_{\mathbb{Q}} : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$\chi_{\mathbb{Q}}(x) := \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q} \end{cases}$$

is discontinuous everywhere. (*Hint: use the sequential definition of density.*)

3.2.3 The Extreme Value Theorem

We now define continuity on closed and bounded intervals (i.e. intervals of the form $[b, c]$ for some $b < c$ in $\mathbb{R}$). Before, we were only working with open intervals. The definition for continuity on such an interval is a natural extension of continuity on open intervals:

Definition 3.19. Let $K = [b, c]$ be a closed and bounded interval for some $b < c$ in $\mathbb{R}$ and let $f : K \rightarrow \mathbb{R}$ be a function. We say that f is *continuous* on K (or just *continuous*) if f is continuous on (b, c) and also

$$\lim_{x \rightarrow b^+} f(x) = f(b), \quad \lim_{x \rightarrow c^-} f(x) = f(c).$$

Lemma 3.20. Let $I \subset \mathbb{R}$ be an open interval and let $K \subset I$ be a closed and bounded interval. If $f : I \rightarrow \mathbb{R}$ is continuous, then $f|_K : K \rightarrow \mathbb{R}$ is continuous.

Proof. Follows from the definition. □

In practice, this is mostly how we will produce continuous functions on closed bounded intervals.

It is not hard to check that all our previous results about continuous functions still apply to functions defined on a closed and bounded interval K , the proofs need only trivial modifications.

Definition 3.21. Let $D \subset \mathbb{R}$ be any subset of $\mathbb{R}$ and let $f : D \rightarrow \mathbb{R}$ be a function. We say that f is *bounded* if the set $\{f(x) : x \in D\}$ is bounded, i.e. there exists $M > 0$ such that $|f(x)| \leq M$ for all $x \in D$.

Lemma 3.22. *Let $K = [b, c]$ be a closed and bounded interval. If $f : K \rightarrow \mathbb{R}$ is continuous, then f is bounded.*

Proof. Suppose for contradiction that f is not bounded. Then for each $M > 0$, there exists $x \in K$ such that $f(x) > M$. In particular, for $n \in \mathbb{N}$ define $M_n = n$. Then by definition we can construct a sequence (x_n) in K inductively such that $f(x_n) > n$ for all $n \in \mathbb{N}$.

Now, (x_n) is bounded, since $x_n \in K$ for all $n \in \mathbb{N}$, so by the Bolzano-Weierstraß Theorem in $\mathbb{R}$ (Theorem 2.32), there exists a convergent subsequence (x_{n_j}) with $\lim_{j \rightarrow \infty} x_{n_j} = L$. However, for all $j \in \mathbb{N}$, $b \leq x_{n_j} \leq c$, so by the Order Limit Theorem (Theorem 2.8), $b \leq L \leq c$, so $L \in K$. By the sequential characterisation of continuity in $\mathbb{R}$ (Theorem 3.17), it follows that $\lim_{j \rightarrow \infty} f(x_{n_j}) = f(L)$. However, this is a contradiction because by construction, for each $j \in \mathbb{N}$, $f(x_{n_j}) > n_j \geq j$, so $f(x_{n_j}) \rightarrow \infty$ as $j \rightarrow \infty$, so it cannot converge. Therefore, f is bounded. $\square$

Remark. This is essentially a topological result. Compact sets are closed (we will define this precisely later, but this means for any convergent sequence in the set, the limit cannot “escape” the set, i.e. cannot lie outside of the set), and the set is bounded. Intuitively, this means there is “no escape” for the function, if it is continuous (and therefore preserves this compact property).

If the interval is open, say $(0, 1)$, the function $f(x) = 1/x$ is continuous but unbounded because a convergent sequence $x_n \in (0, 1)$ can get arbitrarily close to 0 without ever equalling zero, allowing the function to blow up at 0. If the interval is unbounded, say $[0, \infty)$, $f(x) = x$ is continuous but unbounded. A sequence (x_n) contained in $[0, \infty)$ can escape to infinity, allowing the function to escape to infinity also. Compactness means that the *continuous* function’s values have nowhere to escape to, because continuity preserves compactness, as we will see later. Below, we prove this for compact intervals, though continuity also preserves compact sets in general.

Definition 3.23. Let $D \subset \mathbb{R}$ be any subset of $\mathbb{R}$ and let $f : D \rightarrow \mathbb{R}$ be a bounded function. We define

$$\sup_{x \in D} f(x) := \sup\{f(x) : x \in D\} \quad \text{and} \quad \inf_{x \in D} f(x) := \inf\{f(x) : x \in D\}.$$

Theorem 3.24 (Extreme Value Theorem). *Let $K = [b, c]$ be a closed, bounded (compact) interval, and let $f : K \rightarrow \mathbb{R}$ be continuous. Then f*

attains its supremum and infimum. That is, there exists $d, e \in K$ such that

$$f(d) = \sup_{x \in K} f(x) \quad \text{and} \quad f(e) = \inf_{x \in K} f(x).$$

Proof. We will prove that f attains its supremum; the proof for attaining its infimum is similar and is left as an exercise.

Write $s = \sup_{x \in K} f(x)$, and suppose for contradiction that f does *not* attain its supremum, i.e. $f(x) \neq s$ for all $x \in K$. Then the following function is well-defined:

$$g : K \rightarrow \mathbb{R}, \quad x \mapsto \frac{1}{s - f(x)}.$$

Since f is continuous, and $s - f(x) \neq 0$ for all $x \in K$, g is continuous by the Algebra for Continuous Functions (Theorem 3.12), so by Lemma 3.22, g is bounded.

However, by definition of supremum, for each $n \in \mathbb{N}$ there exists $x_n \in K$ such that $s - \frac{1}{n} < f(x_n) < s$. This implies that, for $n \in \mathbb{N}$,

$$g(x_n) = \frac{1}{s - f(x_n)} > \frac{1}{s - (s - \frac{1}{n})} = n,$$

so $g(x_n) \rightarrow \infty$ as $n \rightarrow \infty$, which contradicts g bounded. $\square$

3.2.4 The Intermediate Value Theorem

Theorem 3.25 (Intermediate Value Theorem). *Let $K = [b, c]$ be a closed and bounded interval and suppose $f : K \rightarrow \mathbb{R}$ is continuous. Then*

1. *If $f(b) \leq f(c)$ and A is any real number such that $f(b) \leq A \leq f(c)$, then there exists $a \in K$ such that $f(a) = A$.*
2. *If $f(c) \leq f(b)$ and A is any real number such that $f(c) \leq A \leq f(b)$, then there exists $a \in K$ such that $f(a) = A$.*

Proof. We prove the first case. Define the set $S := \{x \in K : f(x) \leq A\}$. Since $f(b) \leq A$, $b \in S$ so S is non-empty, and $S \subseteq K$ so S is bounded. Therefore, by the completeness of $\mathbb{R}$, $a := \sup S$ exists. We claim that $f(a) = A$; suppose for contradiction that $f(a) \neq A$.

Case 1: $f(a) < A$. First we note that $f(c) \geq A$, but by assumption $f(a) < A$, so $a \neq c$, i.e. $a < c$. Since f is continuous, if we pick $\varepsilon = \frac{A - f(a)}{2}$, there exists $\delta > 0$ such that for any $x \in (a - \delta, a + \delta) \cap K$, we have

$$|f(x) - f(a)| < \varepsilon = \frac{A - f(a)}{2},$$

i.e. for $x \in (a - \delta, a + \delta) \cap K$, we have

$$f(x) < f(a) + \varepsilon = f(a) + \frac{A - f(a)}{2} = \frac{f(a) + A}{2} < A,$$

so $x \in (a - \delta, a + \delta) \cap K \implies x \in S$. In particular, since $a < c$, we can find a $\delta' > 0$ small enough such that $I := [a, a + \delta') \subset K$ and $x \in I \implies x \in S$ by the above argument. Therefore, picking any $t \in (a, a + \delta')$, we have that $t \in S$ but $t > a$, contradicting the fact that a is an upper bound for S .

Case 2: $f(a) > A$. First we note that $f(b) \leq A$ but by assumption $f(a) > A$, so $a \neq b$, i.e. $a > b$. Since f is continuous, if we pick $\varepsilon = \frac{f(a) - A}{2}$, there exists $\delta > 0$ such that for any $x \in (a - \delta, a + \delta) \cap K$, we have

$$|f(x) - f(a)| < \varepsilon = \frac{f(a) - A}{2},$$

i.e. for $x \in (a - \delta, a + \delta) \cap K$, we have

$$f(x) > f(a) - \varepsilon = f(a) - \frac{f(a) - A}{2} = \frac{f(a) + A}{2} > A,$$

so $x \in (a - \delta, a + \delta) \cap K \implies x \notin S$. In particular, since $a > b$, we can find a $\delta' > 0$ small enough such that $I := (a - \delta', a] \subset K$ and $x \in I \implies x \notin S$ by the previous argument. Therefore, for every $x > a - \delta'$, $f(x) > A$, so $S \subseteq [b, a - \delta']$, so $a - \delta'$ is an upper bound for S , contradicting that a is the least upper bound for S .

In both cases, we reach a contradiction, so $f(a) = A$.

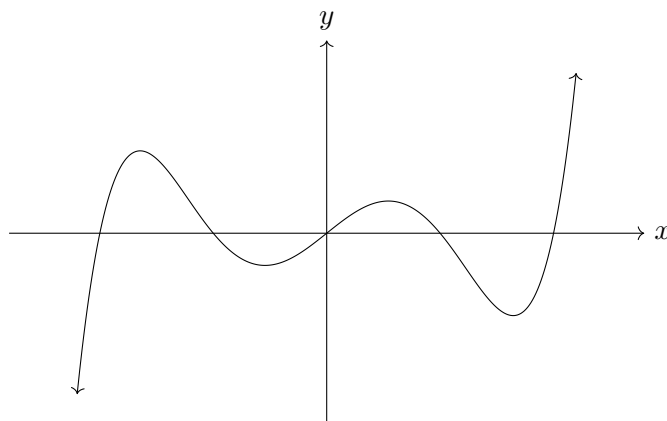
To prove the second case, note that if $f(c) \leq f(b)$, and A is any real number such that $f(c) \leq A \leq f(b)$, then $-f(b) \leq -A \leq -f(c)$ and $-f$ is continuous by the Algebra for Continuous Functions (Theorem 3.12), so by the first case there exists $a \in K$ such that $-f(a) = -A$, i.e. $f(a) = A$. $\square$

Example 3.26 (Fixed Point Application). Let $f : [0, 1] \rightarrow [0, 1]$ be continuous. We can prove f has a fixed point (a point c where $f(c) = c$).

Define $g(x) = f(x) - x$. By the Algebra for Continuous Functions (Theorem 3.12), g is continuous. Since $f(0) \in [0, 1]$, $g(0) = f(0) - 0 \geq 0$. Since $f(1) \in [0, 1]$, $g(1) = f(1) - 1 \leq 0$. Since g is continuous, the Intermediate Value Theorem (IVT) guarantees a point $c \in [0, 1]$ such that $g(c) = 0$, i.e. $f(c) = c$.

Example 3.27. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^5 + x - 1$. Then $f(0) = -1 < 0$ and $f(1) = 1 > 0$, so by the Intermediate Value Theorem there exists $c \in (0, 1)$ such that $f(c) = 0$, i.e. there is a root of f in $(0, 1)$.

In fact, we can generalise this result for odd degree polynomials. Intuitively, the tails of an odd degree polynomial must diverge in opposite directions to each other because of the odd leading power of x :



Corollary 3.28. *Let $P : \mathbb{R} \rightarrow \mathbb{R}$ be any polynomial of odd degree. Then P has at least one root.*

Proof. Let $P : \mathbb{R} \rightarrow \mathbb{R}$ be defined by

$$P(x) = a_{2n+1}x^{2n+1} + a_{2n}x^{2n} + \cdots + a_1x + a_0,$$

where $a_i \in \mathbb{R}$ are constants with $a_{2n+1} \neq 0$, and n is some natural number. Without loss of generality, assume $a_{2n+1} > 0$ (if not, then replace P with $-P$ which has a root if and only if P does). We write $Q(x) = a_{2n}x^{2n} + \cdots + a_1x + a_0$.

Consider evaluating Q on the natural numbers. Then $Q(i)/i^{2n+1} \rightarrow 0$ as $i \rightarrow \infty$, so if i is large enough then $|Q(i)/i^{2n+1}| < a_{2n+1}$ and so $|Q(i)| < a_{2n+1}i^{2n+1}$. Hence

$$P(i) = a_{2n+1}i^{2n+1} + Q(i) \geq a_{2n+1}i^{2n+1} - |Q(i)| > 0,$$

and

$$P(-i) \leq -a_{2n+1}i^{2n+1} + |Q(i)| < 0,$$

for sufficiently large i . Now fix j sufficiently large such that the above inequalities hold; then P is continuous on $[-j, j]$ with $P(-j) < 0$ and $P(j) > 0$, so P has a root in $(-j, j)$. $\square$

Theorem 3.29. *Let I be an interval. If $f : I \rightarrow \mathbb{R}$ is continuous and injective (one-to-one) on I , then f is either strictly increasing or strictly decreasing on I .*

3.3 Topology of $\mathbb{R}$

Definition 3.30 (ε -neighborhood). Let $z \in \mathbb{C}$ and $\varepsilon > 0$. An ε -neighborhood of z is the set

$$V_\varepsilon(z) := \{w \in \mathbb{C} : |w - z| < \varepsilon\}.$$

Definition 3.31 (Open Set in $\mathbb{C}$). A set $O \subset \mathbb{C}$ is *open* if for all $z \in O$, there exists $\varepsilon > 0$ such that $V_\varepsilon(z) \subset O$.

Remark. Intuitively, this means that if O is an open set, then every point has a small “neighborhood” that is entirely contained within O , i.e. the point has “room”, it is not right at the boundary, say.

Example 3.32. For $a < b$ in $\mathbb{R}$, the open interval $(a, b) = \{x \in \mathbb{R} : a < x < b\}$ is an open set (exercise: show this), however $[a, b]$ is not a closed set.

Lemma 3.33. Let $n \in \mathbb{N}$ and suppose $O_1, \dots, O_n \subset \mathbb{C}$ are open. Then

$$\bigcap_{i=1}^n O_i \subset \mathbb{C}$$

is open.

Proof. Put $O := \bigcap_{i=1}^n O_i$ and let $z \in O$ be arbitrary. Then for each i , there exists ε_i such that $V_{\varepsilon_i}(z) \subset O_i$. Take $\varepsilon := \min_i \{\varepsilon_i\}$. Then we have $V_\varepsilon(z) \subset O_i$ for each i , so $V_\varepsilon(z) \subset O$, so O is open. $\square$

Exercise. Find an example of a collection of open sets $\{O_n\}_{n \in \mathbb{N}}$ in $\mathbb{R}$ such that $\bigcap_{n \in \mathbb{N}} O_n$ is not open.

Lemma 3.34. Let $\{O_\alpha\}_{\alpha \in I}$ be an arbitrary collection of open sets in $\mathbb{C}$. Then

$$\bigcup_{\alpha \in I} O_\alpha \subset \mathbb{C}$$

is open.

Proof. Exercise. $\square$

Definition 3.35 (Closed Set in $\mathbb{C}$). A subset $A \subset \mathbb{C}$ is *closed* if for every convergent sequence (a_n) contained in A we have

$$\lim_{n \rightarrow \infty} a_n \in A.$$

Lemma 3.36. Let $A_1, \dots, A_n \subset \mathbb{C}$ be closed. Then

$$\bigcup_{i=1}^n A_i \subset \mathbb{C}$$

is closed.

Proof. Exercise. □

Lemma 3.37. Let $\{A_\alpha\}_{\alpha \in I}$ be an arbitrary collection of closed sets in $\mathbb{C}$. Then

$$\bigcap_{\alpha \in I} A_\alpha \subset \mathbb{C}$$

is closed.

Proof. Exercise. □

Exercise. Find an example of a collection of closed sets $\{A_n\}_{n \in \mathbb{N}}$ in $\mathbb{R}$ such that $\bigcup_{n \in \mathbb{N}} A_n$ is not closed.

Definition 3.38 (Bounded Set in $\mathbb{C}$). A subset $B \subset \mathbb{C}$ is *bounded* if there exists $R \in \mathbb{R}_{>0}$ such that

$$B \subset \{z \in \mathbb{C} : |z| \leq R\}.$$

Definition 3.39 (Compact Set in $\mathbb{C}$). A subset $K \subset \mathbb{C}$ is *compact* if

1. K is closed;
2. K is bounded.

Lemma 3.40. *Let $K_1, \dots, K_n \subset \mathbb{C}$ be compact. Then*

$$\bigcup_{i=1}^n K_i \subset \mathbb{C}$$

is compact. Moreover, the intersection of an arbitrary collection of compact subsets of $\mathbb{C}$ is compact.

Proof. Exercise. □

There is a useful characterisation of compact sets using the Bolzano-Weierstraß Theorem:

Theorem 3.41. *A subset $K \subset \mathbb{C}$ is compact if and only if for every sequence (z_n) in K , there exists a convergent subsequence (z_{n_j}) such that*

$$\lim_{j \rightarrow \infty} z_{n_j} \in K.$$

Proof. Suppose $K \subset \mathbb{C}$ is compact, and let (z_n) be an arbitrary sequence in K . Since K is bounded, (z_n) is bounded so by the Bolzano-Weierstraß Theorem in $\mathbb{C}$ (Theorem 2.43), there exists a convergent subsequence (z_{n_j}) of (z_n) such that $\lim_{j \rightarrow \infty} z_{n_j} =: z$ exists. Since K is closed, $z \in K$.

Conversely, suppose that for every sequence (z_n) in K , there exists a convergent subsequence (z_{n_j}) such that $z := \lim_{j \rightarrow \infty} z_{n_j} \in K$. We now need to argue that K is closed and bounded.

K bounded: suppose for contradiction that K is unbounded. Then we could choose a sequence (z_n) in K such that $|z_n| \rightarrow \infty$ as $n \rightarrow \infty$, which has no convergent subsequence in $\mathbb{C}$, contradicting our assumption, so K must be bounded.

K closed: let (z_n) be a convergent sequence in K such that $z_n \rightarrow z$ as $n \rightarrow \infty$. We need to show that $z \in K$. By our assumption, there exists a convergent subsequence (z_{n_j}) of (z_n) converging to some $w \in K$. However, convergent sequences have unique limits so $z = w$ and thus $z \in K$, so K is closed.

Therefore, K is compact. □

Theorem 3.42 (Continuity preserves compact sets). *Let $K \subset \mathbb{C}$ and $f : K \rightarrow \mathbb{C}$ be continuous. Then the image*

$$f(K) = \{f(z) : z \in K\} \subset \mathbb{C}$$

is compact.

Proof. Let (w_n) be a sequence in $f(K)$ with, and let (z_n) be the sequence such that for all $n \in \mathbb{N}$, $f(z_n) = w_n$. Since K is compact, by Theorem 3.41, there exists a convergent subsequence (z_{n_j}) of (z_n) such that $\lim_{j \rightarrow \infty} z_{n_j} = z$ for some $z \in K$. By the sequential characterisation of continuity, it follows that

$$\lim_{j \rightarrow \infty} w_{n_j} = \lim_{j \rightarrow \infty} f(z_{n_j}) = f(z),$$

so (w_n) has a convergent subsequence (w_{n_j}) whose limit $f(z)$ is in K , so by Theorem 3.41 again, $f(K)$ is compact. $\square$

Theorem 3.43. *Let $K \subset \mathbb{C}$ be compact and let $f : K \rightarrow \mathbb{R}$ be continuous. Then there exist $z_1, z_2 \in K$ such that*

$$f(z_1) \leq f(z) \leq f(z_2) \quad \forall z \in K.$$

In other words, f attains both its minimum and maximum on K .

Proof. By Theorem 3.42, $f(K)$ is compact, so in particular $f(K)$ is bounded, so the real numbers

$$m := \inf f(K), \quad M := \sup f(K)$$

exist. Since $f(K)$ is closed as well, we have $m, M \in f(K)$ (to see this, note that for any bounded set $S \subset \mathbb{R}$, by definition of supremum and infimum, there are convergent sequences (x_n) and (y_n) such that $x_n \rightarrow \inf S$ and $y_n \rightarrow \sup S$, so for compact sets $\inf S, \sup S$ must be in S). Therefore, there exist $z_1, z_2 \in K$ such that $f(z_1) = m$ and $f(z_2) = M$, which completes the proof. $\square$

The statement of Theorem 3.43 cannot be directly generalised to functions with codomain $\mathbb{C}$. This is because the complex numbers do not possess a natural total ordering compatible with field operations. Inequalities such as $f(z_1) \leq f(z) \leq f(z_2)$ are undefined when the outputs are complex numbers; therefore, a complex-valued function cannot attain a “maximum” or “minimum” value in the traditional sense (think about a circle in the complex plane; every point on this circle has equal magnitude).

However, the topological core of the theorem, that continuous functions map compact sets to compact sets, does apply. We can use this to prove a highly useful variation of the Extreme Value Theorem in $\mathbb{C}$ by looking at the *modulus* (magnitude) of the complex function:

Theorem 3.44 (Extreme Value Theorem in $\mathbb{C}$). *Let $K \subset \mathbb{C}$ be compact and $f : K \rightarrow \mathbb{C}$ be continuous. Then $f(K)$ is bounded in $\mathbb{C}$, and there exist $z_1, z_2 \in K$ such that*

$$|f(z_1)| \leq |f(z)| \leq |f(z_2)| \quad \forall z \in K.$$

Proof. First we note that by Theorem 3.42, $f(K)$ is compact, so in particular is bounded. Now, define $h : K \rightarrow \mathbb{R}$ by taking the modulus of our original function:

$$h(z) = |f(z)|$$

The function h can be viewed as the composition of f and the absolute value function $g : \mathbb{C} \rightarrow \mathbb{R}$ defined by $g(w) = |w|$. We are given that $f : K \rightarrow \mathbb{C}$ is continuous. Furthermore, the absolute value function $g : \mathbb{C} \rightarrow \mathbb{R}$ is continuous (this follows from the reverse triangle inequality, $||w_1| - |w_2|| \leq |w_1 - w_2|$). Since the composition of continuous functions is continuous (by Theorem 3.14), $h(z)$ is continuous on K .

By Theorem 3.43, there exist $z_1, z_2 \in K$ such that:

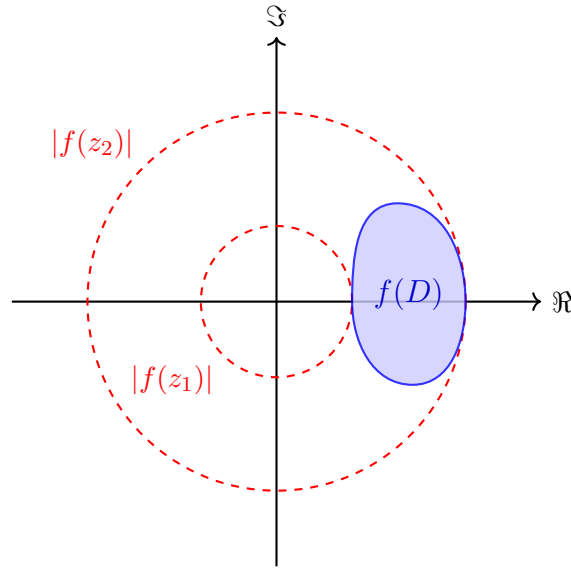
$$h(z_1) \leq h(z) \leq h(z_2) \quad \text{for all } z \in K$$

Substituting $|f(z)|$ back in for $h(z)$, we obtain the desired inequality for the extreme values of the modulus:

$$|f(z_1)| \leq |f(z)| \leq |f(z_2)| \quad \text{for all } z \in K.$$

□

Remark. Both the compactness of K and the continuity of f are essential hypotheses of this theorem.



Example 3.45. The following examples satisfy both hypotheses of the Extreme Value Theorem in $\mathbb{C}$ (Theorem 3.44): the domain $K \subset \mathbb{C}$ is compact (closed and bounded), and the function f is continuous.

1. Let $f(z) = z^2$ defined on the closed unit disk, $K = \{z \in \mathbb{C} : |z| \leq 1\}$. Because K is closed and bounded, it is compact. The polynomial f is continuous everywhere. The modulus is $|f(z)| = |z^2| = |z|^2$. The minimum modulus is 0, attained at $z_1 = 0$. The maximum modulus is 1, attained anywhere on the boundary (e.g., $z_2 = 1$ or $z_2 = i$). The image $f(K)$ is the closed unit disk itself, which is bounded.
2. Let $f(z) = e^z$ defined on the rectangle $K = \{x + iy : 0 \leq x \leq 1, 0 \leq y \leq \pi\}$. K is compact and f is continuous. The modulus is $|f(z)| = |e^{x+iy}| = |e^x e^{iy}| = e^x$. The minimum modulus is 1 (attained when $x = 0$, anywhere on the left edge), and the maximum modulus is e (attained when $x = 1$, anywhere on the right edge). The image $f(K)$ is bounded between radius 1 and e .

Example 3.46. The following examples demonstrate why both hypotheses in Theorem 3.44 are strictly required.

1. Let $D = (0, 1)$ and $f(x) = x$. The domain is not compact because it does not contain its limit points. The function is continuous, but it never attains a maximum or minimum on this open interval.
2. Let $D = [1, \infty)$ and $f(x) = x$. The domain is closed but extends infinitely. The function is continuous, but the set $f(D)$ is unbounded, meaning no maximum exists.

3. Let $D = [0, 1]$. Define $f(x) = 1/x$ for $x \in (0, 1]$, and $f(0) = 0$. The domain is compact. However, due to the discontinuity at $x = 0$, the function approaches infinity as $x \rightarrow 0$. It does not attain a maximum value.
4. Let $D = \{z \in \mathbb{C} : |z| \leq 1\}$ (the closed unit disk). Define $f(z) = 1/z$ for $z \neq 0$, and set $f(0) = 0$. The domain is compact, but the function is discontinuous at the origin. The magnitude $|f(z)| = 1/|z|$ grows arbitrarily large near 0, so $f(D)$ is unbounded and no maximum modulus exists.

3.4 Uniform Continuity

We now consider a *stronger* notion of continuity, which controls the function's behaviour *globally*, rather than *locally*, like the notion of "normal" continuity we have just been studying (left and right limits matching the function value at a point).

Definition 3.47 (Uniform Continuity). Let $D \subset \mathbb{C}$. A function $f : D \rightarrow \mathbb{C}$ is *uniformly continuous* on D if

$$\forall \varepsilon > 0, \exists \delta > 0, \forall x, y \in D, |x - y| < \delta \implies |f(x) - f(y)| < \varepsilon.$$

Remark. Normal continuity is a local property (defined *at a point* a). The δ depends on both the tolerance ε and the specific location a (i.e., we may write $\delta = \delta(\varepsilon, a)$). If a function gets infinitely steep (like $1/x$ near 0), you need smaller and smaller δ 's as you move, to stay within the same ε tolerance.

Uniform continuity guarantees that the function does not suddenly oscillate very fast when you move to a different part of the domain.

Uniform continuity is a global property (defined *on a set*). It means you can find a single "universal δ " that works for the given ε everywhere on the domain (i.e. we may write $\delta = \delta(\varepsilon)$).

Example 3.48. The function $f : (0, 1) \rightarrow \mathbb{R}$ defined by $f(x) = \frac{1}{x}$ is not uniformly continuous.

Intuitively, near 0 the graph of $f(x) = \frac{1}{x}$ becomes extremely steep: very small changes in x can cause very large changes in $\frac{1}{x}$, suggesting that no single $\delta > 0$ can work for all points in $(0, 1)$ at once.

Indeed, pick $\varepsilon := \frac{1}{2}$, and let $\delta > 0$ be arbitrary. Let $n \in \mathbb{N}$ be large enough such that $0 < \frac{1}{n} < \delta$, and pick $x = \frac{1}{n}$, $y = \frac{1}{2n}$. Then we have $|x - y| = \frac{1}{n} < \delta$, but

$$|f(x) - f(y)| = |n - 2n| = n \geq 1 > \frac{1}{2} = \varepsilon,$$

so f is not uniformly continuous on $(0, 1)$.

Again, the problem in the above example was that f diverges to ∞ at 0 and the domain $(0, 1)$ is not closed, i.e. we were allowed to take points arbitrarily close to 0, allowing the function to grow too rapidly to control.

Therefore, given our brief discussion of compact sets in the previous section, it is natural to suspect that continuity on a compact set is enough to be uniformly continuous. Indeed, we have the following theorem:

Theorem 3.49 (Heine's Theorem). *Let $K \subset \mathbb{C}$ be compact and let $f : K \rightarrow \mathbb{C}$ be continuous. Then f is uniformly continuous on K .*

Proof. Suppose for contradiction that f is not uniformly continuous on K . Let $\varepsilon_0 > 0$ be such that for any $\delta > 0$, there exist $x, y \in K$ such that $|x - y| < \delta$ and $|f(x) - f(y)| \geq \varepsilon_0$.

In particular, for $n \in \mathbb{N}$ we let $\delta_n := \frac{1}{n}$, and for $n \in \mathbb{N}$ we inductively choose $z_n, w_n \in K$ such that $|z_n - w_n| < \delta_n$ and $|f(z_n) - f(w_n)| \geq \varepsilon_0$ by definition.

Since K is compact, (z_n) has a convergent subsequence (z_{n_j}) such that $z := \lim_{j \rightarrow \infty} z_{n_j} \in K$. Furthermore, we have

$$|z_{n_j} - w_{n_j}| < \frac{1}{n_j} \leq \frac{1}{j} \rightarrow 0$$

as $j \rightarrow \infty$, so $w_{n_j} \rightarrow z$ as $j \rightarrow \infty$.

Now, by the sequential characterisation of continuity (Theorem 3.17), since f is continuous we get

$$\lim_{j \rightarrow \infty} f(z_{n_j}) = f(z) \quad \text{and} \quad \lim_{j \rightarrow \infty} f(w_{n_j}) = f(z),$$

so in particular

$$\lim_{j \rightarrow \infty} |f(z_{n_j}) - f(w_{n_j})| = 0.$$

Therefore, for sufficiently large j , we have

$$|f(z_{n_j}) - f(w_{n_j})| < \frac{\varepsilon_0}{2} < \varepsilon_0,$$

which contradicts the construction of (z_n) and (w_n) , which satisfy $|f(z_n) - f(w_n)| \geq \varepsilon_0$ for all $n \in \mathbb{N}$. $\square$

Definition 3.50 (Lipschitz Continuity). Let $D \subset \mathbb{R}$. A function $f : D \rightarrow \mathbb{R}$ is *Lipschitz continuous* if there exists a real constant $L \geq 0$ (the *Lipschitz constant*) such that for all $x, y \in D$,

$$|f(x) - f(y)| \leq L|x - y|.$$

Lipschitz continuity places a strict limit on how “steep” the function can get on D . Indeed, if $x < y$, the Lipschitz condition above rearranges to

$$\left| \frac{f(x) - f(y)}{x - y} \right| \leq L,$$

which places a uniform bound L on the average slope of the function on any interval (x, y) . If a function is differentiable, bounding its derivative ($|f'(x)| \leq L$) guarantees it is Lipschitz (exercise: once you have read the derivative chapter, prove this).

Remark. Lipschitz $\implies$ Uniformly Continuous $\implies$ Continuous, however *none* of the reverse implications hold (exercise: find counterexamples).

Example 3.51. The function $f : [0, 1] \rightarrow \mathbb{R}$ defined by $f(x) = \sqrt{x}$ is continuous. Because $[0, 1]$ is compact, it is uniformly continuous by Heine’s Theorem. However, it is *not* Lipschitz continuous because its derivative $\frac{1}{2\sqrt{x}}$ approaches infinity near $x = 0$, meaning its slope cannot be globally bounded by a single constant L . To prove this rigorously (since we have not defined derivative yet), we argue as follows.

The idea is to fix one point at 0 and let (x_n) be a sequence converging to 0, to get the slope diverging as $n \rightarrow \infty$ (visually, imagine the tangent line at a point $x_0 > 0$ getting steeper and steeper as $x_0 \rightarrow 0$).

Suppose for contradiction there exists $L \geq 0$ such that for all $x, y \in [0, 1]$,

$$|\sqrt{x} - \sqrt{y}| \leq L|x - y|.$$

Let $y = 0$, and for $n \in \mathbb{N}$ define $x_n := \frac{1}{n^2}$. Given $n \in \mathbb{N}$, we have by the Lipschitz condition,

$$\left| \sqrt{\frac{1}{n^2}} - 0 \right| = \frac{1}{n} \leq L \left| \frac{1}{n^2} - 0 \right|,$$

so rearranging we get $n \leq L$ for all $n \in \mathbb{N}$, which is a contradiction (e.g. pick $n = \lceil L \rceil + 1$ since n was arbitrary). Therefore, f is not Lipschitz on $[0, 1]$.

Chapter 4

The Derivative

The notion of the derivative is an intuitive one, and is no doubt familiar to the reader. Nonetheless, we first cover some motivation for this concept.

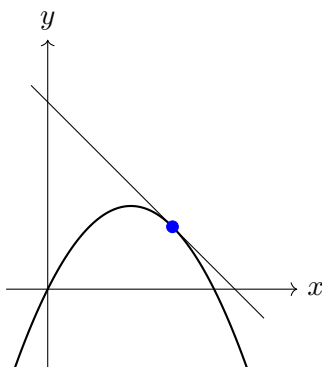
One useful way of interpreting the derivative is that it is the slope of the tangent line of a given differentiable function at a point. Alternatively, it is the best linear approximation of the function at that point, and provides some useful information local information about the function at that point (we will see this later with the *local comparison* theorems).

Example 4.1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = 1 - (x - 1)^2$. By how we define the derivative, $f'(x) = -2(x - 1)$, so $f'(1.5) = -1$. This is the *slope* of the tangent line to the graph of $y = f(x)$ at $x = 1.5$. One can think of this as our best linear guess of how the function is behaving *locally* at $x = 1.5$ (we will extend this interpretation to higher order derivatives with polynomial approximations when we study Taylor's Theorem).

Indeed, the equation of the tangent line to $y = f(x)$ at $x = 1.5$ is

$$y - 0.75 = -(x - 1.5),$$

and we provide an illustration of this below.



The other important way to interpret the derivative is that, if we assign some physical meaning to the output of our function $f(x)$, say displacement,

then the derivative measures the instantaneous rate of change of displacement, otherwise known as instantaneous velocity.

Example 4.2. Let P be a particle travelling in 1 dimension, with displacement given by $x(t) = e^{-t} \sin t$ for $t \geq 0$. Let $\Delta t > 0$ be a short period of time, during which the particle moves. Then the average velocity of the particle in this time is given by

$$\text{average velocity} = \frac{x(t + \Delta t) - x(t)}{\Delta t}, \quad \text{for } t > 0.$$

The natural way to get the instantaneous velocity from this is to take the limit of the expression as $\Delta t \rightarrow 0$, for some fixed t_0 . This will be how we define the *derivative*:

$$\text{velocity at } t_0 = \frac{d}{dx} x(t_0) = \lim_{\Delta t \rightarrow 0} \frac{x(t_0 + \Delta t) - x(t_0)}{\Delta t} = e^{-t_0} (\cos t_0 - \sin t_0).$$

We will now define what it means for a function to be *differentiable* formally.

Definition 4.3. Let I be an open interval and let $f : I \rightarrow \mathbb{R}$ be a function. We say that f is *differentiable* at $a \in I$ if the limit

$$\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

exists. To denote the *derivative* of f at a , we write $f'(a)$.

We say that $f : I \rightarrow \mathbb{R}$ is *differentiable everywhere*, or just *differentiable*, if f is differentiable at every point $a \in I$.

Remark. Notice that this is a two-sided limit, so at the very least f must be well-defined on an open interval containing a , and the slope of f cannot change as we approach a from the left and from the right.

Exercise. Show that it is equivalent to write

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}.$$

4.1 Properties of Differentiable Functions

4.1.1 Basic Results

In fact, being continuous at a is a necessary condition for differentiability (but not sufficient; exercise, give a counterexample).

Theorem 4.4. *If $f : I \rightarrow \mathbb{R}$ is differentiable at $a \in I$ then f is continuous at a .*

Proof. For $x \in I \setminus \{a\}$ we have

$$f(x) - f(a) = \frac{f(x) - f(a)}{x - a}(x - a),$$

so taking the limit of this expression as $x \rightarrow a$ we obtain, using the differentiability of f at a and the Algebra of Limits for functions (Theorem 3.8):

$$\lim_{x \rightarrow a} (f(x) - f(a)) = f'(a) \cdot 0 = 0.$$

Thus f is continuous at a since $\lim_{x \rightarrow a} f(x) = f(a)$. □

Exercise. For which values of $a \in \mathbb{R}$ is the function

$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f(x) = \begin{cases} x^2, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q} \end{cases}$$

differentiable?

We also have an analogue of the Algebra of Limits for differentiable functions:

Theorem 4.5. *Let $f : I \rightarrow \mathbb{R}$ and $g : I \rightarrow \mathbb{R}$ be differentiable at $a \in I$. Then*

1. *The function $f + g : I \rightarrow \mathbb{R}$ is differentiable at $a \in I$ and*

$$(f + g)'(a) = f'(a) + g'(a).$$

2. *The function $fg : I \rightarrow \mathbb{R}$ is differentiable at $a \in I$ and*

$$(fg)'(a) = f'(a)g(a) + f(a)g'(a).$$

3. *If $f(a) \neq 0$ then there exists an open interval $J \subset I$ with $a \in J$ such that the function $\frac{1}{f} : J \rightarrow \mathbb{R}$ is well-defined and differentiable at $a \in I$, with*

$$\left(\frac{1}{f}\right)'(a) = -\frac{f'(a)}{(f(a))^2}.$$

Proof. 1. Follows directly from the Algebra of Limits for functions.

2. For $x \in I \setminus \{a\}$ we have

$$\begin{aligned} \frac{f(x)g(x) - f(a)g(a)}{x - a} &= \frac{f(x)g(x) - f(a)g(x) + f(a)g(x) - f(a)g(a)}{x - a} \\ &= \frac{f(x) - f(a)}{x - a}g(x) + f(a)\frac{g(x) - g(a)}{x - a}. \end{aligned}$$

Using the Algebra of Limits for functions, taking the limit of the above expression as $x \rightarrow a$ yields the result by differentiability of f, g at $a \in I$.

3. By continuity of f at $a \in I$ (by Theorem 4.4), there exists an open interval $J \subset I$ with $a \in J$ such that $f(x) \neq 0$ for all $x \in J$. Therefore, the function $\frac{1}{f} : J \rightarrow \mathbb{R}$ defined by $x \mapsto \frac{1}{f(x)}$ is well-defined. Furthermore, for $x \in J \setminus \{a\}$ we have

$$\frac{\frac{1}{f(x)} - \frac{1}{f(a)}}{x - a} = \frac{\frac{f(a) - f(x)}{f(a)f(x)}}{x - a} = -\frac{1}{f(a)f(x)} \frac{f(x) - f(a)}{x - a}.$$

Using the Algebra of Limits for functions, taking the limit of the above expression as $x \rightarrow a$ yields the result by differentiability of f at $a \in J$. $\square$

Remark. Combining items 2 and 3 in the above theorem, provided that $g(a) \neq 0$ we get the usual quotient rule for the differentiable function $\frac{f}{g}$, on a suitable open interval $J \subset I$ with $a \in J$ (exercise: write out this formula).

Example 4.6. Constant functions are differentiable, with derivative zero. This is trivial by the fact that for any $a, x \in \mathbb{R}$, with $x \neq a$, $f(a) - f(x) = 0$.

The identity function on $\mathbb{R}$ is differentiable with derivative 1 for all $x \in \mathbb{R}$. Indeed, we have for a fixed $a \in \mathbb{R}$, and $x \neq a$,

$$\frac{f(x) - f(a)}{x - a} = \frac{x - a}{x - a} = 1.$$

We deduce by induction using items 1 and 2 above that polynomial functions $a_0 + a_1x + \cdots + a_nx^n$ are differentiable on $\mathbb{R}$. Therefore, rational functions $P/Q : J \rightarrow \mathbb{R}$ for some polynomials P, Q where Q does not vanish on J , are differentiable too.

4.1.2 The Chain Rule

Just like with continuity, compositions of differentiable functions are differentiable:

Theorem 4.7 (Chain Rule). *Let $f : I \rightarrow \mathbb{R}$ be differentiable at $a \in I$. Let $f(I) \subset J$ for some open interval J , and let $g : J \rightarrow \mathbb{R}$ be differentiable at $f(a)$. Then $g \circ f : I \rightarrow \mathbb{R}$ is differentiable at $a \in I$ and*

$$(g \circ f)'(a) = f'(a)g'(f(a)).$$

Proof. Define $\phi : I \rightarrow \mathbb{R}$ by

$$\phi(x) = \begin{cases} \frac{g(f(x)) - g(f(a))}{f(x) - f(a)}, & f(x) \neq f(a), \\ g'(f(a)), & f(x) = f(a). \end{cases}$$

We claim that ϕ is continuous at a . Let $\varepsilon > 0$ be arbitrary. Since g is differentiable at $f(a)$, there exists $\delta > 0$ such that for all $y \in J$ with $0 < |y - f(a)| < \delta$, we have

$$\left| \frac{g(y) - g(f(a))}{y - f(a)} - g'(f(a)) \right| < \varepsilon.$$

Since f is continuous at a there exists $\delta' > 0$ such that for all $x \in I$ with $|x - a| < \delta'$, we have

$$|f(x) - f(a)| < \delta.$$

Therefore, for $x \in I \cap (a - \delta', a + \delta')$, if $f(x) \neq f(a)$ we have

$$|\phi(x) - \phi(a)| = \left| \frac{g(f(x)) - g(f(a))}{f(x) - f(a)} - g'(f(a)) \right| < \varepsilon,$$

and, if $x \in I \cap (a - \delta', a + \delta')$ with $f(x) = f(a)$, then $\phi(x) = g'(f(a))$, so $|\phi(x) - \phi(a)| = 0 < \varepsilon$ in this case too. Therefore, ϕ is continuous at a .

To conclude, we observe that for $x \neq a$, if $f(x) \neq f(a)$ then

$$\begin{aligned} \frac{g(f(x)) - g(f(a))}{x - a} &= \frac{g(f(x)) - g(f(a))}{x - a} \cdot \frac{f(x) - f(a)}{f(x) - f(a)} \\ &= \frac{g(f(x)) - g(f(a))}{f(x) - f(a)} \cdot \frac{f(x) - f(a)}{x - a} \\ &= \phi(x) \frac{f(x) - f(a)}{x - a}. \end{aligned}$$

Furthermore, the above expression holds in the case $f(x) = f(a)$, since both sides become zero. Therefore, the above holds for all $x \in I \setminus \{a\}$. Taking the limit as $x \rightarrow a$, and using the continuity of ϕ at a with $\phi(a) = g'(f(a))$, we get the result. $\square$

Exercise. Show that if f, g are differentiable everywhere and are inverses of each other, then f' and g' are both non-zero, and $f' = \frac{1}{g'(f)}$.

4.1.3 The Local Comparison Theorem

As mentioned at the start of this chapter, the derivative is a useful tool to gather information about the behaviour of a function locally.

Lemma 4.8 (Local Comparison Theorem). *Let $f : I \rightarrow \mathbb{R}$ be a function, and suppose f is differentiable at $c \in I$. Then*

1. *If $f'(c) > 0$ then there exists $\delta > 0$ such that for all $x \in (c - \delta, c)$ and $y \in (c, c + \delta)$, we have*

$$f(x) < f(c) < f(y).$$

2. *If $f'(c) < 0$ then there exists $\delta > 0$ such that for all $x \in (c - \delta, c)$ and $y \in (c, c + \delta)$, we have*

$$f(y) < f(c) < f(x).$$

Proof. We will prove the case $f'(c) > 0$; the proof for case 2 is identical. Since

$$\lim_{t \rightarrow c} \frac{f(t) - f(c)}{t - c} = f'(c) > 0,$$

by definition of the limit there exists $\delta > 0$ such that for all $t \in (c - \delta, c + \delta) \cap I$, we have

$$\frac{f(t) - f(c)}{t - c} > 0.$$

If $t \in (c - \delta, c)$ then $t < c$ so $f(t) - f(c) < 0$, i.e. $f(t) < f(c)$. If $t \in (c, c + \delta)$ then $t > c$ so $f(t) - f(c) > 0$, i.e. $f(t) > f(c)$. $\square$

Remark. This does not mean that the function is monotonic increasing on $(c - \delta, c + \delta)$ (we provide an explicit counterexample below), it just means that *locally*, the function sits below the line $f(c)$ on $(c - \delta, c)$ and above the line $f(c)$ on $(c, c + \delta)$.

Example 4.9. Fix $1 < \alpha < 2$ and define $f : \mathbb{R} \rightarrow \mathbb{R}$ by

$$f(x) = \begin{cases} x + |x|^\alpha \sin\left(\frac{1}{x}\right), & x \neq 0, \\ 0, & x = 0. \end{cases}$$

We first observe that f is differentiable at 0 with a positive derivative; for $x \neq 0$ we have

$$\frac{f(x) - f(0)}{x - 0} = \frac{x + |x|^\alpha \sin\left(\frac{1}{x}\right)}{x} = 1 + |x|^{\alpha-1} \sin\left(\frac{1}{x}\right).$$

Taking the limit as $x \rightarrow 0$ we have that $f'(0) = 1 > 0$.

However, we claim that for any $\varepsilon > 0$, there exist $x, y \in (0, \varepsilon)$ such that $f'(x) < 0 < f'(y)$. Indeed, let $\varepsilon > 0$ be arbitrary. For $x > 0$, by the chain and product rules we have

$$f'(x) = 1 + \alpha x^{\alpha-1} \sin\left(\frac{1}{x}\right) - x^{\alpha-2} \cos\left(\frac{1}{x}\right).$$

To show this oscillation, we evaluate $f'(x)$ at points where the sine term vanishes. Since $\alpha - 2 < 0$, we have $2 - \alpha > 0$, and $(2n\pi)^{2-\alpha} \rightarrow \infty$ as $n \rightarrow \infty$ by Theorem 2.17. We can therefore choose $n \in \mathbb{N}$ large enough such that $\frac{1}{2n\pi} < \varepsilon$ and $(2n\pi)^{2-\alpha} > 1$.

Picking $x = \frac{1}{2n\pi} \in (0, \varepsilon)$, we have $\sin\left(\frac{1}{x}\right) = 0$ and $\cos\left(\frac{1}{x}\right) = 1$, yielding

$$f'(x) = 1 + 0 - (2n\pi)^{2-\alpha} = 1 - (2n\pi)^{2-\alpha} < 0.$$

Similarly, picking $y = \frac{1}{(2n+1)\pi} \in (0, \varepsilon)$, we have $\sin\left(\frac{1}{y}\right) = 0$ and $\cos\left(\frac{1}{y}\right) = -1$, yielding

$$f'(y) = 1 + 0 - (-(2n+1)\pi)^{2-\alpha} = 1 + ((2n+1)\pi)^{2-\alpha} > 0.$$

Now, by the Local Comparison Theorem (Lemma 4.8), it follows that f is not monotone on any interval of the form $(0, \varepsilon)$ for $\varepsilon > 0$, concluding our proof that this is a counterexample to the claim that $f'(c) > 0$ at a point implies monotonicity.

4.1.4 Increasing and Decreasing Functions

We can use the derivative to show monotonicity of differentiable functions:

Theorem 4.10. *Let $f : I \rightarrow \mathbb{R}$ be differentiable.*

1. *If $f'(x) > 0$ for all $x \in I$ then f is strictly increasing.*
2. *If $f'(x) < 0$ for all $x \in I$ then f is strictly decreasing.*

Proof. First we prove the case where $f'(x) > 0$ for all $x \in I$. Suppose for contradiction there exist $s < t$ in I such that $f(s) \geq f(t)$. We consider the set $S := \{x \in I : x > s, f(x) \leq f(t)\}$. Since $t \in S$, S is non-empty and $S \subset I$ so S is bounded, so by the completeness of $\mathbb{R}$, we may define $\alpha := \inf S$.

We claim that $\alpha > s$. Indeed, by local comparison at s , $f'(s) > 0$ implies that there exists $\delta > 0$ such that for all $x \in (s, s + \delta)$, $f(s) < f(x)$. In particular, if $x \in (s, s + \delta)$ then $x \notin S$, so $\alpha \geq s + \delta > s$, as required. Furthermore, since f is differentiable at α , it is continuous there, so $f(\alpha) \leq$

$f(t)$ (to see this, let x_n be a sequence in S converging to α , and use the Order Limit Theorem for sequences, Theorem 2.8, and sequential continuity, to conclude that the limit $f(\alpha)$ satisfies the required inequality). Therefore, $\alpha \in S$.

Finally, to get the desired contradiction we apply the Local Comparison Theorem (Lemma 4.8) at α ; since $f'(\alpha) > 0$, applying local comparison implies that there exists $\rho > 0$ such that for all $x \in (\alpha - \rho, \alpha)$, $f(x) < f(\alpha) \leq f(t)$, implying that for any $x \in (\alpha - \rho, \alpha)$ with $x > s$, we have $x \in S$, but $x < \alpha$, which contradicts the fact that α is a lower bound for S .

To prove the case where $f'(x) < 0$ for all $x \in I$, we note that by Theorem 4.5 $-f$ is differentiable, and $-f'(x) > 0$ for all $x \in I$, so applying the previous item, $-f$ is strictly increasing on I , so $-f(s) < -f(t)$ for all $s < t$ in I , i.e. $f(s) > f(t)$ for all $s < t$ in I , which proves that f is strictly decreasing. $\square$

Exercise. Does the converse of Theorem 4.10 hold? In other words, if f is strictly increasing (or decreasing), does it necessarily hold that $f'(x) > 0$ for all $x \in I$ (or $f'(x) < 0$ for all $x \in I$)? Provide a proof or counterexample.

Exercise. Show that if $f : I \rightarrow \mathbb{R}$ is continuously differentiable (that is, f is differentiable and f' is continuous) and $f'(a) \neq 0$ for some $a \in I$, then there exists $\delta > 0$ such that f is injective on $(a - \delta, a + \delta) \subset I$.

4.2 Critical Points

The derivative can also be used to locate *candidates* for local and global extrema of a differentiable function.

Definition 4.11 (Local Extrema). Let $f : I \rightarrow \mathbb{R}$ be a function.

1. We say that f attains a *local maximum* at $c \in I$ if there exists $\delta > 0$ such that for all $x \in (c - \delta, c + \delta) \subset I$, $f(x) \leq f(c)$.
2. We say that f attains a *local minimum* at $c \in I$ if there exists $\delta > 0$ such that for all $x \in (c - \delta, c + \delta) \subset I$, $f(x) \geq f(c)$.

One makes the obvious modifications to the above definitions for the notions of *global* extrema.

Definition 4.12 (Critical Point). Let $f : I \rightarrow \mathbb{R}$ be differentiable. We say that $c \in I$ is a *critical point* of f if $f'(c) = 0$.

Remark. Critical points do not have to correspond to local extrema.

Theorem 4.13 (Fermat's Theorem). *Let $I \subset \mathbb{R}$ be open, and let $f : I \rightarrow \mathbb{R}$ be a function. If f attains a local maximum or a local minimum at $c \in I$, and f is differentiable at c , then $f'(c) = 0$.*

Proof. We prove the case where f attains a local maximum at $c \in I$, the proof for the local minimum case is identical. Let $c \in I$ be a local maximum of f , and suppose f is differentiable at c . Let $\delta > 0$ be such that for all $x \in (c - \delta, c + \delta) \subset I$, $f(x) \leq f(c)$. Fix $s \in (c - \delta, c)$ and $t \in (c, c + \delta)$. Then we have

$$\frac{f(t) - f(c)}{t - c} \leq 0 \leq \frac{f(c) - f(s)}{c - s}.$$

Taking the limit as $t \rightarrow c^+$, $s \rightarrow c^-$, we have that, since f is differentiable at c , and by the Order Limit Theorem for functions,

$$\lim_{t \rightarrow c^+} \frac{f(t) - f(c)}{t - c} \leq 0 \leq \lim_{s \rightarrow c^-} \frac{f(c) - f(s)}{c - s},$$

so

$$f'(c) \leq 0 \leq f'(c),$$

i.e. $f'(c) = 0$, as required. $\square$

After locating critical points one usually needs more information to decide whether they are maxima, minima, or neither. Furthermore, the classification and existence of global extrema of a function is an even harder problem, requiring the function to satisfy stronger properties.

Remark. If I is a closed interval, the maximum or minimum must occur in the interior of the interval. For example, $f : [0, 1] \rightarrow \mathbb{R}$ defined by $f(x) = x$ has a (global) minimum and a (global) maximum at $x = 0$ and $x = 1$ respectively, but there does not exist $c \in (0, 1)$ such that $f'(c) = 0$.

Exercise. Find a counterexample to the converse of Fermat's Theorem.

4.2.1 A Historical Application: The Law of Light Refraction

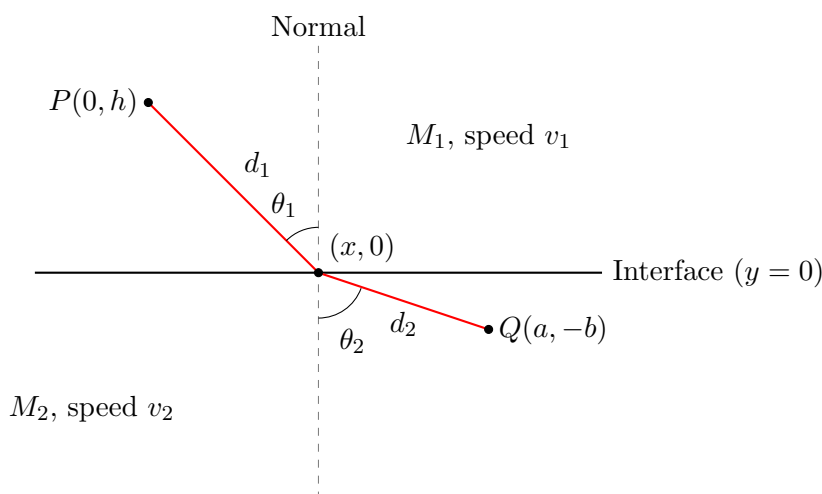
Theorem 4.14 (Fermat's Principle). *The path light takes to travel between two points P and Q is the path of least time.*

Remark. Fermat realised that optical problems can be framed purely in terms of mathematical optimisation. By framing the physics as a problem of finding a global minimum, we can utilise tools from real analysis (specifically, the derivative and critical points) to derive geometric laws.

Theorem 4.15 (Snell's Law). *Let M_1 and M_2 be two media separated by a straight interface. Assume the speed of light is v_1 in M_1 and v_2 in M_2 . If a light ray travels from $P \in M_1$ to $Q \in M_2$ and crosses the interface making angles θ_1 and θ_2 with the normal, then:*

$$\frac{\sin \theta_1}{\sin \theta_2} = \frac{v_1}{v_2}$$

Remark. Fermat realised that Snell's Law can be derived entirely from an optimisation problem. Light does not simply travel in a straight line from P to Q ; it takes an optimal intermediate path to minimise the distance travelled in the slower medium, allowing it to reach the faster medium quicker. This principle allows *any* optical problem to be translated into a mathematical optimisation problem, solvable using the tools of real analysis.



Derivation of Snell's Law. We will show that Fermat's Principle implies Snell's Law. Choose coordinates such that the interface lies on the line $y = 0$. Let the starting point be $P = (0, h) \in M_1$ and the destination be $Q = (a, -b) \in M_2$, with $a, b, h > 0$. Let the light cross the interface at some variable point $(x, 0)$.

The distances travelled in each medium are:

$$d_1(x) = \sqrt{x^2 + h^2}, \quad d_2(x) = \sqrt{(a - x)^2 + b^2}$$

The total travel time $T(x)$ is the sum of the times spent in each medium ($t = d/v$):

$$T(x) = \frac{\sqrt{x^2 + h^2}}{v_1} + \frac{\sqrt{(a - x)^2 + b^2}}{v_2}$$

By Fermat's Principle, light takes the path of least time, so we must find the global minimum of $T(x)$. We begin by differentiating $T(x)$ with respect to x to find the critical points:

$$T'(x) = \frac{x}{v_1 \sqrt{x^2 + h^2}} - \frac{a - x}{v_2 \sqrt{(a - x)^2 + b^2}}$$

Looking at the geometry of the diagram, we can express the trigonometric ratios of the angles θ_1 and θ_2 in terms of our coordinate distances:

$$\sin \theta_1 = \frac{x}{\sqrt{x^2 + h^2}} \quad \text{and} \quad \sin \theta_2 = \frac{a - x}{\sqrt{(a - x)^2 + b^2}}$$

Substituting these into our derivative yields:

$$T'(x) = \frac{\sin \theta_1}{v_1} - \frac{\sin \theta_2}{v_2}$$

Setting $T'(x) = 0$ to find the optimal path gives:

$$\frac{\sin \theta_1}{v_1} = \frac{\sin \theta_2}{v_2} \implies \frac{\sin \theta_1}{\sin \theta_2} = \frac{v_1}{v_2}$$

Finally, we must confirm this critical point is a global minimum. As $x \rightarrow \pm\infty$, the distance $d_1(x)$ grows without bound, meaning $T(x) \rightarrow \infty$. Since $T(x)$ is continuous and differentiable, and blows up at infinity, the single critical point we found must correspond to the global minimum (see below). $\square$

Exercise. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable with a unique critical point $c \in \mathbb{R}$. Show that if $f(x) \rightarrow \infty$ as $x \rightarrow \pm\infty$, then c is a global minimum.

4.2.2 The Brachistochrone Problem

The Brachistochrone problem asks for the curve of fastest descent: Given two points A and B in a vertical plane (where A is higher than B), what is the shape of the wire along which a frictionless bead will slide from A to B in the shortest amount of time under the influence of gravity?

Remark. Johann Bernoulli solved this by noticing an analogy between this mechanical problem and Fermat's Principle in optics. Both problems are fundamentally searching for the "curve of least time."

In mechanics, the conservation of energy dictates the velocity of the falling object at any given height y . Assuming it starts from rest, kinetic energy equals the loss in potential energy, yielding a speed that depends strictly on vertical height:

$$v(y) = \sqrt{2gy}$$

Bernoulli imagined the physical space as an “aquarium” made of discrete horizontal layers. In each layer, the speed is constant, dictated entirely by the conservation of energy equation for that specific height.

By Fermat’s Principle, if the speed in every region is known, the path of least time is completely determined by Snell’s Law at every interface. As the thickness of these horizontal layers approaches zero in the limit, the discrete path dictated by Snell’s Law converges exactly to the continuous Brachistochrone curve.

Derivation of the Brachistochrone Curve. Let the particle start from rest at the origin $(0, 0)$. Let the y -axis point downwards (representing depth) and the x -axis point horizontally.

By the conservation of energy, the kinetic energy gained equals the potential energy lost. Therefore, the velocity v at any depth y is given by:

$$\frac{1}{2}mv^2 = mgy \implies v = \sqrt{2gy}$$

Bernoulli modelled the continuous space as an infinite sequence of infinitesimally thin horizontal optical media. The normal to these horizontal interfaces is the vertical axis. Let θ be the angle between the curve’s tangent and the vertical normal.

From the geometry of the curve $y(x)$, the differential arc length is $ds = \sqrt{dx^2 + dy^2}$. The sine of the angle θ with the vertical is:

$$\sin \theta = \frac{dx}{ds} = \frac{dx}{\sqrt{dx^2 + dy^2}} = \frac{1}{\sqrt{1 + \left(\frac{dy}{dx}\right)^2}} = \frac{1}{\sqrt{1 + (y')^2}}$$

Applying Snell’s Law, the ratio of the sine of the angle to the velocity must be constant along the entire path:

$$\frac{\sin \theta}{v} = C$$

Substituting our expressions for v and $\sin \theta$:

$$\frac{1}{\sqrt{2gy}\sqrt{1 + (y')^2}} = C$$

Squaring both sides and rearranging isolates the differential equation governing the curve:

$$\frac{1}{2gy(1 + (y')^2)} = C^2 \implies y(1 + (y')^2) = \frac{1}{2gC^2}$$

To simplify, let $2a = \frac{1}{2gC^2}$, where a is a positive constant. The equation becomes:

$$y(1 + (y')^2) = 2a$$

Solving for the derivative y' :

$$(y')^2 = \frac{2a - y}{y} \implies \frac{dy}{dx} = \sqrt{\frac{2a - y}{y}}$$

Separating the variables to integrate:

$$dx = \sqrt{\frac{y}{2a - y}} dy$$

To solve this integral, we introduce a parameter t by substituting $y = a(1 - \cos t)$. This substitution is motivated by the domain of y , which must lie between 0 and $2a$. Differentiating gives $dy = a \sin t \, dt$. Substituting these into the integral yields:

$$dx = \sqrt{\frac{a(1 - \cos t)}{a(1 + \cos t)}} (a \sin t) \, dt$$

Using the half-angle trigonometric identities, the term inside the square root simplifies to $\tan^2(t/2)$, but we can also multiply numerator and denominator by $(1 - \cos t)$:

$$dx = \sqrt{\frac{(1 - \cos t)^2}{1 - \cos^2 t}} (a \sin t) \, dt = \frac{1 - \cos t}{\sin t} (a \sin t) \, dt$$

$$dx = a(1 - \cos t) \, dt$$

Integrating this with respect to t gives x :

$$x = \int a(1 - \cos t) \, dt = a(t - \sin t) + K$$

Since the curve starts at the origin $(0, 0)$, when $t = 0$, both $x = 0$ and $y = 0$. This implies the integration constant $K = 0$.

Thus, the path of least time is described by the parametric equations:

$$x(t) = a(t - \sin t)$$

$$y(t) = a(1 - \cos t)$$

These are the exact parametric equations of a cycloid generated by a circle of radius a rolling along the horizontal axis (we will learn more about parametric curves in the “Curves in the Plane” chapter). $\square$

Remark. We took a slight detour into an application of the analysis tools we have learnt (albeit in a slightly less rigorous manner) to demonstrate the practical applications of the results we are proving with care. Note also that the *indefinite integral* can be seen as simply the reverse of differentiation, which we have already defined rigorously, so the above derivation is not too far from the rigour of analysis. We will define *definite integration* rigorously in a later chapter.

4.3 The Mean Value Theorem

The main result we will be proving in this section is the *Mean Value Theorem*, a very important result in analysis. First, we need a result regarding the existence of critical points, Rolle's Theorem. This handles the special case where the function values at the endpoints are equal.

Theorem 4.16 (Rolle's Theorem). *Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . If $f(a) = f(b)$ then there exists $c \in (a, b)$ such that $f'(c) = 0$.*

Proof. By the Extreme Value Theorem (Theorem 3.24), there exists $c \in [a, b]$ such that

$$f(c) = \max_{x \in [a, b]} f(x).$$

If $c \in (a, b)$ then we are done by Fermat's Theorem (Theorem 4.13) since $f'(c) = 0$.

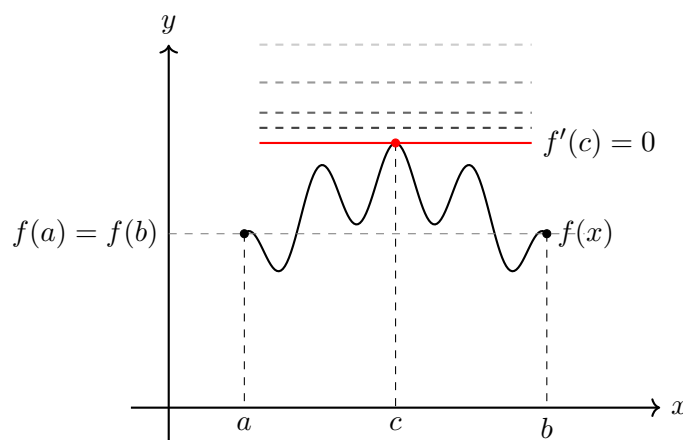
If $c = a$ or $c = b$, then since $f(a) = f(b)$, b is also the global maximum of f . Applying the Extreme Value Theorem again, there exists $d \in [a, b]$ such that

$$f(d) = \min_{x \in [a, b]} f(x).$$

If $d \in (a, b)$, then again we are done by Fermat's Theorem since $f'(d) = 0$.

If $d = a$ or $d = b$, then $\min_{x \in [a, b]} f(x) = \max_{x \in [a, b]} f(x)$, so f is constant on $[a, b]$, in which case $f'(c) = 0$ for every $c \in (a, b)$. This finishes the proof of the theorem. $\square$

Remark. The picture to keep in mind for this theorem is illustrated below. The idea is that if $f(a) = f(b)$, then as you slide a horizontal line down from above (and up from below), it will touch the graph for the first time at some interior point, producing a horizontal tangent.



There is another common scenario that forces a critical point: the derivative being positive at one point and negative at another. In this case, the picture to keep in mind is that of a dome (or bowl depending on the signs).

Theorem 4.17. *Let $f : (a, b) \rightarrow \mathbb{R}$ be differentiable, and suppose there exist $x < y$ in (a, b) such that $f'(x) > 0 > f'(y)$. Then there exists $c \in (x, y)$ such that $f'(c) = 0$.*

Proof. By the Local Comparison Theorem at x (Lemma 4.8), since $f'(x) > 0$ there exists $\delta > 0$ such that for all $t \in (x, x + \delta) \subset (x, y)$, $f(t) < f(x)$. Similarly, by the Local Comparison Theorem at y , since $f'(y) < 0$ there exists $\delta' > 0$ such that for all $t \in (y - \delta', y) \subset (x, y)$, $f(t) > f(y)$. Therefore, f does not attain its global maximum on $[x, y]$ at the endpoints, but since f is continuous on $[x, y]$, by the Extreme Value Theorem there exists $c \in (x, y)$ such that $f(c) = \max_{t \in [x, y]} f(t)$. By Fermat's Theorem (Theorem 4.13), $f'(c) = 0$. $\square$

Theorem 4.18 (Mean Value Theorem). *Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b) . Then there exists $\xi \in (a, b)$ such that*

$$f'(\xi) = \frac{f(b) - f(a)}{b - a}.$$

Proof. We prove this by reducing it to Rolle's Theorem. Define $\ell : [a, b] \rightarrow \mathbb{R}$ by

$$\ell(x) = f(a) + \frac{f(b) - f(a)}{b - a}(x - a).$$

We know that ℓ is continuous on $[a, b]$ and differentiable on (a, b) .

Also, define $h : [a, b] \rightarrow \mathbb{R}$ by $h(x) = f(x) - \ell(x)$. Explicitly:

$$h(x) = f(x) - \left(f(a) + \frac{f(b) - f(a)}{b - a}(x - a) \right)$$

Since $f(x)$ is continuous on $[a, b]$ and differentiable on (a, b) , and ℓ is continuous on $[a, b]$ and differentiable on (a, b) , h is also continuous on $[a, b]$ and differentiable on (a, b) .

Evaluating $h(x)$ at the endpoints, we have

$$h(a) = f(a) - \left(f(a) + \frac{f(b) - f(a)}{b - a}(a - a) \right) = f(a) - f(a) = 0,$$

and

$$h(b) = f(b) - \left(f(a) + \frac{f(b) - f(a)}{b - a}(b - a) \right) = f(b) - (f(a) + f(b) - f(a)) = 0.$$

Since $h(a) = h(b) = 0$, $h(x)$ satisfies all the hypotheses of Rolle's Theorem. Therefore, there exists some $\xi \in (a, b)$ such that $h'(\xi) = 0$.

Differentiating $h(x)$ gives:

$$h'(x) = f'(x) - \frac{f(b) - f(a)}{b - a}.$$

Setting $h'(\xi) = 0$:

$$0 = f'(\xi) - \frac{f(b) - f(a)}{b - a} \implies f'(\xi) = \frac{f(b) - f(a)}{b - a},$$

which completes the proof. $\square$

It is worth noting that Rolle's Theorem is simply a special case of the Mean Value Theorem where the endpoints are at the same height ($f(a) = f(b)$). In this scenario, the average rate of change (the slope of the secant line) is exactly zero, yielding $f'(c) = 0$.

To see the motivation for this elegant proof, notice that the new function $h(x)$ simply measures the *vertical distance* (or error) between the function and the secant line. Because the function intersects the secant line at both endpoints, this vertical distance is exactly zero at $x = a$ and $x = b$ (i.e., $h(a) = h(b) = 0$). We have successfully reduced the general problem into a scenario where Rolle's Theorem directly applies. Finding the point c where $h'(c) = 0$ is exactly finding the point where the vertical distance between the curve and the secant line stops growing and starts shrinking—meaning the function's instantaneous slope is identical to the secant line's slope.

Remark. The quantity $\frac{f(b)-f(a)}{b-a}$ is the slope of the secant line connecting the endpoints $(a, f(a))$ and $(b, f(b))$. The Mean Value Theorem guarantees that if a curve is continuous and smooth, there must be at least one interior point c where the tangent line is perfectly parallel to this secant line. In kinematic terms, if you travel from point A to point B, your instantaneous velocity must exactly equal your average velocity at some point during the trip.

Exercise. Use the MVT to find an easier proof of Theorem 4.10.

Exercise. Let I be an open interval and suppose $f : I \rightarrow \mathbb{R}$ is differentiable. Show that if $f'(x) = 0$ for all $x \in I$ then f is constant on I .

There is also the following generalisation of the MVT, known as Cauchy's Mean Value Theorem (CMVT).

Theorem 4.19 (Cauchy's Mean Value Theorem). *Let $f : [a, b] \rightarrow \mathbb{R}$ and $g : [a, b] \rightarrow \mathbb{R}$ both be continuous on $[a, b]$ and differentiable on (a, b) . Then there exists $\xi \in (a, b)$ such that*

$$(f(b) - f(a))g'(\xi) = (g(b) - g(a))f'(\xi).$$

Proof. We prove this in a similar way to the Mean Value Theorem. Let $h : [a, b] \rightarrow \mathbb{R}$ be defined by

$$h(x) = (f(b) - f(a))g(x) - (g(b) - g(a))f(x)$$

Since f and g are continuous on $[a, b]$ and differentiable on (a, b) , it follows that h , being a linear combination of $f(x)$ and $g(x)$ with constant coefficients, is also continuous on $[a, b]$ and differentiable on (a, b) .

Evaluating h at the endpoints a and b , we get

$$\begin{aligned} h(a) &= (f(b) - f(a))g(a) - (g(b) - g(a))f(a) \\ &= f(b)g(a) - f(a)g(a) - g(b)f(a) + g(a)f(a) \\ &= f(b)g(a) - g(b)f(a), \end{aligned}$$

$$\begin{aligned} h(b) &= (f(b) - f(a))g(b) - (g(b) - g(a))f(b) \\ &= f(b)g(b) - f(a)g(b) - g(b)f(b) + g(a)f(b) \\ &= -f(a)g(b) + g(a)f(b) \\ &= f(b)g(a) - g(b)f(a), \end{aligned}$$

so $h(a) = h(b)$.

Because h satisfies all conditions of Rolle's Theorem on $[a, b]$, there exists $\xi \in (a, b)$ such that $h'(\xi) = 0$.

Taking the derivative of h gives:

$$h'(x) = (f(b) - f(a))g'(x) - (g(b) - g(a))f'(x).$$

Therefore,

$$h'(\xi) = (f(b) - f(a))g'(\xi) - (g(b) - g(a))f'(\xi) = 0,$$

which implies that

$$(f(b) - f(a))g'(\xi) = (g(b) - g(a))f'(\xi),$$

which completes the proof. $\square$

Remark. If additionally we have that $g(b) \neq g(a)$ and g' does not vanish on (a, b) , then the above implies that there exists $\xi \in (a, b)$ such that

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(\xi)}{g'(\xi)},$$

which is a more useful and intuitive form of the theorem (indeed, it is used in proving L'Hôpital's Rule).

4.3.1 Darboux's Theorem

Theorem 4.20 (Darboux's Theorem). *Let $f : (a, b) \rightarrow \mathbb{R}$ be differentiable. Then f' satisfies the intermediate value property, i.e. for any $x < y$ in (a, b) , if η is any real number strictly between $f'(x)$ and $f'(y)$, then there exists $\xi \in (x, y)$ such that $f'(\xi) = \eta$.*

Proof. Assume without loss of generality that $f'(a) < \eta < f'(b)$ (the case where $f'(a) > \eta > f'(b)$ is symmetric, i.e. the argument doesn't change). Define $g : [x, y] \rightarrow \mathbb{R}$ by $g(t) = f(t) - \eta t$. Since f is differentiable on (a, b) , g is continuous on $[x, y]$ and differentiable on (x, y) . Also, for $t \in (x, y)$ we have

$$g'(t) = f'(t) - \eta.$$

Evaluating this derivative at the endpoints gives:

$$g'(x) = f'(x) - \eta < 0 \quad (\text{since } f'(a) < \eta)$$

$$g'(y) = f'(y) - \eta > 0 \quad (\text{since } f'(b) > \eta)$$

By Theorem 4.17, it follows that there exists $\xi \in (x, y)$ such that $g'(\xi) = 0$, i.e. $f'(\xi) = \eta$, which finishes the proof of the theorem. $\square$

Remark. Darboux's Theorem is somewhat surprising, in that it does not require f' to be continuous. We already know from the standard Intermediate Value Theorem that continuous functions cannot “jump over” intermediate values. Darboux's Theorem tells us that *derivatives* also cannot “jump over” intermediate values.

This imposes a massive restriction on how a derivative can behave; derivatives *cannot have jump discontinuities*. Therefore, if a derivative is discontinuous, it must be an essential (oscillatory) discontinuity.

The mere property of being a derivative is strong enough to enforce this. A classic example is the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by:

$$f(x) = \begin{cases} x^2 \sin(1/x), & x \neq 0 \\ 0, & x = 0. \end{cases}$$

Its derivative f' oscillates wildly near 0 and is discontinuous at 0, yet by Darboux's Theorem, f' still successfully takes on every intermediate value in its range.

We also obtain the very important corollary of Darboux's theorem:

Corollary 4.21. *If $f : I \rightarrow \mathbb{R}$ is differentiable and $f'(t) \neq 0$ for all $t \in I$, then f is strictly monotone on I .*

Proof. By Darboux's Theorem, if f' ever took both positive and negative values, it would have to cross 0, which contradicts $f'(t) \neq 0$. Thus, f' is either always strictly positive or always strictly negative, making f strictly monotone by Theorem 4.10. $\square$

Exercise. Let $I \subset \mathbb{R}$ be an open interval and let $f : I \rightarrow \mathbb{R}$ be differentiable. Suppose that $|f'(x)| = 1$ for all $x \in I$. Show that there exists a constant $C \in \mathbb{R}$ such that either

$$f(x) = x + C \quad \text{for all } x \in I,$$

or

$$f(x) = -x + C \quad \text{for all } x \in I.$$

The above fact will be used in the “Curves in the Plane” chapter to show that an arc-length parametrisation of a curve is unique, up to a translation of the parameter (and possibly reversing orientation).

4.4 L'Hôpital's Rule

In this section, we state and prove a very useful tool for finding limits which are in certain so-called “indeterminate forms” (that is to say, the limits themselves are in a form which cannot be directly evaluated using, say, continuity of the function(s) in question). These forms are $\frac{0}{0}$ and $\frac{\pm\infty}{\pm\infty}$ (note that the sign of ∞ does not have to be the same for the numerator and denominator limits). It can be applied as $x \rightarrow c$ for some finite $a \in \mathbb{R}$ (including one-sided limits as $x \rightarrow a^\pm$), or as $x \rightarrow \pm\infty$.

Theorem 4.22 (L'Hôpital's Rule, $\frac{0}{0}$ Case). *Let $a \in \mathbb{R}$, and suppose there exists $\rho > 0$ such that f and g are real-valued functions defined on the domain $D = (a - \rho, a + \rho) \setminus \{a\}$. If*

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0,$$

and the limit

$$\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

exists, then the limit

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$$

exists, and both limits are equal.

Proof. First we note the following implications of our hypotheses. There exists $\delta > 0$ such that:

1. f and g are differentiable on $(a - \delta, a + \delta) \setminus \{a\}$;
2. $g'(x) \neq 0$ on $(a - \delta, a + \delta) \setminus \{a\}$.

Note that (1) and (2) do not need to hold at $x = a$. In fact, f and g do not need to be defined at a for $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ to make sense. Since we are assuming

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0,$$

it makes sense to set $f(a) = g(a) = 0$ (redefining f and g at a if necessary). Now we can say f and g are continuous on $(a - \delta, a + \delta)$ including a , therefore we can use Rolle's theorem and Cauchy's Mean Value Theorem (Theorem 4.19) on closed intervals contained in $(a - \delta, a]$ or in $[a, a + \delta)$.

We claim that $g(x) \neq 0$ for all $x \in (a - \delta, a + \delta) \setminus \{a\}$. Indeed, by assumption $g'(x) \neq 0$ near a , hence g is strictly monotone there by Corollary 4.21, which implies injectivity. Since $g(a) = 0$, it follows that $g(x) \neq 0$ for every $x \neq a$ sufficiently close to a .

Now, since $g(x) \neq 0$ for all $x \neq a$ near a , by Cauchy's Mean Value Theorem, for every $x \in (a, a + \delta)$ (with $\delta > 0$ small) there exists $\alpha_x \in (a, x)$ such that

$$\frac{f'(\alpha_x)}{g'(\alpha_x)} = \frac{f(x) - f(a)}{g(x) - g(a)} = \frac{f(x)}{g(x)}, \quad \text{and} \quad |\alpha_x - a| < |x - a|.$$

Now, let $\varepsilon > 0$ be arbitrary. By assumption, there exists $\delta_0 > 0$ such that

$$0 < |y - a| < \delta_0 \implies \left| \frac{f'(y)}{g'(y)} - L \right| < \varepsilon.$$

Now choose x such that $0 < |x - a| < \delta_0$. Then

$$0 < |\alpha_x - a| < |x - a| < \delta_0,$$

so (taking $y = \alpha_x$) we obtain

$$\left| \frac{f'(\alpha_x)}{g'(\alpha_x)} - L \right| < \varepsilon.$$

Using $\frac{f'(\alpha_x)}{g'(\alpha_x)} = \frac{f'(x)}{g'(x)}$, this gives

$$\left| \frac{f(x)}{g(x)} - L \right| < \varepsilon,$$

hence $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = L$. □ □

Remark. To extend this result to the limits as $x \rightarrow \pm\infty$, we apply the substitution $t = 1/x$. Define $F(t) = f(1/t)$ and $G(t) = g(1/t)$. As $x \rightarrow \pm\infty$, we have $t \rightarrow 0^\pm$.

Note that this substitution requires the one-sided limit version of L'Hôpital's rule, in the case $\frac{0}{0}$. The proof of the initial $x \rightarrow a$ case remains identical for one-sided limits, requiring only obvious modifications to the intervals (e.g., restricting to $(a, a + \delta)$ rather than $(a - \delta, a + \delta)$ for the right-sided limit).

Applying this finite, one-sided case at $a = 0$, we obtain:

$$\lim_{x \rightarrow \pm\infty} \frac{f(x)}{g(x)} = \lim_{t \rightarrow 0^\pm} \frac{F(t)}{G(t)} = \lim_{t \rightarrow 0^\pm} \frac{F'(t)}{G'(t)}$$

Using the chain rule, $F'(t) = f'(1/t)(-1/t^2)$ and $G'(t) = g'(1/t)(-1/t^2)$. The $-1/t^2$ factors cancel out, which yields:

$$\lim_{t \rightarrow 0^\pm} \frac{F'(t)}{G'(t)} = \lim_{t \rightarrow 0^\pm} \frac{f'(1/t)}{g'(1/t)} = \lim_{x \rightarrow \pm\infty} \frac{f'(x)}{g'(x)}$$

This completes the proof for infinite limits.

Theorem 4.23 (L'Hôpital's Rule, $\frac{\infty}{\infty}$ Case). *Let $a \in \mathbb{R}$, and suppose there exists $\rho > 0$ such that f and g are real-valued functions defined on the domain $D = (a - \rho, a + \rho) \setminus \{a\}$. If*

$$\lim_{x \rightarrow a} |f(x)| = \lim_{x \rightarrow a} |g(x)| = \infty,$$

and the limit

$$\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

exists, then the limit

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$$

exists, and both limits are equal.

Proof. Exercise. □

Remark. Just as in the $\frac{0}{0}$ case, we can extend this result to the limits as $x \rightarrow \pm\infty$ by applying the substitution $t = 1/x$. Define $F(t) = f(1/t)$ and

$G(t) = g(1/t)$. As $x \rightarrow \pm\infty$, we have $t \rightarrow 0^\pm$, and our hypothesis becomes $\lim_{t \rightarrow 0^\pm} |F(t)| = \lim_{t \rightarrow 0^\pm} |G(t)| = \infty$.

This substitution again requires the one-sided limit version of L'Hôpital's rule for the finite case. The proof of the initial $x \rightarrow a$ case for $\frac{\infty}{\infty}$ remains identical for one-sided limits, requiring only obvious modifications to the intervals (e.g., restricting to $(a, a + \delta)$ rather than $(a - \delta, a + \delta)$ for the right-sided limit).

Applying this finite, one-sided $\frac{\infty}{\infty}$ case at $a = 0$, we obtain:

$$\lim_{x \rightarrow \pm\infty} \frac{f(x)}{g(x)} = \lim_{t \rightarrow 0^\pm} \frac{F(t)}{G(t)} = \lim_{t \rightarrow 0^\pm} \frac{F'(t)}{G'(t)}$$

Using the chain rule, $F'(t) = f'(1/t)(-1/t^2)$ and $G'(t) = g'(1/t)(-1/t^2)$. The $-1/t^2$ factors cancel out, which yields:

$$\lim_{t \rightarrow 0^\pm} \frac{F'(t)}{G'(t)} = \lim_{t \rightarrow 0^\pm} \frac{f'(1/t)}{g'(1/t)} = \lim_{x \rightarrow \pm\infty} \frac{f'(x)}{g'(x)}$$

This completes the extension to infinite limits for the $\frac{\infty}{\infty}$ case.

Exercise. Evaluate the following limits. You may assume the Fundamental Theorem of Calculus (which will be proved in the integration chapter).

- (a) $\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right)^{\frac{1}{x^2}}$
- (b) $\lim_{x \rightarrow \infty} x \left(\left(1 + \frac{1}{x} \right)^x - e \right)$
- (c) $\lim_{x \rightarrow 0} \frac{1}{x^4} \int_0^x (t - \sin t) dt$

4.5 Second Derivatives

When we say “ $f''(x)$ exists”, we are assuming:

1. $f'(y)$ exists for all y near x (equivalently: there exists an open interval I containing x where f is differentiable);
2. the limit

$$\lim_{y \rightarrow x} \frac{f'(y) - f'(x)}{y - x}$$

exists.

The second derivatives can help us determine the local geometry of f near a critical point (i.e. whether the curve “bending upwards” or “bending downwards” will help us determine the nature of critical points in most cases, excluding the degenerate case when $f''(c) = 0$).

Theorem 4.24 (Second Derivative Test). *Let $f : I \rightarrow \mathbb{R}$ be differentiable and suppose c is a critical point of f , i.e. $f'(c) = 0$. Then*

1. *If $f''(c) > 0$ then f has a local minimum at c .*
2. *If $f''(c) < 0$ then f has a local maximum at c .*

Proof. We prove the case $f''(c) > 0$. The case $f''(c) < 0$ is similar. Let $g(x) = f'(x)$. Then we are assuming $g(c) = 0$ and $g'(c) > 0$. By local comparison (Lemma 4.8), there exists $\delta > 0$ such that

$$\begin{aligned} g(x) &< g(c) = 0 & \forall x \in (c - \delta, c), \\ g(x) &> g(c) = 0 & \forall x \in (c, c + \delta). \end{aligned}$$

Therefore,

$$\begin{aligned} f'(x) &< 0 & \forall x \in (c - \delta, c), \\ f'(x) &> 0 & \forall x \in (c, c + \delta), \end{aligned}$$

so f is monotone decreasing on $(c - \delta, c)$ and monotone increasing on $(c, c + \delta)$. Hence for every $x < c < y$ with $x, y \in (c - \delta, c + \delta)$ we have

$$f(x) > f(c) \quad \text{and} \quad f(y) > f(c),$$

which shows that c is a local minimum for f . □

It is important to remember that evaluating f'' at a single point $x = a$ yields strictly *local* information. A positive second derivative at a critical point guarantees a local minimum, but it tells us absolutely nothing about whether it is a global minimum.

This is best understood through the lens of Taylor series. The values of the derivatives at a point construct a polynomial that matches the function's behaviour only in a small δ -neighborhood around that point. The second derivative term $\frac{f''(a)}{2!}(x - a)^2$ dominates the local shape, forcing the curve to bend upwards (like a parabola) when $f''(a) > 0$.

Remark (The Degenerate Case $f''(c) = 0$). When c is a critical point ($f'(c) = 0$) but the second derivative vanishes (i.e., $f''(c) = 0$), the Second Derivative Test is *inconclusive*.

Intuitively, this means the graph of f is very “flat” at $x = c$. The local quadratic approximation is completely horizontal, failing to capture whether the curve ultimately bends upward or downward. From $f''(c) = 0$ alone, we cannot conclude whether c is a local minimum, a local maximum, or neither (such as a saddle/inflection point). Standard examples at $c = 0$ illustrate all

three possibilities: $f(x) = x^4$ has a minimum, $f(x) = -x^4$ has a maximum, and $f(x) = x^3$ has neither.

To classify the critical point in this degenerate case, we need extra information regarding the behavior of the function near c .

What about the reciprocal direction: if f has a local minimum at a , must we have $f''(a) > 0$?

- Without extra assumptions, this does not even make sense (for instance f may fail to be differentiable).
- Even if we assume $f''(a)$ exists, the conclusion $f''(a) > 0$ is still false in general: for example, f constant; or $f(x) = x^4$, which has a (unique global) minimum at 0 but $f''(0) = 0$.

Theorem 4.25. *Assume $f''(a)$ exists and f has a local minimum at a . Then $f''(a) \geq 0$.*

Proof. We show it by contradiction. Assume:

1. a is a local minimum;
2. $f''(a) < 0$.

Since a is a local minimum and $f'(a)$ exists, we must have $f'(a) = 0$, i.e. a is a critical point. Together with $f''(a) < 0$, the previous theorem implies that a is a local maximum. Therefore a is both a local minimum and a local maximum, so f is locally constant near a . In particular $f''(a) = 0$, which contradicts $f''(a) < 0$. $\square$

4.6 The Inverse Function Theorem

It is common to encounter functions that are locally bijective, that is, there exists a neighborhood of a point $a \in I$ where I is open such that the function restricted to that neighborhood is bijective. For example, even though the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \cos x$ is periodic, and is therefore not bijective on $\mathbb{R}$, when we restrict f to the domain $[0, \pi]$, f is bijective and takes every value in $[-1, 1]$.

It is therefore natural to ask the question: what properties of a bijective function's inverse can we deduce from the properties of the function itself (e.g. in particular continuity and differentiability), and how does, say, the derivative of f^{-1} relate to the derivative of f ? In this section, we answer such questions.

Definition 4.26. Let $f : (a, b) \rightarrow (c, d)$ be a bijection. We say that a function $g : (c, d) \rightarrow (a, b)$ is the *inverse* of f if for all $x \in (a, b)$, $g(f(x)) = x$, and for all $y \in (c, d)$, $f(g(y)) = y$.

One checks that inverses are unique, and therefore we can write f^{-1} to denote *the* inverse of a bijection f .

Strictly monotone functions are natural examples of bijections between intervals. In fact, in the set of continuous functions, these are the *only* bijective functions between intervals. This follows from the Intermediate Value Theorem and is encapsulated in the following exercise.

Exercise. Show that if f is continuous and injective on an interval (x, y) , then f is either strictly increasing or strictly decreasing on that interval.

As a result of the above exercise, continuous bijections map intervals to intervals:

Corollary 4.27. Let $f : [a, b] \rightarrow S$ be a continuous bijection. Then

$$S = [\min\{f(a), f(b)\}, \max\{f(a), f(b)\}].$$

Now, we observe that continuity is preserved by inverses of continuous bijections:

Theorem 4.28. Let $f : (a, b) \rightarrow (c, d)$ be a continuous bijection. Then $f^{-1} : (c, d) \rightarrow (a, b)$ is continuous.

Proof. Let $\varepsilon > 0$ and $y_0 \in (c, d)$ be arbitrary. By Corollary 4.27, since f is continuous it is strictly monotonic. Without loss of generality, assume f is strictly increasing (if f is strictly decreasing, a similar argument applies, or one can consider the increasing function $-f$). Since f is a bijection onto (c, d) , there exists $x_0 \in (a, b)$ such that $f(x_0) = y_0$. Let $\varepsilon' := \min\{\frac{x_0 - a}{2}, \frac{b - x_0}{2}, \varepsilon\}$, so that $(x_0 - \varepsilon', x_0 + \varepsilon') \subset (a, b)$. Since f is continuous and strictly increasing, by Corollary 4.27, it follows that

$$f((x_0 - \varepsilon', x_0 + \varepsilon')) = (f(x_0 - \varepsilon'), f(x_0 + \varepsilon')). \quad (\dagger)$$

In particular, $y_0 \in (f(x_0 - \varepsilon'), f(x_0 + \varepsilon'))$. Now, pick

$$\delta = \min \left\{ \frac{y_0 - f(x_0 - \varepsilon')}{2}, \frac{f(x_0 + \varepsilon') - y_0}{2} \right\}.$$

It follows that for any $y \in (y_0 - \delta, y_0 + \delta)$, we have $y \in (f(x_0 - \varepsilon'), f(x_0 + \varepsilon'))$, from which it follows that

$$f^{-1}(y) \in f^{-1}(f(x_0 - \varepsilon'), f(x_0 + \varepsilon')) = (x_0 - \varepsilon', x_0 + \varepsilon'),$$

by using the preimage of $f((x - \varepsilon', x + \varepsilon'))$ and $(\dagger)$. Hence, since $x_0 = f^{-1}(y_0)$, we have $|f^{-1}(y) - f^{-1}(y_0)| < \varepsilon' \leq \varepsilon$, so f^{-1} is continuous on (c, d) . $\square$

Next, we analyse differentiability. One version of the inverse function theorem is the following:

Theorem 4.29. *Let f be continuous and injective on an open interval I . Assume that f is differentiable at $x_0 \in I$ with $f'(x_0) \neq 0$. Then f^{-1} is differentiable at $f(x_0)$ and $(f^{-1})'(f(x_0)) = \frac{1}{f'(x_0)}$.*

Proof. Let $y_0 = f(x_0)$. We wish to show that the limit

$$\lim_{y \rightarrow y_0} \frac{f^{-1}(y) - f^{-1}(y_0)}{y - y_0}$$

exists, and is equal to $\frac{1}{f'(x_0)}$. Given $y \in f(I) \setminus \{y_0\}$, we have $y = f(x)$ for some $x \in I \setminus \{x_0\}$, and since f is injective, $f(x) \neq f(x_0)$, so

$$\frac{f^{-1}(y) - f^{-1}(y_0)}{y - y_0} = \frac{x - x_0}{f(x) - f(x_0)}. \quad (\star)$$

Now, we need to argue why the limit as $y \rightarrow y_0$ of the LHS of the above equation is the same as the limit as $x \rightarrow x_0$ of the RHS of the equation, which by differentiability of f at x_0 is $\frac{1}{f'(x_0)}$. Let $\varepsilon > 0$ be arbitrary. Then by differentiability of f at x_0 , and the Algebra of Limits for functions since $f'(x_0) \neq 0$, there exists $\delta > 0$ such that for any $x \in (x_0 - \delta, x_0 + \delta) \setminus \{x_0\} \subset I$, we have

$$\left| \frac{1}{\frac{f(x) - f(x_0)}{x - x_0}} - \frac{1}{f'(x_0)} \right| = \left| \frac{x - x_0}{f(x) - f(x_0)} - \frac{1}{f'(x_0)} \right| < \varepsilon.$$

By continuity of f^{-1} (Theorem 4.28), there exists $\rho > 0$ such that for any $y \in (y_0 - \rho, y_0 + \rho) \setminus \{y_0\} \subset f(I)$, we have $f^{-1}(y) \in (x_0 - \delta, x_0 + \delta) \setminus \{x_0\}$.

Therefore, by $(\star)$, for $y \in (y_0 - \rho, y_0 + \rho) \setminus \{y_0\}$, we have (by putting $x = f^{-1}(y)$),

$$\left| \frac{f^{-1}(y) - f^{-1}(y_0)}{y - y_0} - \frac{1}{f'(x_0)} \right| = \left| \frac{x - x_0}{f(x) - f(x_0)} - \frac{1}{f'(x_0)} \right| < \varepsilon,$$

which completes the proof. $\square$

However, we will take the theorem below (which is a corollary of Theorem 4.29 above) as our Inverse Function Theorem, as it is slightly more general.

Theorem 4.30 (Inverse Function Theorem). *Let I be an open interval. Assume $f : I \rightarrow \mathbb{R}$ is continuously differentiable. If $f'(a) \neq 0$ for some $a \in I$, then there exists $\delta > 0$ such that the restriction of f to $(a - \delta, a + \delta)$ is injective. Consequently, f is invertible onto its image $J = f(a - \delta, a + \delta)$, and its inverse $f^{-1} : J \rightarrow (a - \delta, a + \delta)$ is differentiable with $(f^{-1})'(f(t)) = \frac{1}{f'(t)}$ for all $t \in (a - \delta, a + \delta)$.*

Proof. Exercise. □

Remark. The reciprocal relationship in the Inverse Function Theorem has a couple of nice interpretations. It is a direct consequence of basic geometry and the definition of a slope.

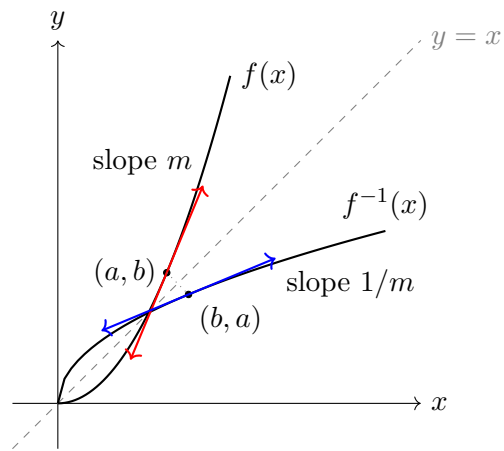
1. *The Geometric Perspective:* The graph of an inverse function f^{-1} is just the graph of f reflected across the diagonal line $y = x$. This reflection physically swaps the horizontal and vertical axes.

The derivative is the slope of the tangent line, which is the ratio of “rise” over “run” ($\frac{\Delta y}{\Delta x}$). When you reflect the graph across $y = x$, the old vertical “rise” becomes the new horizontal “run”, and the old horizontal “run” becomes the new vertical “rise”.

The new slope is therefore $\frac{\Delta x}{\Delta y}$, which is exactly the reciprocal of the original slope $\frac{\Delta y}{\Delta x}$. If a tangent line on f is very steep, its reflection on f^{-1} will be very shallow. Spend some time thinking about a reflection across the line $y = x$, and this should become clear (a picture is also provided below).

2. *The Kinematics Perspective:* Suppose $y = f(x)$ represents your distance travelled (y) at a given time (x). The derivative $f'(x)$ is your velocity, say 5 meters per second.

The inverse function $x = f^{-1}(y)$ flips the question: “How much time x does it take to reach distance y ?” If you are travelling at 5 meters per second, it takes exactly $\frac{1}{5}$ of a second to travel one meter. The rate of change of time with respect to distance is logically the reciprocal of the rate of change of distance with respect to time.



Chapter 5

Convexity

To understand convex functions, we must first define a more general notion of convexity, which is a property of *subsets* of $\mathbb{R}^n$ in general.

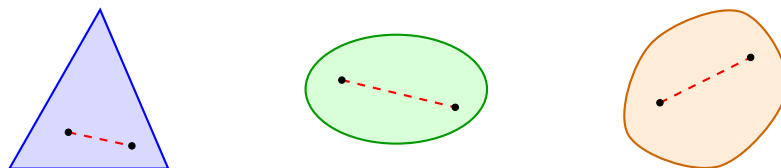
Definition 5.1 (Convex Set). A subset $S \subseteq \mathbb{R}^n$ is *convex* in $\mathbb{R}^n$ if, for any two points $x, y \in S$, the straight line segment connecting them lies entirely within S . Mathematically, for any $x, y \in S$ and for all $t \in [0, 1]$:

$$(1 - t)x + ty \in S.$$

Remark. It is convenient to parameterise the segment from x to y by the function $\gamma : [0, 1] \rightarrow \mathbb{R}^n$ defined by

$$\gamma(t) = (1 - t)x + ty.$$

In $\mathbb{R}$, the only convex sets are intervals (open, closed, or half-open) and single points. Some examples of convex sets in $\mathbb{R}^2$ (which are important for defining the notion of *convex functions* $\mathbb{R} \rightarrow \mathbb{R}$) are illustrated below.



Definition 5.2 (Epigraph). Let $I \subset \mathbb{R}$ be an interval and let $f : I \rightarrow \mathbb{R}$ be a function. The *epigraph* of f is the subset of $\mathbb{R}^2$ defined by

$$\text{epi}(f) := \{(x, y) \in I \times \mathbb{R} : y \geq f(x)\} \subset \mathbb{R}^2.$$

The following is a *geometric* definition of convex functions.

Definition 5.3 (Convex Function). We say that a function $f : I \rightarrow \mathbb{R}$ is *convex* if $\text{epi}(f)$ is a convex set in $\mathbb{R}^2$.

Intuitively, the region above the graph has no “dents”. Unwinding this definition yields two concrete inequalities which are often easier to use in practice.

5.1 Jensen’s Inequality

Theorem 5.4 (Jensen’s Inequality for Convex Functions). *Let $I \subset \mathbb{R}$ be an interval. A function $f : I \rightarrow \mathbb{R}$ is convex on I if and only if for all $a < b \in I$, and for all $t \in [0, 1]$,*

$$f((1-t)a + tb) \leq (1-t)f(a) + tf(b).$$

Proof. First, assume $f : I \rightarrow \mathbb{R}$ satisfies the epigraph definition of convexity (Definition 5.3). Let $a < b$ in I be arbitrary, and consider the points $x = (a, f(a)) \in \text{epi}(f)$ and $y = (b, f(b)) \in \text{epi}(f)$. By convexity of $\text{epi}(f)$, using Definition 5.1 for convex sets, we see that the line segment between x and y lies in $\text{epi}(f)$, but for $t \in [0, 1]$ we have

$$(1-t)x + ty = ((1-t)a + tb, (1-t)f(a) + tf(b)),$$

so the y -coordinate in the above point on the line segment must lie above the graph f at $x = (1-t)a + tb$. In other words, we have

$$f((1-t)a + tb) \leq (1-t)f(a) + tf(b),$$

which proves the $(\Rightarrow)$ direction.

Conversely, assume Jensen’s inequality for convex functions; we wish to show that f satisfies the epigraph definition of convexity, Definition 5.3, i.e. we need to show that $\text{epi}(f)$ is a convex set in $\mathbb{R}^2$. Let $x, y \in \text{epi}(f)$ be arbitrary, and write $x = (x_1, y_1)$, $y = (x_2, y_2)$, so $y_1 \geq f(x_1)$ and $y_2 \geq f(x_2)$. Therefore, for $t \in [0, 1]$ we have

$$(1-t)y_1 + ty_2 \geq (1-t)f(x_1) + tf(x_2) \geq f((1-t)x_1 + tx_2),$$

where the last inequality is by Jensen’s inequality. This implies that

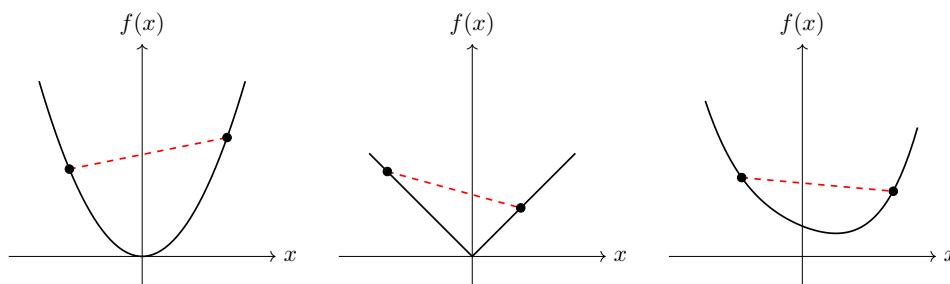
$$(1-t)x + ty \in \text{epi}(f),$$

which finishes the proof. $\square$

The above theorem is an if and only if statement, so can be taken as an equivalent definition of a convex function. Jensen's inequality for convex functions simply states that the graph of the function always lies *below* the chord connecting any two points on the graph.

Remark. One can think of a convex function as being shaped like a bowl; consider a convex function defined on $[0, \infty)$: it is fundamentally trying to bend upwards, though it doesn't have to. It is either trying to bend upwards for all $x > 0$, but fails, in which case it is a decreasing function for $x > 0$, or does eventually start pointing upwards, and continues in this way, as visualised above in the examples. Or it simply starts off as increasing, and must continue in this way to remain convex (in fact, we will see later that the slope itself must be increasing too, for a differentiable convex function).

Some examples of convex functions are illustrated below. As seen in Theorem 5.4, Jensen's inequality for convex functions is equivalent to our epigraph definition and simply states that the graph of the function sits below the chord drawn between any two points on the graph.



Definition 5.5 (Strictly Convex Function). A function $f : I \rightarrow \mathbb{R}$ is *strictly convex* on I if, for all $a < b \in I$, and for all $t \in (0, 1)$:

$$f((1-t)a + tb) < (1-t)f(a) + tf(b).$$

Remark. Strict convexity is a stronger property for a function $f : I \rightarrow \mathbb{R}$ to satisfy, and through it we see the relevance of our study of convexity: convexity is related to minimisation problems. Intuitively you can see that if a strictly convex function is differentiable, and if it has a critical point in some place, then it is unique and it is a global minimiser, because once the function starts bending upwards, it must keep increasing if it is convex everywhere (we will see this later).

It is often convenient to have an explicit function defined on I , rather than a parameterisation, when working with Jensen's inequality and chords:

Definition 5.6 (Chord line). Let $f : I \rightarrow \mathbb{R}$ be a function and let $a < b$ in I . We denote by $\ell_{a,b} : \mathbb{R} \rightarrow \mathbb{R}$ the affine function whose graph is the straight line through $(a, f(a))$ and $(b, f(b))$:

$$\ell_{a,b}(x) = \frac{f(b) - f(a)}{b - a}(x - a) + f(a).$$

Lemma 5.7. Let $f : I \rightarrow \mathbb{R}$ be a function. The following are equivalent:

1. f is convex on I
2. For all $a < b$ in I and for all $x \in [a, b]$,

$$f(x) \leq \ell_{a,b}(x).$$

Proof. Exercise. □

5.2 Slope Monotonicity

While the graph of a convex function lies *below* (or on) the chord inside the interval $[a, b]$, the *opposite* is true outside the interval; the graph lies above the extension of the same chord. Draw a few pictures to convince yourself of this.

Lemma 5.8 (Chord Extension Inequality). Let $f : I \rightarrow \mathbb{R}$ be convex. For any $a, b \in I$ with $a < b$, let $\ell_{a,b}(x)$ be the line passing through $(a, f(a))$ and $(b, f(b))$. Then for all $x \in I \setminus [a, b]$, the graph lies above the chord:

$$f(x) \geq \ell_{a,b}(x)$$

Proof. Suppose $x > b$. Since $a < b < x$, the point b lies strictly between a and x . Therefore, we can express b as a convex combination of a and x . For some $t \in \mathbb{R}$, we have

$$b = (1 - t)a + tx.$$

Solving for t , we get $t = \frac{b-a}{x-a}$. Because b is strictly between a and x , we know $t \in (0, 1)$.

Applying Jensen's inequality on the chord from a to x , we get

$$f(b) \leq (1 - t)f(a) + tf(x).$$

Isolating $f(x)$ yields

$$\begin{aligned} tf(x) &\geq f(b) - (1-t)f(a) \\ f(x) &\geq \frac{1}{t}f(b) - \frac{1-t}{t}f(a) \end{aligned}$$

By using our definition of t , we get

$$f(x) \geq \frac{x-a}{b-a}f(b) - \frac{x-b}{b-a}f(a) \quad (\dagger)$$

To finish, we verify this matches the equation of the secant line $\ell_{a,b}(x)$. The standard point-slope form for the line through a and b from Definition 5.6 is:

$$\ell_{a,b}(x) = \frac{f(b) - f(a)}{b - a}(x - a) + f(a),$$

and rearranging we have

$$\begin{aligned} \ell_{a,b}(x) &= f(a) \left(1 - \frac{x-a}{b-a}\right) + f(b) \left(\frac{x-a}{b-a}\right) \\ &= f(a) \left(\frac{b-a-x+a}{b-a}\right) + f(b) \left(\frac{x-a}{b-a}\right) \\ &= -f(a) \left(\frac{x-b}{b-a}\right) + f(b) \left(\frac{x-a}{b-a}\right), \end{aligned}$$

which matches the inequality $(\dagger)$. Hence we have

$$f(x) \geq \ell_{a,b}(x),$$

which completes the proof. $\square$

Lemma 5.9 (Slope Monotonicity Lemma). *Let $f : I \rightarrow \mathbb{R}$ be a convex function. Then for any $a < b < c$ in I , the slopes of the chords are monotonically increasing:*

$$\frac{f(b) - f(a)}{b - a} \leq \frac{f(c) - f(b)}{c - b}.$$

If f is strictly convex, these inequalities are strict.

Proof. Note that for every $x \in [b, c]$ we have

$$\ell_{a,b}(x) \leq f(x) \leq \ell_{b,c}(x).$$

Indeed, the inequality $\ell_{a,b}(x) \leq f(x)$ holds because $x \geq b$ lies outside $[a, b]$, so we can apply the chord extension inequality to the chord through $(a, f(a))$

and $(b, f(b))$. The inequality $f(x) \leq \ell_{b,c}(x)$ holds by convexity on $[b, c]$. Therefore, for all $x \in [b, c]$,

$$\ell_{a,b}(x) \leq \ell_{b,c}(x).$$

We now write both lines in terms of $x - b$ (this makes the comparison transparent):

$$\ell_{a,b}(x) = \frac{f(b) - f(a)}{b - a}(x - b) + f(b), \quad \ell_{b,c}(x) = \frac{f(c) - f(b)}{c - b}(x - b) + f(b).$$

Since $\ell_{a,b}(x) \leq \ell_{b,c}(x)$ for all $x \in [b, c]$ and $x - b \geq 0$ on this interval (with $x - b > 0$ as soon as $x > b$), we conclude that the coefficient of $(x - b)$ on the left is at most the coefficient on the right, namely

$$\frac{f(b) - f(a)}{b - a} \leq \frac{f(c) - f(b)}{c - b}.$$

This completes the proof. $\square$

Corollary 5.10. *Let $f : I \rightarrow \mathbb{R}$ be convex. Then for any $a < b < c < d$ in I , we have*

$$\frac{f(b) - f(a)}{b - a} \leq \frac{f(d) - f(c)}{d - c}.$$

Proof. Apply Lemma 5.9 to $a < b < c$ and then to $b < c < d$:

$$\frac{f(b) - f(a)}{b - a} \leq \frac{f(c) - f(b)}{c - b} \leq \frac{f(d) - f(c)}{d - c}.$$

$\square$

Theorem 5.11. *Let $I \subset \mathbb{R}$ be an open interval, and let $f : I \rightarrow \mathbb{R}$ be convex. If f is differentiable at two points $s < t$ in I then*

$$f'(s) \leq f'(t).$$

Proof. Fix $s < \alpha < t$ and $t < \beta$ such that $\alpha, \beta \in I$. By Corollary 5.10, we have

$$\frac{f(\alpha) - f(s)}{\alpha - s} \leq \frac{f(\beta) - f(t)}{\beta - t}.$$

Treating β as fixed, and taking the limit as $\alpha \rightarrow s^+$ in the above inequality, and using the differentiability of f at s , we get

$$f'(s) \leq \frac{f(\beta) - f(t)}{\beta - t},$$

and letting $\beta \rightarrow t^+$ in the above inequality, and using the differentiability of f at t yields $f'(s) \leq f'(t)$, which completes the proof. $\square$

Corollary 5.12 (Monotonicity of the Derivative). *Let f be differentiable on an open interval I . Then f is convex on I if and only if its derivative f' is increasing on I . Furthermore, f is strictly convex on I if and only if f' is strictly increasing on I .*

Remark (The Second Derivative and Strict Convexity). If f is twice-differentiable, f being convex is equivalent to $f''(x) \geq 0$ for all $x \in I$. However, if f is strictly convex, it is a false statement that $f''(x) > 0$ everywhere. The standard counterexample is $f(x) = x^4$. It is strictly convex everywhere, yet $f''(0) = 0$.

5.3 Regularity of Convex Functions

Theorem 5.13 (Continuity of Convex Functions). *Let $I \subset \mathbb{R}$ be an open interval. If $f : I \rightarrow \mathbb{R}$ is convex on I , then f is continuous on I .*

Proof. Let $a \in I$ be arbitrary; we wish to show that f is continuous at a . First, since I is an open interval we can let $\delta > 0$ be such that $[a - \delta, a + \delta] \subset I$. Fix $t \in (a - \delta, a)$. By convexity of f on $[a - \delta, a]$, we have

$$f(t) \leq \ell_{a-\delta,a}(t).$$

By the Chord Extension Inequality (Lemma 5.8) on $[a, a + \delta]$, we also have

$$\ell_{a,a+\delta}(t) \leq f(t).$$

Therefore, for all $x \in (a - \delta, a)$, we have

$$\ell_{a,a+\delta}(x) \leq f(x) \leq \ell_{a-\delta,a}(x).$$

Both chord lines are continuous and by construction satisfy $\ell_{a-\delta,a}(a) = \ell_{a,a+\delta}(a) = f(a)$, so by the squeeze theorem for limits of functions,

$$\lim_{x \rightarrow a^-} f(x) = f(a).$$

For the right limit, the argument is identical; indeed, using the same argument as above, for every $x \in (a, a + \delta)$ we have

$$\ell_{a-\delta,a}(x) \leq f(x) \leq \ell_{a,a+\delta}(x),$$

and letting $x \rightarrow a^+$ in the above inequality and using the continuity of the chord lines yields $\lim_{x \rightarrow a^+} f(x) = f(a)$, so f is continuous at a . $\square$

The proof of the above theorem is key: near an interior point a , convexity traps the graph between two straight lines that meet at $(a, f(a))$.

An important subtlety of the proof is that the hypothesis of a being interior was crucial so that we could extend the chords. Because of that the argument does not work at the end points. Indeed, as the next example shows, the result is not true at the boundary of an interval.

Example 5.14. The requirement that I be an *open* interval is strictly necessary. A function can be convex on a closed interval but discontinuous at the boundary. For example, consider $f : [0, 1] \rightarrow \mathbb{R}$ defined by $f(x) = 0$ for $x \in (0, 1)$ and $f(0) = 1, f(1) = 1$. This function satisfies the convexity inequality everywhere on $[0, 1]$ but is clearly discontinuous at the endpoints. Thus, for a general interval I , we can only guarantee that f is continuous on the *interior* of I .

Remark. Convexity only implies continuity (on open intervals); it does not imply differentiability. The classic counterexample is $f(x) = |x|$, which is strictly convex on $\mathbb{R}$ but not differentiable at $x = 0$.

When a convex function *is* differentiable, its local geometry becomes highly constrained:

Theorem 5.15 (Tangent Line Characterisation). *If $f : I \rightarrow \mathbb{R}$ is convex and differentiable at an interior point $x_0 \in I$, then the graph of f sits globally above its tangent line at x_0 :*

$$f(x) \geq f(x_0) + f'(x_0)(x - x_0) \quad \text{for all } x \in I$$

Proof. Let $x > x_0$. By the Slope Monotonicity Lemma, for any fixed $t \in (x_0, x)$, the slopes of the chords satisfy:

$$\frac{f(t) - f(x_0)}{t - x_0} \leq \frac{f(x) - f(x_0)}{x - x_0}$$

Taking the limit as $t \rightarrow x_0^+$ (which exists because f is differentiable at x_0), we get

$$f'(x_0) \leq \frac{f(x) - f(x_0)}{x - x_0},$$

since limits preserve non-strict inequalities. Multiplying by $(x - x_0) > 0$ and rearranging yields $f(x) \geq f(x_0) + f'(x_0)(x - x_0)$. The argument for $x < x_0$ is symmetric. $\square$

Remark. So convexity can be rephrased as: every tangent line (where it exists) is a global under-estimate for the function. Intuitively, it can be seen as a “supporting line” on which the function sits.

5.3.1 Minimisation

Convexity provides incredibly strong guarantees about optimisation and bounds.

Theorem 5.16 (Uniqueness of Critical Points). *Let $f : I \rightarrow \mathbb{R}$ be strictly convex and differentiable on an open interval I . Then f has at most one critical point on I . Moreover, if $f'(c) = 0$ for some $c \in I$, then c is the unique global minimiser of f on I .*

Proof. Suppose $f'(c) = 0$. By Theorem 5.15, we have

$$f(x) \geq f(c) + f'(c)(x - c) = f(c) + 0 = f(c)$$

Because f is strictly convex, this inequality is strict for $x \neq c$, making c the unique global minimum. Furthermore, because f is strictly convex, f' is strictly increasing. A strictly increasing function can intersect the axis $f'(x) = 0$ at most once, meaning there can be at most one critical point. $\square$

Theorem 5.17. *Let $I \subset \mathbb{R}$ be a bounded interval and let $f : I \rightarrow \mathbb{R}$ be a convex function. Then f is bounded below.*

Proof. Fix two points s, t in the interior of I such that $s < t$. We can partition the interval I into the closed sub-interval $[s, t]$ and the remaining outside portion $I \setminus [s, t]$.

First, consider the interval $[s, t]$. Since f is convex, by Theorem 5.13, f is continuous on the interior of I . Since s and t are in the interior of I , the closed interval $[s, t]$ lies entirely within the interior of I . Therefore, f is continuous on $[s, t]$. By the Extreme Value Theorem (Theorem 3.24), f attains a minimum on this closed, bounded interval. Thus, there exists some $M_1 \in \mathbb{R}$ such that:

$$f(x) \geq M_1 \quad \text{for all } x \in [s, t].$$

Next, consider the region outside this sub-interval, $I \setminus [s, t]$. Let $\ell_{s,t}(x)$ be the secant line passing through the points $(s, f(s))$ and $(t, f(t))$. By the Chord Extension Inequality (Lemma 5.8), the graph of a convex function lies above its extended secant chords outside the interval $[s, t]$. Therefore:

$$f(x) \geq \ell_{s,t}(x) \quad \text{for all } x \in I \setminus [s, t].$$

Since I is a bounded interval, x cannot approach $\pm\infty$. The secant line $\ell_{s,t}(x)$ is a linear function, and any linear function restricted to a bounded domain is bounded below. Thus, there exists some $M_2 \in \mathbb{R}$ such that $\ell_{s,t}(x) \geq M_2$ for all $x \in I$. This implies:

$$f(x) \geq M_2 \quad \text{for all } x \in I \setminus [s, t].$$

Finally, let $M = \min\{M_1, M_2\}$. It follows that $f(x) \geq M$ for all $x \in I$. Therefore, f is bounded below on I . $\square$

Exercise. Let $f : [0, \infty) \rightarrow \mathbb{R}$ be convex and bounded. Show that $\lim_{x \rightarrow \infty} f(x)$ exists and is finite.

Exercise. In the setting of the above exercise, assume in addition that f is differentiable on $(0, \infty)$. Prove that $\lim_{x \rightarrow \infty} f'(x) = 0$.

Chapter 6

The Integral

The concept of *area* is a simple and intuitive one, yet formalising it presents some subtle challenges.

The simplest definition area, and the most intuitive one, is that of the rectangle, which is simply the base of the rectangle multiplied by the height: $b \times h$. It can be seen as the number of unit squares that fit inside the rectangle.

However, for a curved region (say, a circle) the idea “counting how many rectangles fit inside” is not so straightforward. The integral formalises the geometric concept of the area under a curve. To do this rigorously, we approximate the area using sums of rectangular areas from below and above, and analyse the limits of these approximations as they become more and more precise. We formalise this notion via *Darboux* integration (equivalently, this is sometimes called *Riemann* integration; ultimately, these two concepts are equivalent).

6.1 Lower and Upper Sums

Definition 6.1 (Partition). Let $[a, b]$ be a closed bounded interval. A *partition* of $[a, b]$ is a finite set of points $P = \{x_0, x_1, \dots, x_n\}$ such that

$$a = x_0 < x_1 < \dots < x_n = b.$$

To denote a partition P of $[a, b]$ we will write

$$P = \{a = x_0 < x_1 < \dots < x_{n-1} < x_n = b\},$$

or simply

$$P = \{x_0, x_1, \dots, x_n\}.$$

Definition 6.2 (Upper and Lower Sums). Let $f: [a, b] \rightarrow \mathbb{R}$ be a bounded function and let $P = \{x_0, \dots, x_n\}$ be a partition of $[a, b]$. The *lower sum* of f for P is

$$L(f, P) := \sum_{i=1}^n m_i(x_i - x_{i-1}),$$

where $m_i := \inf_{x \in [x_{i-1}, x_i]} f(x)$.

The *upper sum* of f for P is

$$U(f, P) := \sum_{i=1}^n M_i(x_i - x_{i-1}),$$

where $M_i := \sup_{x \in [x_{i-1}, x_i]} f(x)$.

Remark. Observe the first necessary condition imposed for integrability: the function f *must be bounded*. We use sup and inf (rather than max and min) because a merely bounded function is not guaranteed to attain its extreme values on a sub-interval. It only attains these if it is continuous. Boundedness ensures that M_i and m_i are finite, well-defined real numbers, making the upper and lower sums well-defined.

Lemma 6.3. Let P and P' be partitions of $[a, b]$. If $P \subseteq P'$ (P' is a refinement of P), then

$$L(f, P) \leq L(f, P') \quad \text{and} \quad U(f, P') \leq U(f, P).$$

Proof. Since partitions are finite, it suffices to prove the case where we add *one* extra point to the partition. This is because any refinement P' of P can be obtained as a sequence of refinements

$$P = P_0 \subset P_1 \subset \dots \subset P_k = P'$$

in which each step $P_i \subset P_{i+1}$ consists of refining by exactly one point (in particular, $k = |P'| - |P|$).

Therefore, let

$$P = \{x_0, x_1, \dots, x_{\ell-1}, x_\ell, \dots, x_{n-1}, x_n\} \quad \text{and} \quad P' = P \cup \{y\},$$

where $x_{\ell-1} < y < x_\ell$. Now, the only sub-intervals over which the upper and lower sums will change in the refinement P' are $[x_{\ell-1}, y]$ and $[y, x_\ell]$; the

other terms in the sums $L(f, P')$ and $U(f, P')$ will remain the same as in $L(f, P)$ and $U(f, P)$. Now, since

$$[x_{\ell-1}, y] \subset [x_{\ell-1}, x_\ell] \quad \text{and} \quad [y, x_\ell] \subset [x_{\ell-1}, x_\ell],$$

it follows that, by using standard properties of the supremum and infimum (in particular, if A and B are non-empty, bounded subsets of $\mathbb{R}$ and $A \subseteq B$ then $\sup A \leq \sup B$ and $\inf A \geq \inf B$):

$$m' := \inf_{x \in [x_{\ell-1}, y]} f(x) \geq \inf_{x \in [x_{\ell-1}, x_\ell]} f(x) \quad \text{and} \quad m'' := \inf_{x \in [y, x_\ell]} f(x) \geq \inf_{x \in [x_{\ell-1}, x_\ell]} f(x),$$

and similarly

$$M' := \sup_{x \in [x_{\ell-1}, y]} f(x) \leq \sup_{x \in [x_{\ell-1}, x_\ell]} f(x) \quad \text{and} \quad M'' := \sup_{x \in [y, x_\ell]} f(x) \leq \sup_{x \in [x_{\ell-1}, x_\ell]} f(x).$$

Using these inequalities, and the previous observation that these are the only sub-intervals which change in the lower and upper sums, it follows that

$$\begin{aligned} L(f, P') - L(f, P) &= m'(y - x_{\ell-1}) + m''(x_\ell - y) - m_\ell(x_\ell - x_{\ell-1}) \\ &\geq m_\ell(y - x_{\ell-1}) + m_\ell(x_\ell - y) - m_\ell(x_\ell - x_{\ell-1}) \\ &= 0. \end{aligned}$$

The same argument shows that $U(f, P') - U(f, P) \leq 0$, which completes the proof for a single point refinement, and as mentioned at the start of the proof, this is sufficient to imply the statement for an arbitrary refinement $P \subseteq P'$. $\square$

Remark. The intuition here is that by adding more points to the partition, the approximation cannot get worse; it can only improve. Mathematically, the lower sums grow as you extend to refinements, and the upper sums shrink. L and U are forced closer together, indicating the approximation to the true area is getting better.

Theorem 6.4. For any two partitions P and Q of $[a, b]$,

$$L(f, P) \leq U(f, Q).$$

Proof. It is difficult to compare $L(f, P)$ and $U(f, Q)$ directly with basic tools because the partitions are different. We resolve this by bringing both P and Q to a common partition, specifically their union $R := P \cup Q$. Because $P \subseteq R$ and $Q \subseteq R$, R is a refinement of both. By the previous monotonicity lemma (Lemma 6.3), we have

$$L(f, P) \leq L(f, R) \leq U(f, R) \leq U(f, Q).$$

By transitivity, $L(f, P) \leq U(f, Q)$. $\square$

6.2 The Darboux Integral

Theorem 6.4 implies that *every* lower sum is less or equal than *every* upper sum (as we would expect). Therefore,

$$\sup\{L(f, P) : P \text{ partition of } [a, b]\} \leq \inf\{U(f, P) : P \text{ partition of } [a, b]\}.$$

In general, these two numbers might be far apart:

Exercise. Show that for the function $f : [0, 1] \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

we have

$$\sup\{L(f, P) : P \text{ partition of } [a, b]\} < \inf\{U(f, P) : P \text{ partition of } [a, b]\}.$$

However, when these numbers coincide, we say that f is (Darboux) integrable, and the common number is the value of the *integral*:

Definition 6.5. A bounded function $f : [a, b] \rightarrow \mathbb{R}$ is *Darboux integrable* if

$$\sup_P \{L(f, P)\} = \inf_P \{U(f, P)\}$$

where the supremum and infimum are taken over all possible partitions P of $[a, b]$. This common value is denoted by the singular symbol

$$\int_a^b f(x) \, dx.$$

In this course, because we are not going to discuss other definitions of the integral (e.g. the *Lebesgue integral*), we will simply say that f is integrable if the above definition is satisfied for a function $f : [a, b] \rightarrow \mathbb{R}$.

Remark. The symbol $\int_a^b f(x) \, dx$ is not a collection of operators acting together; it represents a single number. The subtlety of using sup and inf over sets of partitions is that it automatically handles all possible partitions, side-stepping the need to formally define a “limit” of partitions.

In practice, we want a nicer criterion to show that a given function is integrable, since the above requires knowledge of upper and lower sums over *all* partitions of a given interval. Luckily, we have an ε criterion for integrability from standard properties of inf and sup:

Theorem 6.6. *A bounded function $f : [a, b] \rightarrow \mathbb{R}$ is integrable if and only if for all $\varepsilon > 0$, there exists a partition P of $[a, b]$ such that*

$$U(f, P) - L(f, P) < \varepsilon.$$

Proof. Let

$$A = \{L(f, P) : P \text{ is a partition of } [a, b]\},$$

and

$$B = \{U(f, P) : P \text{ is a partition of } [a, b]\}.$$

By Theorem 6.4, $a \leq b$ for all $a \in A$ and $b \in B$, guaranteeing $\sup A \leq \inf B$.

($\Leftarrow$) Assume that for all $\varepsilon > 0$, there exists a partition P such that $U(f, P) - L(f, P) < \varepsilon$. For this partition P , we have $\sup A \geq L(f, P)$ and $\inf B \leq U(f, P)$. Therefore,

$$0 \leq \inf B - \sup A \leq U(f, P) - L(f, P) < \varepsilon.$$

Since this holds for all $\varepsilon > 0$, we must have $\inf B - \sup A = 0$, so $\sup A = \inf B$, meaning f is integrable.

($\Rightarrow$) Assume f is integrable, so $\sup A = \inf B = I$. Let $\varepsilon > 0$ be arbitrary. By definition of supremum and infimum, there exists a lower sum $L(f, P_1)$ and an upper sum $U(f, P_2)$ such that

$$I - \frac{\varepsilon}{2} < L(f, P_1) \leq I \leq U(f, P_2) < I + \frac{\varepsilon}{2}.$$

Let $P = P_1 \cup P_2$. Since P is a refinement of both P_1 and P_2 , $L(f, P_1) \leq L(f, P)$ and $U(f, P) \leq U(f, P_2)$. Therefore,

$$I - \frac{\varepsilon}{2} < L(f, P) \leq U(f, P) < I + \frac{\varepsilon}{2}.$$

Subtracting the inequalities yields $U(f, P) - L(f, P) < \varepsilon$. $\square$

Remark. The idea for the above theorem is that, for a (Darboux) integrable function, we can always make the upper and lower approximation arbitrarily close, by choosing an appropriate partition. This is incredibly useful in practice: instead of trying to understand the entire set of upper and lower sums simultaneously, you only need to construct *one* partition where the error is less than ε to conclude that a function is integrable.

As usual, connected to this ε criterion is a natural (and equally as useful) *sequential* criterion for integrability, which is perhaps more familiar if you have seen *Riemann sums* in a non-rigorous calculus course (perhaps even at A-Level or equivalent). The proof is left as an exercise:

Exercise. (a) Prove that a bounded function f is integrable on $[a, b]$ if and only if there exists a sequence of partitions $(P_n)_{n=1}^{\infty}$ satisfying

$$\lim_{n \rightarrow \infty} [U(f, P_n) - L(f, P_n)] = 0,$$

and in this case $\int_a^b f(x) dx = \lim_{n \rightarrow \infty} U(f, P_n) = \lim_{n \rightarrow \infty} L(f, P_n)$.

(b) For each n , let P_n be the partition of $[0, 1]$ into n equal subintervals. Find formulas for $U(f, P_n)$ and $L(f, P_n)$ if $f(x) = x$. The formula

$$1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}$$

will be useful.

(c) Use the sequential criterion for integrability from (a) to show directly that $f(x) = x$ is integrable on $[0, 1]$ and compute $\int_0^1 f(x) dx$.

6.3 Basic Properties of the Integral

A nice fact we can prove which also demonstrates the power of choosing appropriate partitions is the following rough estimate of an integral:

Lemma 6.7 (Rough Integral Estimate). *Let $f : [a, b] \rightarrow \mathbb{R}$ be integrable. Then*

$$\inf_{x \in [a, b]} f(x)(b-a) \leq \int_a^b f(x) dx \leq \sup_{x \in [a, b]} f(x)(b-a).$$

Proof. We pick the trivial partition of $[a, b]$, $P = \{a, b\}$. Then,

$$L(f, P) = \inf_{x \in [a, b]} f(x)(b-a),$$

and

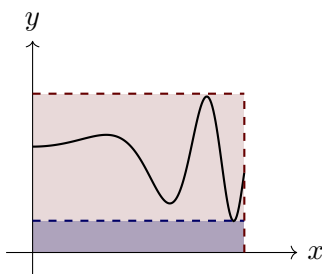
$$U(f, P) = \sup_{x \in [a, b]} f(x)(b-a).$$

Furthermore, the integral $\int_a^b f(x) dx$ by definition always lies between any lower sum and any upper sum, so we have

$$\inf_{x \in [a, b]} f(x)(b-a) \leq \int_a^b f(x) dx \leq \sup_{x \in [a, b]} f(x)(b-a).$$

□

Remark. Geometrically, we are estimating the integral with a *pair* of rectangles, rather than a more sophisticated bound which may use more rectangles, or different shapes like trapeziums (recall the trapezium rule from A-Level):



Theorem 6.8 (Basic Properties of the Integral). *Let f and g be integrable on $[a, b]$. Then*

1. $f + g$ is integrable on $[a, b]$, and $\int_a^b (f(x) + g(x)) \, dx = \int_a^b f(x) \, dx + \int_a^b g(x) \, dx$.
2. For all $\lambda \in \mathbb{R}$, λf is integrable on $[a, b]$, and $\int_a^b \lambda f(x) \, dx = \lambda \int_a^b f(x) \, dx$.
3. fg is integrable on $[a, b]$.
4. If $f(x) \leq g(x)$ for all $x \in [a, b]$, then $\int_a^b f(x) \, dx \leq \int_a^b g(x) \, dx$.
5. For $c \in (a, b)$,

$$f \text{ integrable on } [a, b] \iff f \text{ integrable on } [a, c] \text{ and } [c, b],$$

$$\text{and in that case } \int_a^b f(x) \, dx = \int_a^c f(x) \, dx + \int_c^b f(x) \, dx.$$

Proof. 1. Let $\varepsilon > 0$ be arbitrary. By the ε criterion for integrability (Theorem 6.6), there are partitions P_1 and P_2 of $[a, b]$ such that

$$U(f, P_1) - L(f, P_1) < \frac{\varepsilon}{2} \quad \text{and} \quad U(g, P_2) - L(g, P_2) < \frac{\varepsilon}{2}.$$

Now, let $P = P_1 \cup P_2$, and write $P = \{x_0, \dots, x_n\}$. For every $1 \leq i \leq n$ we have

$$\begin{aligned} M_i'' &:= \sup_{x \in [x_{i-1}, x_i]} (f + g)(x) \leq \sup_{x \in [x_{i-1}, x_i]} f(x) + \sup_{x \in [x_{i-1}, x_i]} g(x) \\ &= M_i + M_i', \end{aligned}$$

and

$$\begin{aligned} m_i'' &:= \inf_{x \in [x_{i-1}, x_i]} (f + g)(x) \geq \inf_{x \in [x_{i-1}, x_i]} f(x) + \inf_{x \in [x_{i-1}, x_i]} g(x) \\ &= m_i + m_i'. \end{aligned}$$

Therefore, summing these inequalities over $1 \leq i \leq n$ we get

$$\begin{aligned} U(f + g, P) - L(f + g, P) &= \sum_{i=1}^n (M_i'' - m_i'')(x_i - x_{i-1}) \\ &\leq \sum_{i=1}^n (M_i + M_i' - m_i - m_i')(x_i - x_{i-1}) \\ &= (U(f, P) - L(f, P)) + (U(g, P) - L(g, P)) \\ &\leq (U(f, P_1) - L(f, P_1)) + (U(g, P_2) - L(g, P_2)) \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon. \end{aligned}$$

Thus $f + g$ is integrable on $[a, b]$. Furthermore, it follows from the supremum and infimum inequalities at the start that

$$L(f, P) + L(g, P) \leq \int_a^b (f(x) + g(x)) \, dx \leq U(f, P) + U(g, P),$$

and by definition we have

$$L(f, P) + L(g, P) \leq \int_a^b f(x) \, dx + \int_a^b g(x) \, dx \leq U(f, P) + U(g, P).$$

Therefore, both $\int_a^b (f(x) + g(x)) \, dx$ and $\int_a^b f(x) \, dx + \int_a^b g(x) \, dx$ are bounded in the same range, which has a width of ε . However, since ε was chosen to be arbitrary, it follows from

$$\left| \int_a^b (f(x) + g(x)) \, dx - \left(\int_a^b f(x) \, dx + \int_a^b g(x) \, dx \right) \right| < \varepsilon$$

that

$$\int_a^b (f(x) + g(x)) \, dx = \int_a^b f(x) \, dx + \int_a^b g(x) \, dx,$$

which completes the proof of (1). □

2. Let $\varepsilon > 0$ and $\lambda \in \mathbb{R}$ be arbitrary.

Case A: $\lambda = 0$. Then $f(x) = 0$ for all $x \in [a, b]$, so choosing the partition $P = \{a, b\}$ yields $U(f, P) - L(f, P) = 0 < \varepsilon$, so λf is integrable in this case and since for any partition the upper and lower sums are zero, it follows by definition that

$$\int_a^b \lambda f(x) \, dx = \int_a^b 0 \, dx = 0,$$

which verifies the trivial case for $\lambda = 0$.

Case B: $\lambda \neq 0$. By the ε criterion for integrability, there exists a partition $P = \{x_0, \dots, x_n\}$ of $[a, b]$ such that

$$U(f, P) - L(f, P) < \frac{\varepsilon}{|\lambda|}.$$

Now, by a standard property of infimum and supremum (which you should verify), for any $1 \leq i \leq n$ we have

$$\sup_{x \in [x_{i-1}, x_i]} \lambda f(x) - \inf_{x \in [x_{i-1}, x_i]} \lambda f(x) = |\lambda| (M_i - m_i),$$

which implies that

$$U(\lambda f, P) - L(\lambda f, P) \leq |\lambda| (U(f, P) - L(f, P)) < \varepsilon,$$

so λf is integrable and by the same argument as in the proof of (1), it follows that

$$\int_a^b \lambda f(x) dx = \lambda \int_a^b f(x) dx,$$

which completes the proof of (2). $\square$

3. First we will show that if $f : [a, b] \rightarrow \mathbb{R}$ is integrable, then f^2 is integrable. Suppose f is integrable on $[a, b]$. Since f is integrable, it is bounded, so there exists $M > 0$ such that $|f(x)| \leq M$ for all $x \in [a, b]$.

Let $\varepsilon > 0$. Then, by integrability of f and Theorem 6.6, there exists a partition P of $[a, b]$ such that

$$U(f, P) - L(f, P) < \frac{\varepsilon}{2M}.$$

Let $P = \{a = x_0 < x_1 < \dots < x_n = b\}$. For any subinterval $[x_{i-1}, x_i]$ (where $1 \leq i \leq n$) and for all $s, t \in [x_{i-1}, x_i]$, we have:

$$\begin{aligned} |f(s)^2 - f(t)^2| &= |f(s) + f(t)| |f(s) - f(t)| \\ &\leq (|f(s)| + |f(t)|) |f(s) - f(t)| \\ &\leq 2M \left(\sup_{x \in [x_{i-1}, x_i]} f(x) - \inf_{x \in [x_{i-1}, x_i]} f(x) \right). \end{aligned}$$

By taking the supremum over all $s, t \in [x_{i-1}, x_i]$ on the left-hand side, we obtain the difference between the supremum and infimum of f^2 on that interval:

$$\sup_{x \in [x_{i-1}, x_i]} f^2(x) - \inf_{x \in [x_{i-1}, x_i]} f^2(x) \leq 2M \left(\sup_{x \in [x_{i-1}, x_i]} f(x) - \inf_{x \in [x_{i-1}, x_i]} f(x) \right).$$

Since this holds for all subintervals, we can sum over all i :

$$\begin{aligned}
 U(f^2, P) - L(f^2, P) &= \sum_{i=1}^n \left(\sup_{x \in [x_{i-1}, x_i]} f^2(x) - \inf_{x \in [x_{i-1}, x_i]} f^2(x) \right) (t_i - t_{i-1}) \\
 &\leq \sum_{i=1}^n 2M \left(\sup_{x \in [x_{i-1}, x_i]} f(x) - \inf_{x \in [x_{i-1}, x_i]} f(x) \right) (t_i - t_{i-1}) \\
 &= 2M(U(f, P) - L(f, P)) \\
 &< 2M \left(\frac{\varepsilon}{2M} \right) \\
 &= \varepsilon.
 \end{aligned}$$

Thus, f^2 is integrable on $[a, b]$. Now, to prove the full statement for (3), suppose f and g are integrable on $[a, b]$. Then

- $f + g$ is integrable (known from item 1 in this theorem).
- f^2 and g^2 are integrable.
- $(f + g)^2$ is integrable.

Furthermore, we can write the product as:

$$2fg = (f + g)^2 - f^2 - g^2$$

Since the sum and difference of integrable functions are integrable (by linearity of integration), $2fg$ is integrable. Therefore, $fg = \frac{1}{2}(2fg)$ is also integrable, by item 2 in this theorem (with $\lambda = \frac{1}{2}$). This completes the proof of (3). $\square$

4. Let $P = \{x_0, x_1, \dots, x_n\}$ be an arbitrary partition of $[a, b]$. Since $f(x) \leq g(x)$ for all $x \in [a, b]$, it follows that on any subinterval $[x_{i-1}, x_i]$, the infimum of f must be less than or equal to the infimum of g . That is:

$$\inf_{x \in [x_{i-1}, x_i]} f(x) \leq \inf_{x \in [x_{i-1}, x_i]} g(x).$$

Since this holds for all $1 \leq i \leq n$, summing over all i yields

$$\sum_{i=1}^n \left(\inf_{x \in [x_{i-1}, x_i]} f(x) \right) (x_i - x_{i-1}) \leq \sum_{i=1}^n \left(\inf_{x \in [x_{i-1}, x_i]} g(x) \right) (x_i - x_{i-1}),$$

which implies that

$$L(f, P) \leq L(g, P).$$

Since g is integrable, we know that any lower sum of g is bounded above by its integral. Therefore:

$$L(f, P) \leq L(g, P) \leq \int_a^b g(x) \, dx.$$

This shows that the number $\int_a^b g(x) dx$ is an upper bound for the set of all lower sums of f , since P was arbitrary. Because the integral of f is defined as the supremum (the least upper bound) of its lower sums, it cannot exceed this upper bound, by definition of supremum. Thus

$$\int_a^b f(x) dx \leq \int_a^b g(x) dx,$$

which completes the proof of (4). $\square$

5. Fix $c \in (a, b)$. ($\Rightarrow$) Suppose f is integrable on $[a, b]$, and let $\varepsilon > 0$ be arbitrary. Let P be a partition of $[a, b]$ such that

$$U(f, P) - L(f, P) < \varepsilon.$$

Without loss of generality, we may assume $c \in P$ (if not, replace P by $P \cup \{b\}$; refining can only decrease U and increase L , so we would still have the same inequality above). Therefore, we can split P into $P' := P \cap [a, c]$ which is a partition of $[a, c]$ and $P'' := P \cap [c, b]$, which is a partition of $[c, b]$. Since $P = P' \cup P''$, we have

$$L(f, P) = L(f, P') + L(f, P'') \quad \text{and} \quad U(f, P) = U(f, P') + U(f, P''),$$

which implies that (since there are less positive terms in the sum)

$$U(f, P') - L(f, P') \leq U(f, P) - L(f, P) < \varepsilon,$$

and

$$U(f, P'') - L(f, P'') \leq U(f, P) - L(f, P) < \varepsilon.$$

Therefore, by the ε -criterion for integrability (Theorem 6.6), it follows that f is integrable on $[a, c]$ and $[c, b]$.

($\Leftarrow$) Suppose f is integrable on $[a, c]$ and $[c, b]$, and let $\varepsilon > 0$ be arbitrary. By the ε -criterion for integrability, we may choose partitions P', P'' of $[a, c]$ and $[c, b]$ respectively such that

$$U(f|_{[a,c]}, P') - L(f|_{[a,c]}, P') < \frac{\varepsilon}{2} \quad \text{and} \quad U(f|_{[c,b]}, P'') - L(f|_{[c,b]}, P'') < \frac{\varepsilon}{2}.$$

Now, let $P = P' \cup P''$, which is a partition of $[a, b]$. Then

$$U(f, P) = U(f, P') + U(f, P'') \quad \text{and} \quad L(f, P) = L(f, P') + L(f, P''),$$

so

$$U(f, P) - L(f, P) = (U(f, P') - L(f, P')) + (U(f, P'') - L(f, P'')) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Thus f is integrable on $[a, b]$.

Finally, to prove the integral equality, note that for $\varepsilon > 0$ arbitrary we can write

$$\left| \int_a^b f(x) \, dx - \left(\int_a^c f(x) \, dx + \int_c^b f(x) \, dx \right) \right| < \varepsilon,$$

since

$$U(f, P) - \left(\int_a^c f + \int_c^b f \right) = \left(U(f, P') - \int_a^c f \right) + \left(U(f, P'') - \int_c^b f \right),$$

and both errors on the RHS can be made arbitrarily small (respectively for lower sums too). $\square$

Another extremely important and useful property of the integral is left as an exercise:

Exercise (Integral Triangle Inequality). Let $f : [a, b] \rightarrow \mathbb{R}$ be integrable. Show that $|f|$ is integrable and

$$\left| \int_a^b f(x) \, dx \right| \leq \int_a^b |f(x)| \, dx.$$

(Hint: you may wish to use the reverse triangle inequality.)

Remark. The equality

$$\sup_{s, t \in [x_{i-1}, x_i]} |f(s)^2 - f(t)^2| = \sup_{x \in [x_{i-1}, x_i]} f^2(x) - \inf_{x \in [x_{i-1}, x_i]} f^2(x),$$

which was used in the proof of item 3 of the above theorem (Theorem 6.8) relies on a standard, though non-trivial, property of the supremum and infimum. Specifically, for any bounded real-valued function g on a set I , the supremum of the distance between any two function values is equal to the difference between the supremum and the infimum of the function on that set:

$$\sup_{s, t \in I} |g(s) - g(t)| = \sup_{x \in I} g(x) - \inf_{x \in I} g(x)$$

To see why, note that for any $s, t \in I$, we have $g(s) \leq \sup_{x \in I} g(x)$ and $g(t) \geq \inf_{x \in I} g(x)$, meaning $|g(s) - g(t)| \leq \sup_{x \in I} g(x) - \inf_{x \in I} g(x)$. Taking the supremum over all s, t yields the $\leq$ inequality. The reverse inequality follows from the definition of supremum and infimum, as we can choose s and t such that $g(s)$ and $g(t)$ are arbitrarily close to the supremum and infimum, respectively. Applying this general property with $g = f^2$ and $I = [x_{i-1}, x_i]$ justifies the step.

Remark. The idea for the inequalities $\sup(f + g) \leq \sup f + \sup g$ and $\inf(f + g) \geq \inf f + \inf g$, used in the proof of item 1 in the above theorem, is as follows: when we take the supremum of $f + g$, we are taking the *pointwise*

supremum, whereas the expression $\sup f + \sup g$ means that we are searching for the supremum of f and g independently; so unless the supremum of f and g on the interval occur precisely at the same point, we must have $\sup(f + g) < \sup f + \sup g$. This idea can be applied to the respective infimum inequality too.

Properties 1 and 2 in the above Theorem shows that the integral is *linear* on the space of Darboux integrable functions; this is very useful in practice.

Theorem 6.9. *Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous. Then f is integrable.*

Proof. Let $\varepsilon > 0$ be arbitrary. By Heine's Theorem (Theorem 3.49), since f is continuous on a closed and bounded interval $[a, b]$, f is uniformly continuous on $[a, b]$ so there exists $\delta > 0$ such that for all $x, y \in [a, b]$ such that $|x - y| < \delta$, $|f(x) - f(y)| < \frac{\varepsilon}{b-a}$. Now, choose a partition $P = \{x_0, \dots, x_n\}$ such that for every $1 \leq i \leq n$, we have $x_i - x_{i-1} < \delta$. By the Extreme Value Theorem (Theorem 3.24), f attains its supremum and infimum on each sub-interval, so by uniform continuity, for each $1 \leq i \leq n$,

$$M_i - m_i < \frac{\varepsilon}{b-a}.$$

Hence, we have

$$U(f, P) - L(f, P) = \sum_{i=1}^n (M_i - m_i)(x_i - x_{i-1}) < \frac{\varepsilon}{b-a} \sum_{i=1}^n (x_i - x_{i-1}) = \varepsilon,$$

so by the ε -criterion for integrability (Theorem 6.6), f is integrable on $[a, b]$. $\square$

Remark. This proof relied on forcing the *vertical* error of the rectangle approximations to be within $\frac{\varepsilon}{b-a}$, and this is precisely why uniform continuity is required: achieving uniform control of the function on $[a, b]$ is exactly the property of uniform continuity.

To make the above remark more precise, let $P = \{x_0, \dots, x_n\}$ be an arbitrary partition of $[a, b]$. Then

$$U(f, P) - L(f, P) = \sum_{i=1}^n (M_i - m_i)(x_i - x_{i-1}).$$

To *control* the size of $U(f, P) - L(f, P)$ (i.e. the size of our error in trying to approximate the area with rectangles), this depends entirely on the differences $M_i - m_i$, which we can interpret as the variation in the range (the *oscillation*) of the function over the interval $[x_{i-1}, x_i]$. Restricting the

variation of f uniformly over arbitrary intervals is *precisely* what it means to say that f is uniformly continuous on that interval.

However, not all integrable functions have to be continuous; continuity is a sufficient condition, but not a necessary one. For example, consider the following exercise:

Exercise. Show that if a function $f : [a, b] \rightarrow \mathbb{R}$ is continuous except perhaps at finitely many points, then it is integrable.

The idea of the above exercise (*Hint!*) is that rather than controlling the vertical variation of f using continuity, if f only “jumps” (i.e. has a removable or jump discontinuity) at finitely many points, then at these points, despite the oscillation of the function ($M_i - m_i$) being fixed to *at least* the size of the discontinuity jump, we can also restrict the *horizontal* width of the interval around those points, in order to make our error $U(f, P) - L(f, P)$, and construct a partition that way.

The idea is best understood conceptually via Measure Theory (in particular, Lebesgue’s Criterion for Riemann Integrability), but the intuitive way via our rectangle approximations is that because these are finitely many “isolated” points (not in the topological sense, we use the word “isolated” purely in the colloquial sense here) at which the function is discontinuous, we can make our rectangle approximation around those points arbitrarily small (in the limit, they are 1-dimensional strips) which makes it clear that they do not contribute to the *area* underneath the curve, because 1-dimensional objects do not have area.

Again, this will make much more sense when you study Measure Theory, but the proof only requires the ideas discussed here, and the full details are left to the reader.

Exercise. Show that if $f : [a, b] \rightarrow \mathbb{R}$ is a continuous function such that $f(x) \geq 0$ for all $x \in [a, b]$, and $\int_a^b f(x) dx = 0$, then $f(x) = 0$ for all $x \in [a, b]$. Does this still hold if we drop the hypothesis that f is continuous?

6.4 The Fundamental Theorem of Calculus

By Theorem 6.8 (in particular, item 5), we the following function is well-defined. Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous; we then define $F : [a, b] \rightarrow \mathbb{R}$ by

$$F(x) := \int_a^x f(t) dt.$$

Theorem 6.10. *Let $f : [a, b] \rightarrow \mathbb{R}$ be integrable. Then the function $F : [a, b] \rightarrow \mathbb{R}$ defined by $F(x) = \int_a^x f(t) dt$ is Lipschitz continuous on $[a, b]$.*

Proof. Since f is integrable on $[a, b]$, it must be bounded. Therefore, we can let $M > 0$ be such that $|f(t)| \leq M$ for all $t \in [a, b]$. For any $x, y \in [a, b]$ with $x < y$, we examine the difference quotient:

$$\begin{aligned} |F(y) - F(x)| &= \left| \int_a^y f(t) \, dt - \int_a^x f(t) \, dt \right| \\ &= \left| \int_a^x f(t) \, dt + \int_x^y f(t) \, dt - \int_a^x f(t) \, dt \right| \quad (\text{Theorem 6.8 (5)}) \\ &= \left| \int_x^y f(t) \, dt \right|. \end{aligned}$$

By the Integral Triangle Inequality from an exercise and the rough estimate for integrals (Lemma 6.7), we get

$$\left| \int_x^y f(t) \, dt \right| \leq \int_x^y |f(t)| \, dt \leq M(y - x) = M|y - x|.$$

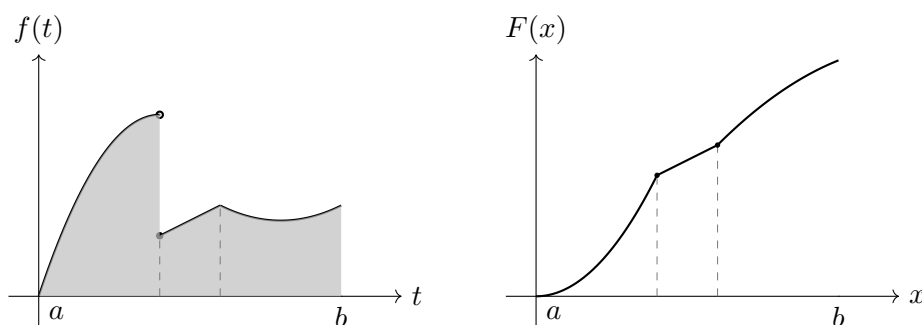
If $x = y$, the inequality holds because the integral becomes zero. Thus, $|F(y) - F(x)| \leq M|y - x|$ for all $x, y \in [a, b]$, which completes the proof. $\square$

Remark. This theorem reveals a fundamental geometric property of integration: the act of accumulating area is *smoothing*.

Because integrable functions are bounded (say, by a maximum magnitude M), the rate at which $F(x)$ accumulates area is strictly capped. The steepness of F can never exceed M . Therefore, no matter how "rough" or erratic the original function f is, its integral F is forced to be tightly constrained.

Consider what happens if the original function f has a severe jump discontinuity. The value of f teleports instantly from one height to another. However, area cannot teleport; it must be gathered progressively. The integral F responds to this jump not by breaking or jumping itself, but simply by abruptly changing its rate of accumulation. The jump discontinuity in f is smoothed out into a *sharp corner* (a kink) in F ; this is illustrated below.

This relates to why Lipschitz continuity is viewed as being *almost differentiable*. A Lipschitz function is globally constrained by a maximum slope, preventing vertical asymptotes or infinite cusps. It is smoothly continuous everywhere, save for isolated sharp corners corresponding to the jumps in f . If we slightly strengthen the condition and demand f be continuous instead of just integrable, those remaining corners are smoothed out entirely, and F becomes fully differentiable everywhere (which leads directly into the Fundamental Theorem of Calculus, which we will prove shortly).



Theorem 6.11 (Fundamental Theorem of Calculus, 1st Version). *Let $f : [a, b] \rightarrow \mathbb{R}$ be integrable and define $F : [a, b] \rightarrow \mathbb{R}$ by $F(x) = \int_a^x f(t) dt$. If f is continuous at $c \in (a, b)$, then F is differentiable at c and*

$$F'(c) = f(c).$$

Proof. Fix $h \neq 0$ such that $c + h \in [a, b]$ (this is possible since $a < c < b$). We have

$$\begin{aligned} \frac{F(c+h) - F(c)}{h} &= \frac{1}{h} \left(\int_a^{c+h} f(t) dt - \int_a^c f(t) dt \right) \\ &= \frac{1}{h} \int_c^{c+h} f(t) dt. \end{aligned}$$

Now, we rewrite the constant $f(c)$ as an integral over the same interval divided by h (since $f(c)$ does not depend on the dummy variable t), giving

$$f(c) = \frac{1}{h} \int_c^{c+h} f(c) dt.$$

It follows that, using standard properties of the integral from the previous section,

$$\begin{aligned} \left| \frac{F(c+h) - F(c)}{h} - f(c) \right| &= \left| \frac{1}{h} \int_c^{c+h} f(t) dt - \frac{1}{h} \int_c^{c+h} f(c) dt \right| \\ &= \left| \frac{1}{h} \int_c^{c+h} (f(t) - f(c)) dt \right| \\ &\leq \frac{1}{|h|} \left| \int_c^{c+h} |f(t) - f(c)| dt \right|. \end{aligned}$$

By definition of continuity at c , for any given $\varepsilon > 0$, there exists a $\delta > 0$ such that for all t :

$$|t - c| < \delta \implies |f(t) - f(c)| < \varepsilon$$

Now, choose h small enough such that $0 < |h| < \delta$ and the interval of width 2δ about c is contained in $[a, b]$. Then, for any t between c and $c + h$, $|t - c| \leq |h| < \delta$. Therefore, $|f(t) - f(c)| < \varepsilon$. Since this holds for any t between c and $c + h$, it follows that

$$\left| \frac{F(c+h) - F(c)}{h} - f(c) \right| < \frac{1}{|h|} \left| \int_c^{c+h} \varepsilon \, dt \right| = \frac{1}{|h|} \varepsilon |h| = \varepsilon,$$

so we have shown that

$$\lim_{h \rightarrow 0} \frac{F(c+h) - F(c)}{h} = f(c),$$

i.e. $F'(c) = f(c)$. □

Notice that we require f to be continuous in this case. If f were not continuous then the supremum and infimum of f on the interval between c and $c + h$ for a small fixed $h \neq 0$ would not necessarily converge to the same value as $h \rightarrow 0$, but these are the left and right limits of the quotient $\frac{F(c+h)-F(c)}{h}$, so it would mean the graph of F has a “kink” (or sharp corner) at c , meaning it is not differentiable there. So integration “smooths out discontinuities” into sharp corners.

Remark. The proof should make clear the intuition behind this theorem (the instantaneous rate of change of area F is just the height of the curve f at that point). If the geometry isn’t clear, one can think of this in terms of accumulation or kinematics. If $f(t)$ is the velocity of a particle over time, then the area underneath the curve represents the total net displacement of the particle, which is $F(t)$. The derivative $F'(t)$ simply asks: *at what rate is this accumulation (which in this case is displacement) changing at the point t ?* Well that is precisely the velocity, $f(t)$.

Corollary 6.12. *Let $f : [a, b] \rightarrow \mathbb{R}$ and $g : [a, b] \rightarrow \mathbb{R}$ be continuous. If g is differentiable on (a, b) and $f(x) = g'(x)$ for all $x \in (a, b)$, then*

$$\int_a^b f(x) \, dx = g(b) - g(a).$$

Proof. Define $F : [a, b] \rightarrow \mathbb{R}$ by

$$F(x) = \int_a^x f(t) \, dt.$$

By the 1st version of the Fundamental Theorem of Calculus (Theorem 6.11), since f is continuous F is differentiable and $F'(x) = f(x)$ for every $x \in (a, b)$. Now, consider the function $H : [a, b] \rightarrow \mathbb{R}$ defined by

$$H(x) = F(x) - g(x).$$

Since F and g are differentiable on (a, b) , so is H and $H'(x) = F'(x) - g'(x) = f(x) - f(x) = 0$ for all $x \in (a, b)$. Therefore, by an exercise in Chapter 4, H is constant on (a, b) , i.e. $H(x) = c$ for every $x \in (a, b)$, for some $c \in \mathbb{R}$. By continuity of F and g , it follows that H must be constant on $[a, b]$ (one can show this by contradiction).

Furthermore, $H(a) = F(a) - g(a) = -g(a)$, and $H(b) = F(b) - g(b)$, but $H(a) = H(b)$ since H is constant on $[a, b]$, from which it follows that

$$\int_a^b f(x) \, dx = F(b) = g(b) - g(a),$$

which completes the proof. $\square$

We now introduce the notion of “swapping the order of integration” which is often not rigorously justified or defined in standard calculus courses, and is just taken as a rule. However, the standard definition of the (Darboux) integral using partitions strictly requires the lower bound to be smaller than the upper bound, because a partition is defined as a strictly increasing sequence of points $a = x_0 < \dots < x_n = b$. Because of this, an integral evaluated “backwards” (from b to a rather than from a to b where $a < b$) has absolutely no meaning under the partition definition. We must define what happens when bounds are varied or swapped. To motivate our definition, we must first define a new integral function which varies the *lower limit* of integration, rather than the upper limit.

Definition 6.13. Let f be integrable on $[a, b]$. For $x \in [a, b]$, the function defined by varying the lower limit of integration is given by:

$$\int_x^b f(t) \, dt := \int_a^b f(t) \, dt - \int_a^x f(t) \, dt.$$

Theorem 6.14. If f is continuous on $[a, b]$, then the function $G : [a, b] \rightarrow \mathbb{R}$ defined by $G(x) := \int_x^b f(t) \, dt$ is differentiable on (a, b) and its derivative is:

$$\frac{dG(x)}{dx} = -f(x).$$

Proof. Using the definition above, we can write $G(x) = \int_a^b f(t) \, dt - \int_a^x f(t) \, dt$. Differentiating both sides with respect to x gives:

$$\frac{dG(x)}{dx} = \frac{d}{dx} \left(\int_a^b f(t) \, dt \right) - \frac{d}{dx} \left(\int_a^x f(t) \, dt \right).$$

Since $\int_a^b f(t) dt$ is a constant scalar value, its derivative is 0. By the 1st version of the Fundamental Theorem of Calculus (Theorem 6.11), $\frac{d}{dx} \left(\int_a^x f(t) dt \right) = f(x)$. Therefore,

$$\frac{dG(x)}{dx} = 0 - f(x) = -f(x).$$

□

Remark. The intuition here is that the derivative is a *rate of change*, a directional quantity.

When we vary the upper limit, the area under the curve is growing, so the derivative is positive. However, if we increase the lower limit, the area is decreasing as x increases. This is because we *shrink* the size of the interval $[x, b]$ when we increase x , whereas it is the exact opposite for $[a, x]$.

Definition 6.15. For $a < b$, we define the integral with swapped limits as:

$$\int_b^a f(t) dt := - \int_a^b f(t) dt.$$

Remark. There are a few ways to see why this definition is natural.

- The integral measures the total accumulated change of a function. Integrating backwards asks for the reverse change, which is algebraically exactly the negative: $F(a) - F(b) = -(F(b) - F(a))$.
- Geometrically, the integral sums the products of height and width ($f(x)\Delta x$). When integrating right-to-left, the step size Δx is strictly negative, flipping the sign of the total sum.
- More conceptually, the integral $\int_a^b f(x) dx$ evaluates a function along a directed path (ordered partition), not over a static set. When we apply the same definition in reverse (i.e. traversing the number line in reverse), the Δx terms become negative.

There is also a second version of the Fundamental Theorem of Calculus, which is a *stronger* statement than Corollary 6.12 because it weakens the required hypotheses. We no longer assume that the derivative g' is continuous, only that it exists and is integrable. Here, we bypass continuity by using the Mean Value Theorem to neatly link the net change of g directly to points $g'(x_i)$ that fit perfectly into the upper and lower partition sums.

Theorem 6.16 (Fundamental Theorem of Calculus, 2nd Version). *If $g : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b) and g' is integrable on $[a, b]$, then*

$$\int_a^b g'(x) \, dx = g(b) - g(a).$$

Proof. Let $P = \{t_0, \dots, t_n\}$ be any partition of $[a, b]$. By the Mean Value Theorem applied to g on each sub-interval $[t_{i-1}, t_i]$, there exists $x_i \in (t_{i-1}, t_i)$ such that

$$g(t_i) - g(t_{i-1}) = g'(x_i)(t_i - t_{i-1}).$$

Summing this from $i = 1$ to n produces a telescoping sum on the left:

$$g(b) - g(a) = \sum_{i=1}^n g'(x_i)(t_i - t_{i-1}).$$

Let m_i and M_i be the infimum and supremum of g' on $[t_{i-1}, t_i]$. Since $m_i \leq g'(x_i) \leq M_i$, we have:

$$m_i(t_i - t_{i-1}) \leq g(t_i) - g(t_{i-1}) \leq M_i(t_i - t_{i-1}).$$

Summing these inequalities from $i = 1$ to n yields:

$$L(g', P) \leq g(b) - g(a) \leq U(g', P).$$

Since g' is assumed to be integrable, for any $\varepsilon > 0$, we can choose a partition P such that $U(g', P) - L(g', P) < \varepsilon$. Because the fixed value $g(b) - g(a)$ sits perfectly between every lower and upper sum, it must be exactly equal to the integral $\int_a^b g'(x) \, dx$. $\square$

Remark. There is a subtle technicality in evaluating $\int_a^b g'(x) \, dx$. If g is only differentiable on the open interval (a, b) (which is how we defined the derivative, we did not define one-sided derivatives), then g' is not defined at the endpoints a and b . However, the partition definition of the integral requires the integrand to be defined on the entire closed interval $[a, b]$.

To resolve this rigorously, we can implicitly extend g' to $[a, b]$ by assigning arbitrary finite values to the endpoints (e.g., $g'(a) := 0$ and $g'(b) := 0$). Because changing a function's value at a finite number of points does not affect its integrability or the value of its integral (one should check this), this slight abuse of notation is perfectly safe. Alternatively, if we assume g has one-sided derivatives at the boundaries on $[a, b]$, g' is natively defined on $[a, b]$.

Remark. While the 1st version of the FTC (Theorem 6.11) shows that the rate of change of an accumulated area is the function itself, this 2nd version (Theorem 6.16) provides another perspective: it tells us that if you accumulate all the local rates of change of a function ($g'(x) dx$) across an interval, they sum to the total net change of that function ($g(b) - g(a)$), which is what the telescoping sum showed algebraically, and the MVT allowed us to perform this for an arbitrary partition, thereby establishing the theorem. As an example, consider the case when $g(b) = g(a)$; then the integral of g' over $[a, b]$ evaluates to zero, which matches our intuition that when you add up all the local rates of change, you get the net change, which is zero in this case. This is what the telescoping sum and the Mean Value Theorem formalised.

With the theorems established in this chapter, some familiar and useful properties of the integral can now be proven. We leave them as exercises for the reader.

Exercise (Integration by Parts). Let $f : [a, b] \rightarrow \mathbb{R}$ and $g : [a, b] \rightarrow \mathbb{R}$ be continuously differentiable. Prove that

$$\int_a^b f(x)g'(x) dx = f(b)g(b) - f(a)g(a) - \int_a^b f'(x)g(x) dx.$$

Exercise (Change of Variables). Let $\varphi : [a, b] \rightarrow \mathbb{R}$ be continuously differentiable, and let $f : \mathbb{R} \rightarrow \mathbb{R}$ be continuous. Prove that

$$\int_a^b f(\varphi(t))\varphi'(t) dt = \int_{\varphi(a)}^{\varphi(b)} f(u) du.$$

6.5 The Logarithm and Exponential Functions

Definition 6.17. Let $L : (0, \infty) \rightarrow \mathbb{R}$ be the function defined by

$$L(x) := \int_1^x \frac{1}{t} dt, \quad x > 0.$$

Remark. For $x > 1$, $L(x)$ represents the strictly positive area under the curve $1/t$ from 1 to x . When $0 < x < 1$, we are integrating “backwards” from a larger lower limit (1) to a smaller upper limit (x). By our established convention for swapping limits, $\int_1^x \frac{1}{t} dt = -\int_x^1 \frac{1}{t} dt$, meaning $L(x)$ is negative the area under $1/t$ on $[x, 1]$.

Theorem 6.18. The function L is differentiable and strictly increasing on $(0, \infty)$.

Proof. The integrand $f(t) = 1/t$ is continuous on $(0, \infty)$. By the Fundamental Theorem of Calculus (1st version, Theorem 6.11), the integral function L is differentiable for all $x > 0$, and its derivative is

$$L'(x) = \frac{1}{x}.$$

Since $x > 0$, we have $L'(x) > 0$ for all $x \in (0, \infty)$. A strictly positive derivative implies that L is strictly increasing by Theorem 4.10. $\square$

Theorem 6.19 (Algebraic Properties of L). *For all $x, y \in (0, \infty)$ and $n \in \mathbb{Z}$:*

1. $L(xy) = L(x) + L(y)$
2. $L(1) = 0$
3. $L(1/x) = -L(x)$
4. $L(x^n) = nL(x)$

Proof. To prove (1), we use the domain additivity of the integral:

$$L(xy) = \int_1^{xy} \frac{1}{t} dt = \int_1^x \frac{1}{t} dt + \int_x^{xy} \frac{1}{t} dt = L(x) + \int_x^{xy} \frac{1}{t} dt.$$

For the second integral, we apply the change of variables theorem. Let $t = xu$, which implies $dt = x du$. The integration bounds change from $t = x \implies u = 1$ to $t = xy \implies u = y$. This yields:

$$\int_x^{xy} \frac{1}{t} dt = \int_1^y \frac{1}{xu} x du = \int_1^y \frac{1}{u} du = L(y).$$

Substituting this back gives $L(xy) = L(x) + L(y)$.

To prove (2), $L(1) = \int_1^1 \frac{1}{t} dt = 0$. Alternatively, using (1), $L(1) = L(1 \cdot 1) = L(1) + L(1) \implies L(1) = 0$.

To prove (3), we use (1) and (2): $0 = L(1) = L(x \cdot \frac{1}{x}) = L(x) + L(1/x) \implies L(1/x) = -L(x)$.

To prove (4), for positive n , repeated application of (1) gives $L(x^n) = L(x \cdot x^{n-1}) = L(x) + (n-1)L(x) = nL(x)$. For negative n , let $m = -n$ where $m > 0$. Using (3), $L(x^{-m}) = L(1/x^m) = -L(x^m) = -mL(x) = nL(x)$. $\square$

Theorem 6.20. *The function L is a bijection from $(0, \infty)$ to $\mathbb{R}$, with limits:*

$$\lim_{x \rightarrow \infty} L(x) = \infty \quad \text{and} \quad \lim_{x \rightarrow 0^+} L(x) = -\infty.$$

Proof. Fix a real number $a > 1$. Because L is strictly increasing, $L(a) > L(1) = 0$. Consider the sequence $x_n = a^n$ as $n \rightarrow \infty$. By our algebraic properties, $L(a^n) = nL(a)$. Since $L(a) > 0$, we have $\lim_{n \rightarrow \infty} nL(a) = \infty$. Because L is strictly monotonic, it must be that $\lim_{x \rightarrow \infty} L(x) = \infty$.

For the limit at zero, let $u = 1/x$. As $x \rightarrow 0^+$, $u \rightarrow \infty$. Using property (3),

$$\lim_{x \rightarrow 0^+} L(x) = \lim_{u \rightarrow \infty} L(1/u) = \lim_{u \rightarrow \infty} -L(u) = -\infty.$$

Since L is strictly increasing, it is injective. Because L is continuous on $(0, \infty)$ with limits $-\infty$ and ∞ at the boundaries of its domain, the Intermediate Value Theorem (Theorem 3.25) guarantees that L takes on every real value. Thus, L is surjective onto $\mathbb{R}$, making it a bijection. $\square$

Definition 6.21. Because $L: (0, \infty) \rightarrow \mathbb{R}$ is a continuous bijection, it has a well-defined inverse. We define the *exponential function* $E: \mathbb{R} \rightarrow (0, \infty)$ to be the inverse of L .

Remark. From this point forward, we use the standard notation for the functions L and E , denoting $L(x)$ by $\log(x)$ and $E(x)$ by $\exp(x)$.

Theorem 6.22. *The function E is differentiable on $\mathbb{R}$, strictly increasing, and satisfies*

$$E'(x) = E(x) \quad \text{with} \quad E(0) = 1.$$

Proof. Because $L(1) = 0$ and E is the inverse of L , we immediately have $E(0) = 1$.

To find the derivative, we use the Inverse Function Theorem (Theorem 4.30). Since L is continuously differentiable and $L'(y) = 1/y \neq 0$ for all $y \in (0, \infty)$, its inverse E is differentiable everywhere on $\mathbb{R}$. Evaluating the derivative gives:

$$E'(x) = \frac{1}{L'(E(x))}.$$

Since $L'(y) = 1/y$, substituting $y = E(x)$ yields $L'(E(x)) = 1/E(x)$. Therefore:

$$E'(x) = \frac{1}{1/E(x)} = E(x).$$

$\square$

Theorem 6.23 (Algebraic Properties of E). *For all $x, y \in \mathbb{R}$, $E(x+y) = E(x)E(y)$.*

Proof. Let $u = E(x)$ and $v = E(y)$, so $x = L(u)$ and $y = L(v)$. Using the algebraic property of L , we have:

$$x + y = L(u) + L(v) = L(uv).$$

Applying E to both sides yields:

$$E(x + y) = E(L(uv)) = uv = E(x)E(y).$$

□

Exercise. Prove the inequality $\log(x) \leq x - 1$ for all $x > 0$. (*Hint: Compare the constant function $f(t) = 1$ and $g(t) = 1/t$ on the interval $[1, x]$ for $x \geq 1$, and adapt the argument for $x \in (0, 1)$.)*

Exercise. Evaluate the limit:

$$\lim_{h \rightarrow 0} \frac{\log(1 + h)}{h}.$$

Exercise. Using the inequality established in the first exercise and the properties of the inverse function, deduce that $\exp(x) \geq 1 + x$ for all $x \in \mathbb{R}$.

Chapter 7

Taylor's Theorem

This chapter is motivated by the question: how can we approximate a function with other, simpler functions, locally? In particular, the approximation of functions by *polynomials*. Polynomials are the natural first choice as they are, in some way, the simplest type of function on $\mathbb{R}$ and have very nice properties.

When discussing Taylor's theorem, we are fundamentally talking about *polynomials*, not power series. A series that defines a function is not just a single series; for every x , you have a different limit. Establishing whether such a series converges, is continuous, or is differentiable requires significant machinery in general, but for functions with a finite Taylor polynomial, these properties are automatic because of the algebraic properties of polynomials.

7.1 Taylor Polynomials

Definition 7.1 (Taylor Polynomial). Let f be n times differentiable at a point a . The n -th order *Taylor polynomial* of f centred at a is:

$$P_{n,a}(x) := \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k.$$

Remark. We can construct a polynomial with prescribed derivatives at a point a by writing it in the “centred” form $\sum_{k=0}^n c_k(x-a)^k$. The derivatives of this polynomial at a directly determine the coefficients $c_k = f^{(k)}(a)/k!$.

Geometrically, by forcing the polynomial and the function to share the exact same value, slope, concavity, and higher-order rates of change at a single point a , we are essentially locking their *local geometries* together. The higher the order n , the more rigid this lock becomes, widening the neighbourhood

where the polynomial stays glued to the function before inevitably peeling away.

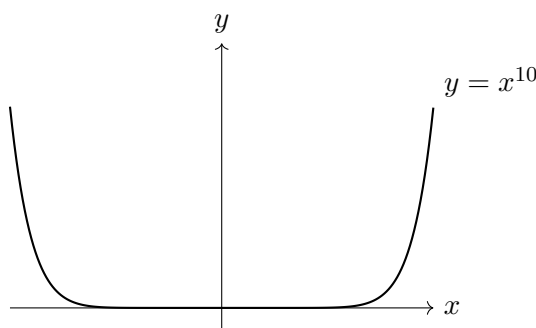
It may seem slightly unnatural to construct a polynomial whose derivatives up to order n to act as a local approximation to a given function, but to see why this works it is useful to consider the *remainder* function for a given function and its Taylor polynomial:

Definition 7.2 (The Remainder). The *remainder* of the Taylor approximation of order n is the difference between the function and its Taylor polynomial:

$$R_{n,a}(x) := f(x) - P_{n,a}(x).$$

The remainder $R_{n,a}(x)$ represents the exact “error” in our approximation. By construction, the function and the polynomial share their first n derivatives at a . Therefore, the remainder function has its first n derivatives vanishing at a .

Geometrically, this corresponds to extreme “flatness” at a (similar to how $y = x^n$ is extremely flat near the origin, for n large - see below). Because the error is practically zero in this flat region, the polynomial “hugs” the function tightly.



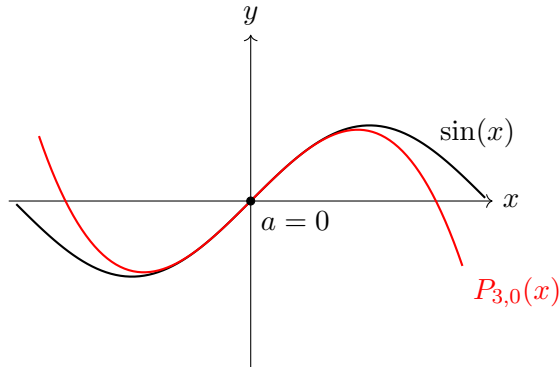
Definition 7.3 (Little- o Notation). Let $k \in \mathbb{N}$. We say that $f \in o(|x - a|^k)$ if

$$\lim_{x \rightarrow a} \frac{f(x)}{|x - a|^k} = 0.$$

Conceptually, the extreme flatness of the remainder is demonstrated in the above Little- o notation. As we will show below, $R_{n,a} \in o(|x - a|^n)$.

This limit explicitly states that as x approaches a , the error $R_{n,a}(x)$ shrinks to zero strictly *faster* than the n -th power of the distance from the

centre. The error is strictly dominated by the polynomial $|x - a|^n$. This is the formal mathematical translation of the polynomial “hugging” the function: the approximation error vanishes so rapidly that it is negligible even when compared to the microscopic scale of $|x - a|^n$.



Theorem 7.4 (Taylor's Theorem with Lagrange Remainder). *Let f be $n + 1$ times differentiable on an open interval containing a and x . Then there exists some θ strictly between a and x such that:*

$$R_{n,a}(x) = \frac{f^{(n+1)}(\theta)}{(n+1)!} (x - a)^{n+1}.$$

Proof. Fix $x \neq a$. Consider the function $F(t)$ defined on the closed interval between a and x :

$$F(t) := f(x) - \sum_{k=0}^n \frac{f^{(k)}(t)}{k!} (x - t)^k.$$

Notice that $F(x) = 0$ and $F(a) = R_{n,a}(x)$. Differentiating $F(t)$ with respect to t causes the sum to telescope, leaving only the final term:

$$F'(t) = -\frac{f^{(n+1)}(t)}{n!} (x - t)^n.$$

Now, define an auxiliary function $G(t)$ to apply Rolle's Theorem:

$$G(t) := F(t) - \left(\frac{x - t}{x - a} \right)^{n+1} F(a).$$

We have $G(x) = 0$ and $G(a) = F(a) - F(a) = 0$. Since G is continuous on the closed interval between a and x and differentiable on the open interval, Rolle's Theorem guarantees a point θ strictly between a and x where $G'(\theta) = 0$.

Differentiating $G(t)$ gives:

$$G'(t) = F'(t) + (n+1) \frac{(x-t)^n}{(x-a)^{n+1}} F(a).$$

Setting $G'(\theta) = 0$ and substituting $F'(\theta)$ yields:

$$\frac{f^{(n+1)}(\theta)}{n!} (x-\theta)^n = (n+1) \frac{(x-\theta)^n}{(x-a)^{n+1}} R_{n,a}(x).$$

Cancelling $(x-\theta)^n$ (since $\theta \neq x$) and rearranging gives the desired form for $R_{n,a}(x)$. $\square$

Remark. The Lagrange form is incredibly convenient for rough approximations. If the $(n+1)$ -th derivative is bounded, the remainder is immediately bounded. However, it has significant drawbacks: the value θ depends on x in a way that is not explicitly understood and is generally not continuous. Therefore, if we want to differentiate the remainder or evaluate it precisely, the Lagrange form is inadequate.

Theorem 7.5 (Taylor's Theorem with Integral Remainder). *Let f be $n+1$ times continuously differentiable on an open interval containing a . For any x in this interval,*

$$R_{n,a}(x) = \int_a^x \frac{f^{(n+1)}(t)}{n!} (x-t)^n dt.$$

Proof. We proceed by mathematical induction on n . For $n=0$, the theorem states $f(x) - f(a) = \int_a^x f'(t) dt$, which is exactly the Fundamental Theorem of Calculus.

Assume the formula holds for some k . We apply integration by parts to the integral remainder for k . Let $u = f^{(k+1)}(t)$ and $v = -\frac{(x-t)^{k+1}}{k+1}$. Then $du = f^{(k+2)}(t) dt$ and $dv = (x-t)^k dt$.

$$\int_a^x \frac{f^{(k+1)}(t)}{k!} (x-t)^k dt = \left[-f^{(k+1)}(t) \frac{(x-t)^{k+1}}{(k+1)!} \right]_a^x + \int_a^x \frac{f^{(k+2)}(t)}{(k+1)!} (x-t)^{k+1} dt.$$

Evaluating the boundary term at x yields 0. Evaluating at a yields $\frac{f^{(k+1)}(a)}{(k+1)!} (x-a)^{k+1}$. Thus,

$$R_{k,a}(x) = \frac{f^{(k+1)}(a)}{(k+1)!} (x-a)^{k+1} + \int_a^x \frac{f^{(k+2)}(t)}{(k+1)!} (x-t)^{k+1} dt.$$

Since $R_{k,a}(x) - \frac{f^{(k+1)}(a)}{(k+1)!} (x-a)^{k+1} = R_{k+1,a}(x)$, the result holds for $n = k+1$. $\square$

Remark. The integral remainder provides an *explicit* formula for the remainder that works for every x . Because the dependence on x is explicitly captured inside the integral (and not hidden in an arbitrary θ), we can use the Fundamental Theorem of Calculus to differentiate the remainder, bound it precisely, and evaluate it anywhere.

7.2 Taylor Series

We have seen that Taylor polynomials are fundamentally local approximations, however there is a special class of functions, called *analytic functions*, whose Taylor polynomials converge to the original function on an interval as $n \rightarrow \infty$. This is quite surprising, as it says that these functions can be recovered *globally* in the form of an “infinite polynomial” from only their derivatives *at a single point*, i.e. from only local information. It may be counter-intuitive that purely local information - derivatives evaluated at a single microscopic point a - can perfectly reconstruct a function's global shape. If you draw an arbitrary curve by hand, knowing its slope at $x = 0$ tells you absolutely nothing about its height at $x = 10$.

However, polynomials are extremely rigid. A polynomial of degree n cannot suddenly change its asymptotic behaviour; its behaviour on the real line is entirely determined by its finite coefficients. In other words, its global behaviour is completely determined by a finite set of points. Because an analytic function is essentially an “infinite polynomial”, it inherits this “rigidity”. If you know a real analytic function and all of its derivatives at a single point, you possess all the information you need to rebuild the function. There is zero room for the function to do anything unexpected globally because its global behaviour is a rigid algebraic consequence of its local derivatives.

Indeed, these *are* the special functions defined by this property; there are many examples, as we will see below, of functions that are in C^∞ (infinitely differentiable) but are not analytic.

Definition 7.6 (Real Analytic Function). We say that a function $f \in C^\infty$ is *real analytic* at a point a if there exists $R > 0$ such that for every $x \in (a - R, a + R)$ we have $\lim_{n \rightarrow \infty} P_{n,a}(x) = f(x)$. In this case, we write

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n.$$

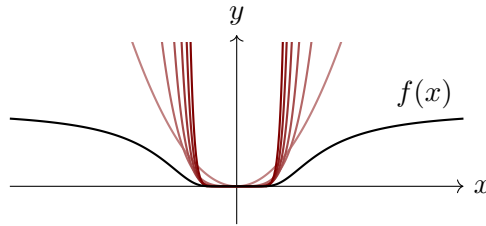
The maximal $R > 0$ for which this property holds is called the *radius of convergence* of the Taylor series.

Intuitively, this means that the function is “rigid” enough that it can

be approached globally (i.e. on an interval) by polynomial functions. This is precisely why the derivatives of the function at one point generate can build the whole function; as we saw in the discussion at the start of this chapter, polynomials are extremely special functions that can be re-centred to any point while remaining invariant, and so the local information of an analytic function inherits this property, such that local information, like derivatives at a point a , can still recover the function globally. Where we “centre” our approximation does not matter, as long as it is inside the interval of convergence; in the limit, the approximation recovers the whole function.

Example 7.7. Not all functions in C^∞ are (real) analytic. The idea is to construct a C^∞ function $f : \mathbb{R} \rightarrow \mathbb{R}$ whose graph, though not horizontal, is “very flat” near 0 in the sense that all of its derivatives vanish at 0. A classic example is the bump function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} \exp(-1/x^2), & x \neq 0, \\ 0, & x = 0. \end{cases}$$



This function is infinitely differentiable everywhere (which one should verify), but $f^{(n)}(0) = 0$ for all $n \in \mathbb{N}$. Therefore, its Taylor polynomial at $a = 0$ is the identically zero function, $P_{n,0}(x) = 0$ for all n .

Intuitively, because this function converges to zero as $x \rightarrow 0$ faster than any polynomial, each power x^n is “too weak” to match the exponential decay of f near 0, so has nothing to contribute to the Taylor polynomial at 0. Precisely, it is because $f \in o(|x|^n)$ for every $n \in \mathbb{N}$; the function itself vanishes faster than any polynomial term x^n can measure, so its best polynomial approximation at the origin is simply the zero polynomial, $P_{n,0}(x) = 0$. The polynomials literally cannot “see” the function’s global growth because it is locally too flat.

Based on the Lagrange form of the remainder in Taylor’s Theorem, we get a useful sufficient condition for the convergence of Taylor series to the original function.

Theorem 7.8. *Let $f \in C^\infty$ be defined on $(a - \rho, a + \rho)$ for some $\rho > 0$. If there exists $C, R > 0$ and $N \in \mathbb{N}$ such that for all $x \in (a - \rho, a + \rho)$*

and all $n \geq N$,

$$|f^{(n+1)}(x)| \leq C \frac{(n+1)!}{R^{n+1}},$$

then the Taylor polynomial $P_{n,a}(x)$ of f converges to $f(x)$ as $n \rightarrow \infty$ for every x satisfying

$$|x - a| < \min\{\rho, R\}.$$

In particular, f is analytic on $|x - a| < \min\{\rho, R\}$.

Proof. Using the Lagrange form of the remainder (Theorem 7.4), we have, for x such that $|x - a| < \min\{\rho, R\}$,

$$R_{n,a}(x) = \frac{f^{(n+1)}(\theta)}{(n+1)!} (x - a)^{n+1},$$

for some θ between a and x . Since $|x - a| < \min\{\rho, R\}$, and θ is between a and x , we also have $|\theta - a| < \min\{\rho, R\}$, so for $n \geq N$, the derivative bound applies at θ also:

$$|R_{n,a}(x)| = \left| \frac{f^{(n+1)}(\theta)}{(n+1)!} (x - a)^{n+1} \right| \leq \frac{C \frac{(n+1)!}{R^{n+1}}}{(n+1)!} |x - a|^{n+1} = C \left(\frac{|x - a|}{R} \right)^{n+1}.$$

Since, in particular, $|x - a| < R$ we know $\frac{|x-a|}{R} < 1$. Therefore, as $n \rightarrow \infty$, the remainder $|R_{n,a}(x)| \rightarrow 0$. This proves that the sequence of Taylor polynomials converges pointwise to $f(x)$. $\square$

Remark. The expression $\frac{(n+1)!}{R^{n+1}}$ is highly advantageous when bounding error because factorials grow significantly faster than exponentials. This affords us a massive amount of “room” to bound the derivatives of functions like $\sin(x)$, $\cos(x)$, and $\exp(x)$, which are indeed analytic functions.

The distinction between infinitely differentiable and analytic is better understood via Complex Analysis (the analysis of $\mathbb{C}$).

In $\mathbb{R}$, being infinitely differentiable is a surprisingly weak geometric condition. Because you can only approach a point from two directions (left or right) on the 1D real line, it is easy to “cheat” and smoothly glue a completely flat line to a curved function (like our bump function) without breaking differentiability.

However, in the complex plane, a function takes inputs from a 2D surface. To be complex-differentiable at a point, the limit of the difference quotient must perfectly agree no matter which of the infinite possible directions you approach from. This geometric constraint is so incredibly rigid that if a complex function is differentiable just *once* on an open set, it is automatically infinitely differentiable and perfectly equal to its Taylor series everywhere in that domain. In the complex world, the “bump function” is impossible.

Exercise. Prove that $\exp(x)$ is real analytic.

Chapter 8

Convergence of Functions

In the last chapter, we implicitly introduced the notion of convergence of *functions to other functions*, during our discussion of Taylor series, which are a very particular type of function sequence (polynomials). This was a concept called *pointwise convergence*, as we will define properly below, and the results we proved were *pointwise* results. This motivates a more general study of convergence of functions, which we will discuss in this chapter.

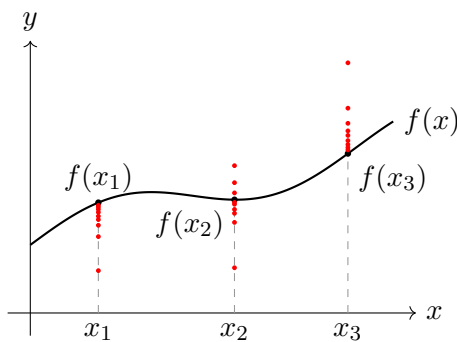
8.1 Pointwise Convergence

Definition 8.1 (Pointwise Convergence). A sequence of functions $f_n : D \rightarrow \mathbb{R}$ converges *pointwise* to a limit function $f : D \rightarrow \mathbb{R}$ if, for every $x \in D$, the sequence of real numbers $(f_n(x))_{n=1}^{\infty}$ converges to $f(x)$. In such a case, we may write $f_n \rightarrow f$ pointwise.

Remark. In quantifiers, this definition translates to:

$$\forall x \in D, \forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall n \geq N, |f_n(x) - f(x)| < \varepsilon.$$

Notice that N may depend on *both* the point x *and* ε , i.e. we may write $N = N(x, \varepsilon)$. It is helpful to keep the following picture in mind:



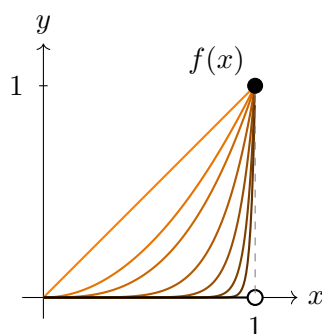
As illustrated above, pointwise convergence means that for every x in the domain, we have a different limit $\lim_{n \rightarrow \infty} f_n(x)$ that converges to some value $f(x)$ (where f is the limit function). For example, if the domain of each f_n is the open interval $(0, 1)$, then we have uncountably many limits to consider.

Pointwise convergence is a weak notion of function convergence because it evaluates the limit completely independently for each isolated coordinate x . It treats a function sequence merely as an unlinked collection of individual point-sequences rather than a cohesive geometric object.

To make these issues concrete, we examine how pointwise convergence can destroy both continuity and integrability.

Example 8.2. Consider the sequence $f_n(x) = x^n$ on the interval $[0, 1]$. Each function in the sequence is continuous. Pointwise evaluation yields the discontinuous step function (by standard properties of the limit from Chapter 2):

$$f(x) = \lim_{n \rightarrow \infty} f_n(x) = \begin{cases} 0, & x \in [0, 1), \\ 1, & x = 1. \end{cases}$$



To force the function values to drop below an error threshold $\varepsilon > 0$ for $x \in (0, 1)$, we require $x^n < \varepsilon$, which dictates our choice of N :

$$N = \left\lceil \frac{\log \varepsilon}{\log x} \right\rceil.$$

Notice that as x approaches 1 from the left, $\log x \rightarrow 0$, forcing $N \rightarrow \infty$. The convergence becomes arbitrarily slow near the boundary.

This shows precisely why the pointwise convergence does not guarantee continuity in the limit in this case: no matter how large we pick an index N , we can always find a point x that “escapes” our attempt to confine the sequence within ε of f . For instance, choosing $x_0 = (1/2)^{1/N} \in [0, 1)$ guarantees that $f_N(x_0) = 1/2$. If we set $\varepsilon = 1/4$, the candidate N fails because the function value $1/2$ has escaped the error bound. One can also see that $\lim_{n \rightarrow \infty} \sup_{x \in [0, 1]} |f_n(x) - f(x)| = 1 \neq 0$, which is the precise reason

pointwise convergence is too weak here (we will explain what this limit means later; intuitively, this shows that the greatest error between the limit function and the sequence functions tends to 1 as $n \rightarrow \infty$, demonstrating this visual notion of the limit “tearing away” from the sequence functions).

Example 8.3. The failure of pointwise limits can be even more severe, breaking Darboux integrability (which is even weaker than continuity!). Consider the sequence of functions $f_n : [0, 1] \rightarrow \mathbb{R}$ defined by, for $n \in \mathbb{N}$,

$$f_n(x) = \begin{cases} 1, & n!x \in \mathbb{Z}, \\ 0, & \text{otherwise.} \end{cases}$$

For any fixed n , $f_n(x)$ evaluates to 1 only at a finite set of rational coordinates. Having only finitely many discontinuities, every f_n is strictly Riemann integrable with $\int_0^1 f_n(x) dx = 0$.

However, as $n \rightarrow \infty$, the factorial scaling $n!$ expands to eventually divide evenly into the denominator of *every* rational number. The pointwise limit is the Dirichlet function:

$$f(x) = \lim_{n \rightarrow \infty} f_n(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

Because the rational numbers are dense in $\mathbb{R}$, this limit function is nowhere continuous and not Darboux integrable.

8.2 Uniform Convergence

8.2.1 Motivation

In order to motivate a stronger (though, not too strong lest the preservation of certain properties like continuity of the sequence become too trivial) definition of function convergence, we must examine what went wrong in the above examples, where the sequence of functions merely converged *pointwise*.

Because the choice of the threshold N depends on x , it permits different points across the domain to converge at “different rates”. For a given (sufficiently large) n , some points may be arbitrarily close to $f(x)$, while other points may not have even started to move meaningfully towards $f(x)$ (if their threshold N^* is much larger than n). In other words, no matter how large n is, we can still find some x where $f_n(x)$ differs a lot from $f(x)$. Thus, if we are given pointwise convergence, there is no guarantee that for very large n , f_n will “look like” f , since there might be some points for which f_n has not started to move towards f .

This lack of a globally enforced pace allows the spatial cohesion of the functions to break apart in the limit (one can think of this as “tearing” the sequence functions apart). Hence, what we need is for f_n to converge to f at

the same pace. This is known as uniform convergence. Before we formally define this concept, we first make an essential definition.

Definition 8.4 (Supremum Norm). Let $f : D \rightarrow \mathbb{R}$ be a bounded function. We define the *supremum norm* of f , written $\|f\|_\infty$, as the following non-negative real number:

$$\|f\|_\infty := \sup\{|f(x)| : x \in D\}.$$

Lemma 8.5. Let $f, g : D \rightarrow \mathbb{R}$ be bounded functions on a domain D , and let $c \in \mathbb{R}$. Then the supremum norm satisfies the following properties:

1. For any $x \in D$, $|f(x)| \leq \|f\|_\infty$.
2. $\|cf\|_\infty = |c|\|f\|_\infty$.
3. Triangle inequality: $\|f + g\|_\infty \leq \|f\|_\infty + \|g\|_\infty$.
4. $\|fg\|_\infty \leq \|f\|_\infty \|g\|_\infty$.

Proof. We prove each property in turn.

(1) By definition, $\|f\|_\infty = \sup_{x \in D} |f(x)|$. Since the supremum of a set is an upper bound for that set, it immediately follows that $|f(x)| \leq \|f\|_\infty$ for all $x \in D$.

(2) Using the definition of the norm and the properties of suprema:

$$\|cf\|_\infty = \sup_{x \in D} |cf(x)| = \sup_{x \in D} (|c||f(x)|).$$

Since $|c| \geq 0$ is a constant, we can bring it out of the supremum:

$$\sup_{x \in D} (|c||f(x)|) = |c| \sup_{x \in D} |f(x)| = |c|\|f\|_\infty.$$

(3) Let $x \in D$ be an arbitrary point. By the standard triangle inequality for real numbers and (1), we have

$$|(f + g)(x)| = |f(x) + g(x)| \leq |f(x)| + |g(x)| \leq \|f\|_\infty + \|g\|_\infty.$$

This shows that the constant value $\|f\|_\infty + \|g\|_\infty$ is an upper bound for the set $\{|(f + g)(x)| : x \in D\}$. Because the supremum is the *least* upper bound, we must have

$$\sup_{x \in D} |(f + g)(x)| \leq \|f\|_\infty + \|g\|_\infty \implies \|f + g\|_\infty \leq \|f\|_\infty + \|g\|_\infty.$$

(4) Let $x \in D$ be an arbitrary point. Using the multiplicative property of absolute values and (1), we get

$$|(fg)(x)| = |f(x)g(x)| = |f(x)||g(x)|.$$

Since absolute values are non-negative, and we know $|f(x)| \leq \|f\|_\infty$ and $|g(x)| \leq \|g\|_\infty$, we can multiply these inequalities to obtain

$$|(fg)(x)| \leq \|f\|_\infty \|g\|_\infty.$$

This shows that $(\|f\|_\infty \|g\|_\infty)$ is an upper bound for the set $\{|(fg)(x)| : x \in D\}$. Taking the supremum over all $x \in D$ yields:

$$\|fg\|_\infty \leq \|f\|_\infty \|g\|_\infty.$$

This completes the proof. $\square$

Definition 8.6 (Uniform Convergence). A sequence of functions $f_n: D \rightarrow \mathbb{R}$ converges *uniformly* to f on D if, for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$ and for all $x \in D$,

$$|f_n(x) - f(x)| < \varepsilon.$$

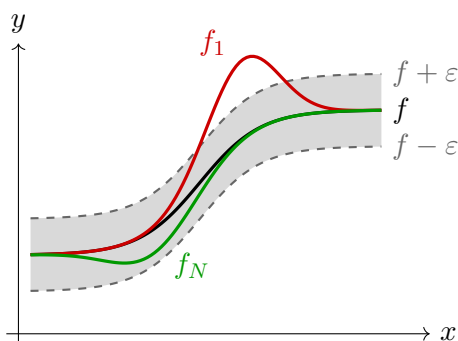
In such a case, we write $f_n \rightrightarrows f$, or simply $f_n \rightarrow f$ uniformly.

Remark. The only difference with pointwise convergence is the order of quantifiers: for uniform convergence, the threshold $N = N(\varepsilon)$ works simultaneously for all $x \in D$. In particular, uniform convergence implies pointwise convergence, but not the other way around.

In terms of intuition, the Wikipedia page on this concept puts it nicely (available at https://en.wikipedia.org/wiki/Uniform_convergence): roughly speaking, $f_n \rightarrow f$ uniformly if the functions f_n uniformly approximate the function f over the whole domain, meaning that all but finitely many of the functions of the sequence lie in a uniform error bar of the original function. Graphically this means that, given any thin band around the graph of f , the graphs of all but finitely many of the functions f_n lie within that thin band.

This is in contrast to pointwise convergence, in which all but finitely many of the functions lie in a thin band at each point, but the finite set of functions which must be excluded in order for that to be the case varies from point to point.

The following diagram should make these ideas clear.



Exercise. Prove that uniform convergence implies pointwise convergence.

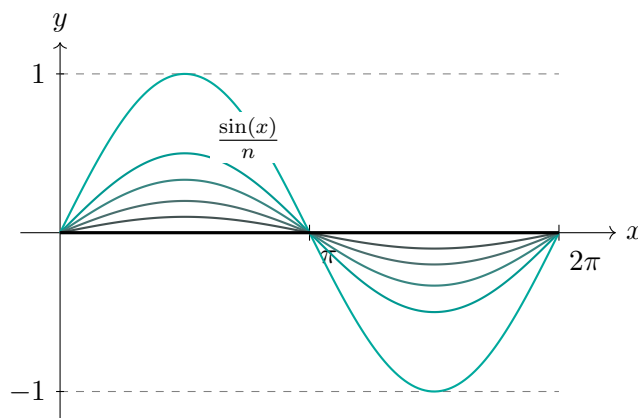
As hinted at in the discussion of pointwise convergence, we may restate the definition of uniform convergence in terms of the supremum norm (which is perhaps a more intuitive way of thinking about the concept):

Lemma 8.7. Let $f_n : D \rightarrow \mathbb{R}$ be a sequence of functions on D . Then

$$f_n \rightrightarrows f \iff \lim_{n \rightarrow \infty} \|f_n - f\|_{\infty} = 0.$$

Proof. Exercise. □

In other words, the maximum error between the function f_n and f goes to zero *uniformly* in the domain D (i.e. independent of where the function is evaluated in the domain), which intuitively means the functions f_n are being “squeezed” together towards the limit function across the whole domain. In this sense, properties like continuity of the limit functions should be preserved in the limit as no “tearing” can take place, unlike in pointwise convergence (we will prove these facts in the next subsection). The idea is demonstrated below with the functions $f : [0, 2\pi] \rightarrow \mathbb{R}$ defined by $f_n(x) = \frac{\sin x}{n}$ for $n \geq 1$. Since $\sin x$ is uniformly bounded by -1 and 1 on $[0, 2\pi]$, we have $f_n \rightrightarrows 0$:



In this example, since the limit function is also 0, this graph also visualises how the error $\|f_n - f\|_\infty \rightarrow 0$ as $n \rightarrow \infty$ uniformly across the whole domain.

8.2.2 Properties of Uniform Convergence

Theorem 8.8. *If $f_n \rightrightarrows f$ on $[a, b]$ and each f_n is continuous on $[a, b]$ then f is continuous on $[a, b]$.*

Proof. We will show that f is uniformly continuous on $[a, b]$. Let ε be arbitrary. By uniform convergence, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, $|f(t) - f_n(t)| < \frac{\varepsilon}{3}$ for all $t \in [a, b]$.

Since f_N is continuous on a compact interval $[a, b]$, it is uniformly continuous on $[a, b]$ so there exists $\delta = \delta(N, \varepsilon)$ such that for all $s, t \in [a, b]$, $|s - t| < \delta \implies |f_N(s) - f_N(t)| < \frac{\varepsilon}{3}$.

Therefore, if $x, y \in [a, b]$ are such that $|x - y| < \delta$, we have

$$\begin{aligned} |f(x) - f(y)| &= |f(x) - f_N(x) + f_N(x) - f_N(y) + f_N(y) - f(y)| \\ &\leq |f(x) - f_N(x)| + |f_N(x) - f_N(y)| + |f_N(y) - f(y)| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} \\ &= \varepsilon. \end{aligned}$$

Thus, f is uniformly continuous on $[a, b]$, so in particular f is continuous on $[a, b]$. $\square$

Theorem 8.9. *If $f_n \rightrightarrows f$ on $[a, b]$ and f_n are integrable on $[a, b]$ then f is integrable and*

$$\int_a^b f(x) \, dx = \lim_{n \rightarrow \infty} \int_a^b f_n(x) \, dx.$$

Proof. We will first show that f is integrable using the ε -criterion for (Darboux) integrability (Theorem 6.6). First we must check that f is bounded: since f_n is integrable on $[a, b]$ for each $n \in \mathbb{N}$, each f_n must be bounded and by uniform convergence there exists $N \in \mathbb{N}$ such that $\|f_n - f\|_\infty < 1$. Then for all $x \in [a, b]$ we have

$$|f(x)| \leq |f(x) - f_N(x)| + |f_N(x)| < 1 + |f_N(x)|,$$

so f is bounded.

Now, let $\varepsilon > 0$ be arbitrary. By uniform convergence fix n sufficiently large such that

$$\|f_n - f\|_\infty < \frac{\varepsilon}{4(b-a)}.$$

Since f_n is integrable, by Theorem 6.6 there exists a partition $P = \{x_0, \dots, x_n\}$ of $[a, b]$ such that

$$U(f_n, P) - L(f_n, P) < \frac{\varepsilon}{2}.$$

For each subinterval $[x_{i-1}, x_i]$, where $1 \leq i \leq n$, we write

$$m_i^n = \inf_{x \in [x_{i-1}, x_i]} f_n(x) \quad \text{and} \quad M_i^n = \sup_{x \in [x_{i-1}, x_i]} f_n(x),$$

and the usual notation m_i, M_i for the function f . Our goal is now to bound the oscillation of f , $M_i - m_i$, by f_n 's oscillation, plus some error term, which we will achieve by using the uniform convergence of f_n to f . From the uniform convergence bound above, for every $x \in [a, b]$ we have

$$f_n(x) - \frac{\varepsilon}{4(b-a)} < f(x) < f_n(x) + \frac{\varepsilon}{4(b-a)},$$

and using standard properties of suprema and infima, taking sup and inf over these inequalities yields

$$m_i \geq m_i^n - \frac{\varepsilon}{4(b-a)} \quad \text{and} \quad M_i \leq M_i^n + \frac{\varepsilon}{4(b-a)},$$

from which it follows that

$$M_i - m_i \leq M_i^n - m_i^n + \frac{\varepsilon}{2(b-a)}.$$

Summing this inequality over all $1 \leq i \leq n$, we get

$$\begin{aligned} U(f, P) - L(f, P) &= \sum_{i=1}^n (M_i - m_i)(x_i - x_{i-1}) \\ &\leq \sum_{i=1}^n (M_i^n - m_i^n)(x_i - x_{i-1}) + \sum_{i=1}^n \frac{\varepsilon}{2(b-a)}(x_i - x_{i-1}) \\ &= U(f_n, P) - L(f_n, P) + \frac{\varepsilon}{2} \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon, \end{aligned}$$

so f is integrable on $[a, b]$ by Theorem 6.6.

To show the integral equality, using standard properties of the integral, we have for $n \in \mathbb{N}$

$$\begin{aligned} \left| \int_a^b f_n(x) \, dx - \int_a^b f(x) \, dx \right| &= \left| \int_a^b (f_n(x) - f(x)) \, dx \right| \\ &\leq \int_a^b |f_n(x) - f(x)| \, dx \\ &\leq (b-a) \|f_n - f\|_\infty, \end{aligned}$$

so since $\|f_n - f\|_\infty \rightarrow 0$ by uniform convergence of f_n to f , and $(b-a)$ is finite, this upper bound can be made arbitrarily small for sufficiently large n , so by definition of sequence convergence (Definition 2.2), we have

$$\lim_{n \rightarrow \infty} \int_a^b f_n(x) \, dx = \int_a^b f(x) \, dx.$$

□

Example 8.10. We can use Theorem 8.9 to evaluate limits of integrals that would otherwise be computationally tedious. Consider the sequence of functions on the closed interval $[0, 1]$:

$$f_n(x) = \frac{x}{1 + nx^2}.$$

First, we find the pointwise limit. For $x = 0$, $f_n(0) = 0$ for all n . For any fixed $x \in (0, 1]$, the denominator grows linearly with n , so $\lim_{n \rightarrow \infty} f_n(x) = 0$. Thus, $f_n \rightarrow 0$ pointwise.

To apply Theorem 8.9, we must verify that this convergence is *uniform*. We find the maximum of $f_n(x)$ using standard calculus:

$$f'_n(x) = \frac{(1 + nx^2) - x(2nx)}{(1 + nx^2)^2} = \frac{1 - nx^2}{(1 + nx^2)^2}.$$

Setting the derivative to zero yields a critical point at $x = 1/\sqrt{n}$; to confirm this is the global maximum on $[0, 1]$, observe the sign of the derivative: for $0 \leq x < 1/\sqrt{n}$, we have $1 - nx^2 > 0$, so $f'_n(x) > 0$ (the function is strictly increasing). For $x > 1/\sqrt{n}$, we have $1 - nx^2 < 0$, so $f'_n(x) < 0$ (the function is strictly decreasing).

Because f_n strictly increases to this point and strictly decreases away from it, the critical point is the maximum (first derivative test). Therefore, evaluating the function here gives the supremum:

$$\|f_n - 0\|_\infty = f_n\left(\frac{1}{\sqrt{n}}\right) = \frac{1/\sqrt{n}}{1 + 1} = \frac{1}{2\sqrt{n}}.$$

Because $\lim_{n \rightarrow \infty} \frac{1}{2\sqrt{n}} = 0$, the sequence converges uniformly to the zero function ($f_n \Rightarrow 0$).

By Theorem 8.9, we are now mathematically guaranteed that the limit of the integrals is the integral of the limit:

$$\lim_{n \rightarrow \infty} \int_0^1 \frac{x}{1 + nx^2} dx = \int_0^1 0 dx = 0.$$

We can verify this by integrating the sequence directly. Using the substitution $u = 1 + nx^2$ (with the change of variables theorem for integrals from a previous exercise), we get

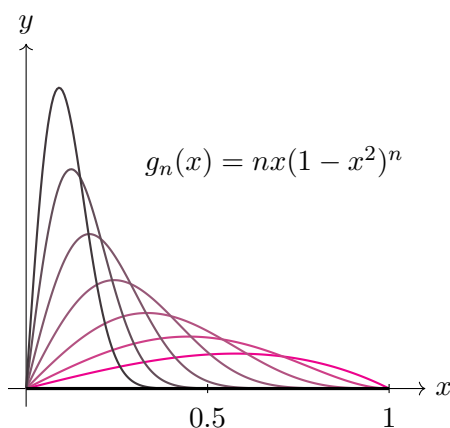
$$\int_0^1 \frac{x}{1 + nx^2} dx = \left[\frac{\ln(1 + nx^2)}{2n} \right]_0^1 = \frac{\ln(1 + n)}{2n}.$$

By L'Hôpital's rule (or comparing growth rates), $\lim_{n \rightarrow \infty} \frac{\ln(1+n)}{2n} = 0$, which indeed matches the theorem's prediction.

Example 8.11. It is also instructive to see what can go wrong when the hypothesis of uniform convergence fails. Consider the sequence of functions $g_n : [0, 1] \rightarrow \mathbb{R}$ defined by

$$g_n(x) = nx(1 - x^2)^n.$$

For $x = 0$, $g_n(0) = 0$. For any $x \in (0, 1]$, the term $(1 - x^2)$ is strictly between 0 and 1. Because exponential decay dominates polynomial growth, $\lim_{n \rightarrow \infty} nx(1 - x^2)^n = 0$. Thus, g_n converges pointwise to the zero function.



If we naively interchange the limit and the integral, we would expect the area to be 0. Let us explicitly compute the integral using the substitution $u = 1 - x^2$ (again using the change of variables theorem for integrals from a previous exercise):

$$\int_0^1 nx(1 - x^2)^n dx = \frac{n}{2} \int_0^1 u^n du = \frac{n}{2} \left[\frac{u^{n+1}}{n+1} \right]_0^1 = \frac{n}{2(n+1)}.$$

Taking the limit as $n \rightarrow \infty$ yields:

$$\lim_{n \rightarrow \infty} \int_0^1 g_n(x) \, dx = \lim_{n \rightarrow \infty} \frac{n}{2n+2} = \frac{1}{2}.$$

The limit of the integrals ($1/2$) does *not* equal the integral of the limit (0).

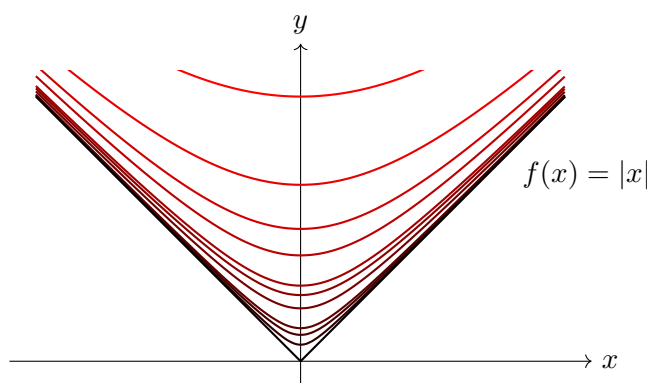
This failure occurs because the convergence $g_n \rightarrow 0$ is pointwise, not uniform. As n increases, the function forms an increasingly thin “spike” of area converging to $1/2$ that slides towards $x = 0$. While the spike eventually passes any fixed point $x > 0$ (causing the pointwise limit to be 0), its maximum height grows towards infinity, violating the uniform bound required by Theorem 8.9, thereby allowing the area, $\int_0^1 g_n$, in the limit as $n \rightarrow \infty$ to be non-zero (one can think of this as a balance between $x \rightarrow 0$ and the height going to infinity, settling on a finite number, $1/2$; indeed, this is represented in the formula for the area, $\frac{n}{2n+2}$).

One may wonder whether differentiability of the sequence functions passes to the limit function when convergence is uniform. With some extra hypotheses this is true, however uniform convergence of the sequence of *derivatives* is required, rather than just uniform convergence of the original differentiable functions.

One way to see why this theorem does not work without uniform convergence of the derivatives is that sharp corners (where a function is not differentiable because the left and right slopes exist, but are not equal) can be “smoothed out” by sequences of functions, all of which are differentiable everywhere. The following example makes this clear.

Example 8.12. Consider the sequence of functions $g_n : [-1, 1] \rightarrow \mathbb{R}$ defined by

$$g_n(x) = \sqrt{x^2 + \frac{1}{n}}.$$



Every function in this sequence is continuously differentiable everywhere with

$$g'_n(x) = \frac{x}{\sqrt{x^2 + \frac{1}{n}}},$$

for each $x \in [-1, 1]$.

We claim that $g_n \rightrightarrows g$ where $g : [-1, 1] \rightarrow \mathbb{R}$ is defined by $g(x) = |x|$. Indeed, for every $x \in [-1, 1]$ and $n \geq 1$ we have

$$0 \leq g_n(x) - g(x) = \sqrt{x^2 + \frac{1}{n}} - |x| = \frac{1}{n} \sqrt{x^2 + \frac{1}{n}} + |x| \leq \frac{1/n}{1/\sqrt{n}} = \frac{1}{\sqrt{n}},$$

so taking the supremum over $[-1, 1]$ we get, for each $n \geq 1$,

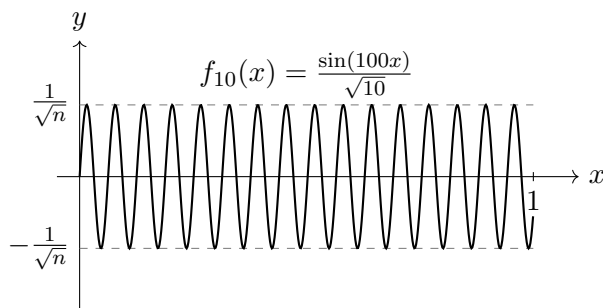
$$\|g_n - g\|_\infty \leq \frac{1}{\sqrt{n}},$$

and $1/\sqrt{n} \rightarrow 0$ as $n \rightarrow \infty$, so $g_n \rightrightarrows g$.

This demonstrates that a uniform limit of differentiable functions need not be differentiable.

Example 8.13. Even in cases where the uniform limit function remains differentiable, its derivative may have no relationship to the derivatives of the original sequence (i.e. the limit may not commute with the derivative). Let $f_n : [0, 1] \rightarrow \mathbb{R}$ be defined by

$$f_n(x) = \frac{\sin(n^2 x)}{n}.$$



Because $|\sin(n^2 x)| \leq 1$, the amplitude of these functions is bounded by $1/n$, meaning f_n converges uniformly to the identically zero function, $f(x) = 0$. This limit function is trivially differentiable everywhere, with $f'(x) = 0$.

However, if we examine the sequence of derivatives, we find $f'_n(x) = n \cos(n^2 x)$. Evaluated at the origin, $f'_n(0) = n$, which diverges to infinity as $n \rightarrow \infty$. Thus, $\lim_{n \rightarrow \infty} f'_n(x) \neq f'(x)$.

This pathological example shows that uniform convergence controls only the amplitude of the functions (to be uniformly within ε of the limit function, past some point $N \in \mathbb{N}$), not their slopes. The functions f_n , in this case, are uniformly squeezed to zero while the increasing oscillations lead to unbounded slopes.

Hence, in order to guarantee that a limit function is differentiable, and to safely commute the limit and derivative operations, we require a stronger theorem.

Theorem 8.14. *Let $f_n: [a, b] \rightarrow \mathbb{R}$ be a sequence of continuously differentiable functions. Assume that:*

1. *The sequence of derivatives f'_n converges uniformly to a function g on $[a, b]$.*
2. *There exists at least one point $x_0 \in [a, b]$ such that the sequence of numbers $f_n(x_0)$ converges in $\mathbb{R}$.*

Then there exists a function f such that $f_n \rightarrow f$ uniformly on $[a, b]$, and

$$f'(x) = g(x) \quad \text{for all } x \in (a, b).$$

In particular, f is differentiable and $f'(x) = \lim_{n \rightarrow \infty} f'_n(x)$ for all $x \in (a, b)$.

Proof. Since each f_n is continuously differentiable on $[a, b]$, the Fundamental Theorem of Calculus gives

$$f_n(x) = f_n(x_0) + \int_{x_0}^x f'_n(t) \, dt$$

for all $x \in [a, b]$. By hypothesis, the sequence of real numbers $(f_n(x_0))$ converges to some limit L , and $f'_n \rightarrow g$ uniformly. Because each f'_n is continuous and converges uniformly to g , the limit function g must also be continuous, and therefore integrable on $[a, b]$. We may thus define our candidate limit function $f: [a, b] \rightarrow \mathbb{R}$ as

$$f(x) := L + \int_{x_0}^x g(t) \, dt.$$

By the Fundamental Theorem of Calculus (Theorem 6.11), since g is continuous (as a uniform limit of continuous functions), f is differentiable and $f'(x) = g(x)$ for all $x \in (a, b)$.

It remains to prove that $f_n \rightarrow f$ uniformly on $[a, b]$. For any $x \in [a, b]$, we evaluate the difference

$$|f_n(x) - f(x)| = \left| f_n(x_0) - L + \int_{x_0}^x (f'_n(t) - g(t)) \, dt \right|.$$

Applying the triangle inequality and bounding the integral by the supremum norm of the integrand yields

$$\begin{aligned} |f_n(x) - f(x)| &\leq |f_n(x_0) - L| + \left| \int_{x_0}^x |f'_n(t) - g(t)| dt \right| \\ &\leq |f_n(x_0) - L| + \|f'_n - g\|_\infty |x - x_0|. \end{aligned}$$

Since $|x - x_0| \leq b - a$ for all $x \in [a, b]$, taking the supremum over the interval gives

$$\|f_n - f\|_\infty \leq |f_n(x_0) - L| + (b - a)\|f'_n - g\|_\infty.$$

As $n \rightarrow \infty$, we have $f_n(x_0) \rightarrow L$ and $\|f'_n - g\|_\infty \rightarrow 0$. Therefore, $\|f_n - f\|_\infty \rightarrow 0$, which proves that f_n converges uniformly to f on $[a, b]$. $\square$

Remark. The key idea in the proof is the Fundamental Theorem of Calculus (Theorem 6.11), which allows us to reconstruct a function globally from a single starting value and its rate of change by integration.

The point x_0 acts as an “anchor” to guarantee the existence of a limit function; because $f_n(x_0)$ converges, this pins the sequence down and prevents the functions from simply drifting vertically to infinity (for example, without this hypothesis the sequence of constant functions $f_1 = 1, f_2 = 2, \dots$ have derivatives which converge uniformly to the zero function, but f_n does not converge to any function). From this point x_0 , we integrate the derivatives f'_n outwards to build the rest of the function f_n , and we are guaranteed that there exists a limit function with the properties we desire by uniform convergence.

Exercise. Show that if $f_n \rightarrow f$ and $g_n \rightarrow g$ uniformly on $[a, b]$, and f_n, g_n are all continuous, then $f_n g_n \rightarrow fg$.

Chapter 9

Series of Functions

Using the tools developed in the previous chapter, we may now discuss the concept of series of functions, for example Taylor series are series of polynomials x^n . More importantly, we may now consider when such series converge uniformly to a limit function, defined in terms of the series. From our discussion of Taylor series, we introduced analytic functions which converge *pointwise* to a limit function $\sum_{n=0}^{\infty} a_n x^n$ on some interval of $\mathbb{R}$. By the end of this chapter, we will be able to analyse when such Taylor series converge uniformly.

If (φ_n) is a sequence of functions on $[a, b]$ then we define the *partial sums*

$$f_n(x) := \sum_{k=1}^n \varphi_k(x), \quad x \in [a, b].$$

Definition 9.1 (Uniform Convergence of Series). We say that the series $\sum_{n=1}^{\infty} \varphi_n$ *converges uniformly* on $[a, b]$ to a function f if the sequence (f_n) converges uniformly on $[a, b]$. In that case, we write

$$f(x) = \sum_{n=1}^{\infty} \varphi_n(x),$$

meaning that $f_n \rightarrow f$ uniformly.

The definition above is just the usual definition of uniform convergence, applied to the sequence of partial sums (f_n) . In particular, we can apply the theorems from last section. In particular, we have the following three corollaries from results established in the previous chapter.

Corollary 9.2. *Let (φ_n) be a sequence of continuous functions on $[a, b]$. If $f_n := \sum_{k=1}^n \varphi_k \Rightarrow f$ then f is continuous.*

Proof. Since each φ_n is continuous, each f_n is continuous by the Algebra of Continuous Functions (Theorem 3.12). Since $f_n \rightarrow f$ uniformly, Theorem 8.8 implies that $f = \sum_{n=1}^{\infty} \varphi_n$ is continuous on $[a, b]$. $\square$

Corollary 9.3. *Let (φ_n) be a sequence of integrable functions on $[a, b]$. If $f_n := \sum_{k=1}^n \varphi_k \Rightarrow f$ then f is integrable and*

$$\int_a^b f(x) \, dx = \int_a^b \sum_{n=1}^{\infty} \varphi_n(x) \, dx = \sum_{n=1}^{\infty} \int_a^b \varphi_n(x) \, dx.$$

Proof. Since each φ_n is integrable, by linearity of the integral each f_n is integrable and, for each $n \in \mathbb{N}$,

$$\int_a^b f_n(x) \, dx = \sum_{k=1}^n \int_a^b \varphi_k(x) \, dx.$$

Applying Theorem 8.9 by uniform convergence of f_n to f gives

$$\int_a^b f(x) \, dx = \lim_{n \rightarrow \infty} \int_a^b f_n(x) \, dx = \lim_{n \rightarrow \infty} \sum_{k=1}^n \int_a^b \varphi_k(x) \, dx = \sum_{n=1}^{\infty} \int_a^b \varphi_n(x) \, dx.$$

$\square$

Corollary 9.4. *Let (φ_n) be a sequence of functions in $C^1([a, b])$, and suppose:*

1. *For some $x_0 \in [a, b]$, the series $\sum_{n=1}^{\infty} \varphi_n(x_0)$ converges;*
2. *The series of derivatives $\sum_{n=1}^{\infty} \varphi'_n$ converges uniformly on $[a, b]$.*

Then the partial sums $f_n := \sum_{k=1}^n \varphi_k$ converge uniformly on $[a, b]$ to a function f , which is differentiable on (a, b) and satisfies

$$f'(x) = \sum_{n=1}^{\infty} \varphi'_n(x) \quad \forall x \in (a, b),$$

where the series on the right is understood as the uniform limit of its partial sums.

9.1 Weierstraß M -Test

In practice, it is often inconvenient to verify uniform convergence directly from the definition, so we would like a sufficient condition that implies uniform convergence of series that is easier to work with. The Weierstraß M -test provides a simple sufficient criterion: if each term (or each derivative term) can be dominated by a numerical series that converges, then the relevant uniform convergence follows automatically.

Definition 9.5 (Uniformly Cauchy sequence). Let $f_n : D \rightarrow \mathbb{R}$ be a sequence of functions. We say that the sequence of functions (f_n) is *uniformly Cauchy* if

$$\forall \varepsilon > 0, \exists N \in \mathbb{N}, \forall m, n \geq N, \|f_m - f_n\|_\infty < \varepsilon.$$

Lemma 9.6. *Let $f_n : D \rightarrow \mathbb{R}$ be a sequence of functions. Then (f_n) converges uniformly if and only if (f_n) is uniformly Cauchy.*

Proof. ($\Rightarrow$) Suppose $f_n \rightarrow f$ uniformly. Given $\varepsilon > 0$, by definition there exists $N \in \mathbb{N}$ such that for all $n \geq N$, $\|f_n - f\|_\infty < \frac{\varepsilon}{2}$. Therefore, by the triangle inequality for the supremum norm we have, for $m, n \geq N$,

$$\|f_m - f_n\|_\infty \leq \|f_m - f\|_\infty + \|f_n - f\|_\infty < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

($\Leftarrow$) Conversely, suppose (f_n) is uniformly Cauchy. Then in particular, the sequence $(f_n(x))$ is Cauchy for all $x \in D$ and since $\mathbb{R}$ is complete, $f_n(x)$ therefore converges for all $x \in D$. Construct $f : D \rightarrow \mathbb{R}$ by letting $f(x) = \lim_{n \rightarrow \infty} f_n(x)$ for all $x \in D$. We now want to show that $f_n \rightarrow f$ uniformly. Given $\varepsilon > 0$, let $N \in \mathbb{N}$ be such that for all $m, n \geq N$ we have $\|f_m - f_n\|_\infty < \frac{\varepsilon}{2}$. In particular, fixing $x \in D$ and $n \in \mathbb{N}$ and letting $m \rightarrow \infty$ yields

$$|f_n(x) - f(x)| \leq \frac{\varepsilon}{2},$$

since $f_m(x) \rightarrow f(x)$ as $m \rightarrow \infty$. Thus for all $x \in D$, $|f_n(x) - f(x)| \leq \frac{\varepsilon}{2} < \varepsilon$, so $f_n \rightarrow f$ uniformly. $\square$

This is an important result. If we are given a concrete sequence of functions, then the usual way to show it converges is to compute the pointwise limit and then prove that the convergence is uniform. However, if we are dealing with sequences of functions in general, this is less likely to work. Instead, it is often much easier to show that a sequence of functions is uniformly convergent by showing it is uniformly Cauchy.

Theorem 9.7 (Weierstraß M -test). *Let (φ_n) be a sequence of functions on $[a, b]$ and define $f_n := \sum_{k=1}^n \varphi_k$ as the partial sums. Suppose there exists numbers $M_n \geq 0$ such that $\|\varphi_n\|_\infty \leq M_n$, and $\sum_{n=1}^\infty M_n$ converges. Then $f_n \Rightarrow f$ on $[a, b]$.*

Proof. First we must show that there exists a limit function, f , since we do not yet have a candidate limit. We do this by using the Cauchy criterion for sequences of functions. Given natural numbers $m < n$ and $x \in [a, b]$, we have, by the triangle inequality and hypothesis,

$$|f_m(x) - f_n(x)| = \left| \sum_{k=n+1}^m \varphi_k(x) \right| \leq \sum_{k=n+1}^m |\varphi_k(x)| \leq \sum_{k=n+1}^m M_k \leq \sum_{k=n+1}^\infty M_k.$$

Since $\sum_{n=1}^\infty M_n$ converges, we know that the tail $\sum_{k=n}^\infty M_k \rightarrow 0$ as $n \rightarrow \infty$, so given $\varepsilon > 0$, $N \in \mathbb{N}$ can be chosen sufficiently large such that for all $n \geq N$, $\sum_{k=n+1}^\infty M_k < \varepsilon$. Therefore, (f_n) is uniformly Cauchy so by Lemma 9.6, $f_n \rightarrow f$ uniformly on $[a, b]$. $\square$

Example 9.8. Let $\varphi_n : [0, 1] \rightarrow \mathbb{R}$ be defined by $\varphi_n(x) = \frac{\cos(nx)}{n^2}$. Then $\|\varphi_n\|_\infty \leq \frac{1}{n^2}$ for each $n \in \mathbb{N}$, and the series $\sum_{n=1}^\infty \frac{1}{n^2}$ converges by the p -series test since $p = 2 > 1$. Therefore, the partial sums $f_n := \sum_{k=1}^n \varphi_k$ converge uniformly to $f := \sum_{n=1}^\infty \varphi_n$ on $[0, 1]$. The limit function $f : [0, 1] \rightarrow \mathbb{R}$ defined by $f(x) = \sum_{n=1}^\infty \frac{\cos(nx)}{n^2}$ is called a *Fourier series*.

By Corollary 9.4, this Fourier series can be differentiated term-by-term to yield a new Fourier series (since $\cos$ differentiates to $-\sin$), $g : [0, 1] \rightarrow \mathbb{R}$ defined by

$$g'(x) = f(x) = - \sum_{n=1}^\infty \frac{\sin(nx)}{n}.$$

Note: Fourier series and their computation is explored in Calculus and Applications (MATH40004).

9.2 Uniform Convergence of Power Series

A particularly interesting kind of series is a power series, and we have already met these in Chapter 7, and proved some results about them. However, as mentioned in the introduction to this chapter, our results were pointwise results, discussing how the series behaves at a particular point $x \in \mathbb{R}$. Here we will quickly look into how the power series behaves as a function of x . In particular, we want to know whether it converges uniformly.

Theorem 9.9. Let $\sum_{n=0}^{\infty} c_n(x-a)^n$ be a (real) power series. Then there exists a unique number $R \in [0, +\infty]$ (called the radius of convergence) such that

1. If $|x-a| < R$, then $\sum c_n(x-a)^n$ converges absolutely;
2. If $|x-a| > R$, then $\sum c_n(x-a)^n$ diverges;
3. If $R > 0$ and $0 < r < R$ then $\sum c_n(x-a)^n$ converges uniformly on $[a-r, a+r]$.

Proof. Items (1) and (2) were proven in Chapter 2. For (3), note that from (1), taking $x = a-r$ shows that $\sum |c_n|r^n$ is convergent. But if $x \in [a-r, a+r]$ then $|x-a| \leq r$ so

$$|c_n(x-a)^n| \leq |c_n|r^n,$$

so the result follows by the Weierstraß M -test (Theorem 9.7) with $M_n = |c_n|r^n$ which is independent of x . $\square$

Note that these results hold for complex power series as well. Also, uniform convergence need not hold on the entire interval of convergence, as the following example will demonstrate.

Example 9.10. Consider the power series $\sum x^n$. The interval of convergence of this power series is $(-1, 1)$, however it does not converge *uniformly* on $(-1, 1)$. For $m < n$, the tail

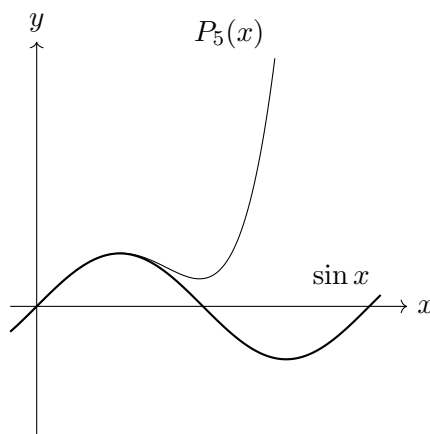
$$\sum_{k=m}^n x^k = x^m \sum_{k=0}^{n-m} x^k \geq \frac{x^m}{1-x},$$

for $x \in (-1, 1)$. Since uniform convergence for series of functions relies on this tail going to zero uniformly in x , but the above expression on the right hand side can be made arbitrarily small by choosing x close to 1, we do not have uniform convergence on $(-1, 1)$.

Remark. The idea for why power series do not necessarily converge on their entire *open* interval of convergence is the same as the idea for why uniform continuity may only work on a compact interval and not the full open interval, for example the function $x \mapsto \frac{1}{x}$ on $(0, 1)$ is not uniformly continuous, but is uniformly continuous on $[r, 1)$ for any $0 < r < 1$. It is due to the fact that, with open intervals, there are limit points which are not in the interval itself, where points x in the interval can “escape” to. If these points are where a function has a singularity (for example $\frac{1}{1-x}$ in the above example, near $x = 1$), then we cannot control *uniformly* in x in the open interval, because by virtue of the interval being *open*, one can always choose a point closer to

the singularity, breaking the notion *uniform* convergence (we are required to find some $N \in \mathbb{N}$, that works for all x in the interval, whereas pointwise convergence relies on choosing an $N \in \mathbb{N}$ that may depend on ε and x).

Another way to see this, specifically for Taylor series, is that polynomials have tails that blow up to $\pm\infty$ eventually, so if this is the Taylor series of, say, a bounded function on $\mathbb{R}$, we cannot possibly expect the Taylor series to converge uniformly on $\mathbb{R}$, because for each finite $N \in \mathbb{N}$ we choose, the Taylor polynomial approximation will blow up eventually for some large x , but the function is bounded, so while pointwise convergence holds here, uniform convergence does not, because no matter which $N \in \mathbb{N}$ we choose, there is always a large x such that the Taylor polynomial has blown up and has “stripped away” from the target function. This is illustrated below with the function $x \mapsto \sin x$ on $\mathbb{R}$.



9.3 Beyond the Weierstraß M -Test

One should also remember that the M -test is only a sufficient condition; uniform convergence may still hold if no summable supremum bound exists.

For instance, two different mechanisms by which we can construct sequences of functions which converge uniformly are:

- *Geometric structure*: the functions in the sum have (almost) disjoint supports, so that at each point only few functions are “active”;
- *Cancellation*: (e.g. alternating signs) where large terms cancel in the partial sums.

The next two examples illustrate these two phenomena.

Example 9.11 (Disjoint supports: uniform convergence without a summable bound on $|\phi'_n|$). We will work on the compact interval $[0, 1]$. Define

$\text{dist}(t, \{1, 2\}) := \min\{|t-1|, |t-2|\}$ to be the shortest distance from $t \in [1, 2]$ to the points in the set $\{1, 2\}$, and define the continuous “tent” function $\tau : \mathbb{R} \rightarrow \mathbb{R}$ by

$$\tau(t) := \begin{cases} \text{dist}(t, \{1, 2\}), & t \in [1, 2], \\ 0, & t \notin [1, 2]. \end{cases}$$

Therefore, the *support* of τ (that is, the set of values of t for which $\tau(t) \neq 0$) is contained in $[1, 2]$, and the supremum of τ is $\frac{1}{2}$.

Now we will construct our sequence of functions. For $n \geq 1$, define

$$\psi_n(x) := \frac{2}{n} \tau(2^{n+1}x), \quad x \in [0, 1],$$

and set

$$\phi_n(x) := \int_0^x \psi_n(t) dt,$$

so that in particular $\phi'_n = \psi_n$ by the Fundamental Theorem of Calculus (Theorem 6.11).

Now, notice that since τ vanishes outside of the interval $[1, 2]$, the support of each ψ_n is the set of values of x for which

$$1 < 2^{n+1}x < 2,$$

i.e. $(2^{-(n+1)}, 2^{-n})$ is the support of ψ_n . These intervals are pairwise disjoint.

Uniform convergence of the derivative series. Because $\sup |\tau| = \frac{1}{2}$, we have $\sup |\psi_n| = \frac{1}{n}$ by construction. Fixing $N \geq 1$, for any $x \in [0, 1]$ we see that at most one term $\psi_n(x)$ with $n \geq N+1$ is non-zero, so for all $m > N$ we have

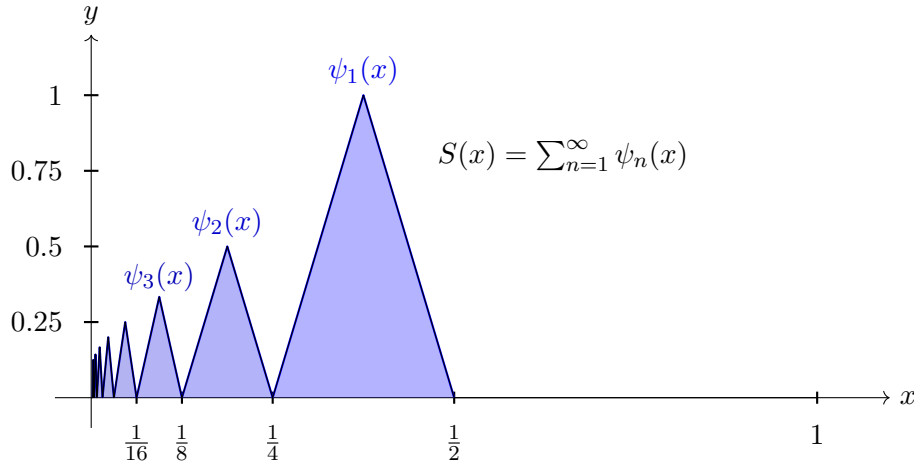
$$\left| \sum_{n=N+1}^m \phi'_n(x) \right| = \left| \sum_{n=N+1}^m \psi_n(x) \right| \leq \sup_{n \geq N+1} |\psi_n| = \frac{1}{N+1}.$$

Thus the derivative series $\sum \phi'_n$ is uniformly Cauchy, so it converges uniformly on $[0, 1]$.

The uniform limit function is visualised below; the idea behind the construction is that each $x \in [0, 1]$ belongs to precisely one interval of the form $(2^{-(n+1)}, 2^{-n})$, on which only the function ψ_n is supported, so the infinite sum reduces to $\psi_n(x)$. The process of uniform convergence, here, acts as a “glue” stitching infinitely many localized functions ψ_n together across the domain.

The geometric separation of the functions guarantees that uniform convergence relies exclusively on the spatial disjointness of the functions; because at most one term in the sum is non-zero for any x , the absolute value of the tail of the series reduces to the absolute value of a single active term. The looseness of the M -test arises because replacing $|\psi_n(x)|$ with $\sup |\psi_n|$ discards the spatial dependence of each function. The final bound represents

the sum of the global suprema regardless of where those values are attained. In this case we were able to construct a uniformly convergent function where this sup-norm summation is $\sum \frac{1}{n}$, which famously diverges.



Uniform convergence of the original series. Each ϕ_n is increasing and non-negative, so $\sup |\phi_n| = \phi(1)$. Moreover,

$$\sup |\phi_n| = \int_0^1 \psi_n(t) dt = \frac{2}{n} \int_0^1 \tau(2^{n+1}t) dt,$$

and with the substitution $u = 2^{n+1}t$ we obtain

$$\sup |\phi_n| = \frac{2}{n} 2^{-(n+1)} \int_1^2 \text{dist}(u, \{1, 2\}) du.$$

Now, by the formulae for areas of triangles, the value of the integral is $\frac{1}{4}$ (tent of base 1, height $\frac{1}{2}$), so $\sup |\phi_n| = \frac{2^{-(n+2)}}{n}$. Since $\sum \frac{2^{-(n+2)}}{n}$ converges by the ratio test, the Weierstraß M -test (Theorem 9.7) implies that $\sum \phi_n$ converges uniformly on $[0, 1]$.

Example 9.12 (Cancellation: uniform convergence although $\sum \sup |\phi_n|$ and $\sum \sup |\phi'_n|$ diverge). Again, we work on $[0, 1]$. For $n \geq 1$, define

$$\phi_n(x) := (-1)^{n-1} \frac{x}{n+x}, \quad x \in [0, 1].$$

Then $\phi_n \in C^1([0, 1])$ and

$$\phi'_n(x) = (-1)^{n-1} \frac{n}{(n+x)^2}.$$

Uniform convergence of $\sum \phi_n$ (alternating series mechanism). Let $u_n(x) := \frac{x}{n+x}$. For each fixed $x \in [0, 1]$, the sequence $u_n(x)$ is decreasing in n and $u_n(x) \rightarrow 0$ as $n \rightarrow \infty$. Moreover,

$$0 \leq u_n(x) \leq \sup_{t \in [0, 1]} u_n(t) = \frac{1}{n+1},$$

so using the alternating series estimate gives, for every $N \geq 1$ and every $x \in [0, 1]$,

$$\left| \sum_{n=N+1}^{\infty} \phi_n(x) \right| \leq u_{N+1}(x) \leq \frac{1}{N+2}.$$

Hence, $\sum \phi_n$ converges uniformly on $[0, 1]$.

Uniform convergence of $\sum \phi'_n$. Set $v_n(x) := \frac{n}{(n+x)^2}$. For each fixed $x \in [0, 1]$, the function $s \mapsto \frac{s}{(s+x)^2}$ is decreasing for $s \geq x$; in particular, $v_n(x)$ is decreasing in n for all $n \geq 2$. Also, $0 \leq v_n(x) \leq v_n(0) = \frac{1}{n}$, so $v_n(x) \rightarrow 0$ uniformly. Applying the alternating series estimate again to $\sum_{n=2}^{\infty} (-1)^{n-1} v_n(x)$ yields, for every $N \geq 2$ and $x \in [0, 1]$,

$$\left| \sum_{n=N+1}^{\infty} \phi'_n(x) \right| \leq v_{N+1}(x) \leq \frac{1}{N+1},$$

so $\sum \phi'_n$ converges uniformly on $[0, 1]$.

Failure of the M-test. We have

$$\sup |\phi_n| = \sup_{x \in [0,1]} \frac{x}{n+x} = \frac{1}{n+1} \quad \text{and} \quad \sup |\phi'_n| = \sup_{x \in [0,1]} \frac{n}{(n+x)^2} = \frac{1}{n},$$

so both $\sum \sup |\phi_n|$ and $\sum \sup |\phi'_n|$ diverge, so the Weierstraß M -test again does not apply, despite uniform convergence of both series.

Finally, note also that Since $\sum \phi_n(0) = 0$ and $\sum \phi'_n$ converges uniformly, Corollary 9.4 still applies: the sum $f := \sum \phi_n$ is differentiable on $(0, 1)$ and $f' = \sum \phi'_n$ there.

Chapter 10

The Trigonometric Functions

Chapter 11

Curves in the Plane